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UNIVERSITÀ DI ROMA

Spacecraft Design Project

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AODCS

Analysis and Design for the 6U HyonoSat CubeSat

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1. Introduction

1.1. CubeSat History

The CubeSat form is a standardized layout of satellite that has begun revolutionizing access to space. Basically, it is made up of multiples U units (10×10×10 cm cubic units) and its main purpose is for scientific experimentation and low-cost mission. Indeed, its relatively low costs and brief design process made it the perfect means for testing new technologies and introduce students in the space systems' design.

1.2. HyonoSat CubeSat

HyonoSat, for which the AODCS has been designed, is a 6U CubeSat aiming at obtaining precise spectral information on land, water resources and atmosphere, by using the strengths of hyperspectral data, the primary payload shall consist therefore in a Hyperspectral sensor. HyonoSat will also be capable of evaluating the Total Electron Content (TEC) in the high Ionosphere by using two accurate GPS receivers.

1.3. AODCS motivation and objectives

The purpose of this design is development and verification of AODCS algorithm for CubeSat HyonoSat. AODCS is one of the essential subsystems of CubeSat for successful mission. This subsystem plays major rule for survival from external disturbances which generates disturbance torques to the CubeSat.

Necessity of AODCS could be summed up to three reasons. First reason is due to communication problem. Since, HyonoSat use directional S-band antenna for payload and omnidirectional UHF antenna to send mission data to ground station, the bottom face which adheres S-band antenna should aim at the ground station while passing by. The second reason is due to GPS receiver. Since HyonoSat is in 400-600[km] height, GPS receiver is adhered in upper face in order to get GPS signals from GPS. Therefore, HyonoSat should always maintain nadir pointing in order to get mission data and transmit it successfully to the ground station. The third reason is the presence of the primary payload that is the Hyperspectral camera which has to point toward the Earth.

It is almost impossible to realize space environment in ground to verify AODCS algorithm. Therefore, space environment simulator is used for verification. The purpose of space environment simulator is to verify MATLAB based AODCS algorithm and extend it to OBC (On-Board Computer) based AODCS algorithm afterwards. The proposed algorithm should meet AODCS requirements considering operation scenario for low earth orbit. The space mission simulation software that was

used in this design were 2: STK (Satellite Tool Kit) and GMAT (General Mission Analysis Tool).

The main objective of our work is to model the AODCS subsystem for the HyonoSat mission. Our subsystem is divided into two large areas: orbit and attitude. For the first area, through sensors, AODCS is able to gather environment data and is capable of determine and maintain a Sun-synchronous orbit by operating its actuators. Like in most Earth-observing missions, this type of orbit places the satellite in constant sunlight allowing him to perform imagery in a well-lit Earth whilst producing its electrical power through its solar panels. Moreover, the subsystem, in addition to determining and maintaining the Sun-synchronous orbit, is able to perform both passive and active deorbiting to satisfy the requirement of 25 years in orbit. For the second area, the subsystem is able to determine and control the attitude of the CubeSat during its entire operational life, acquiring information from on-board sensors and activating the actuators to keep pointing to the ground and counteract the disturbances encountered in orbit.

2. Mathematical Modelling

Before starting with the proper AODCS design, lets introduce some mathematical models used for the dynamics and kinematics of the spacecraft, as well as the models and assumptions used for the environment and the inertia of the spacecraft itself.

2.1. Reference frames

It is important to define various reference frames when working with relative motion of two objects. Three types of coordinate frames which are particularly useful for orbiting satellites are inertial frames, orbital frames, and body-fixed coordinate frames. The frames used in the research are described below. Each coordinate frame has advantages and disadvantages and moving from one frame to another must be taken carefully.

2.1.1. Earth Centered Inertial frame

The typical inertial frame used in satellite attitude models is Earth Centered Inertial (ECI) frame. The coordinate system has the origin at the centre of the Earth where the z-axis points in the direction of true North, the x-axis points in the direction of the vernal equinox, and the y-axis completes the right-hand orthogonality rule. While this is not a truly inertial frame as it precesses over a long period of time, in the case of most satellites it can be considered inertial.

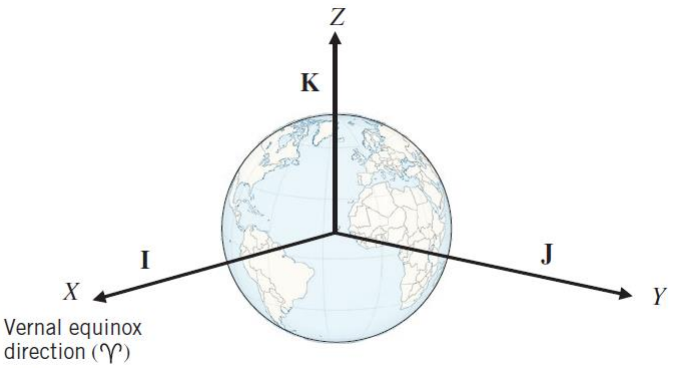


Figure 2.1: Earth Centered Inertial Frame

2.1.2. Earth-Centered Earth-Fixed Frame

Another common reference frame is Earth Centered Earth Fixed which is similar to the ECI frame except it is non-inertial as it fixed to the Earth. The z-axis is co-aligned in

the ECI and ECEF frame, but the x-axis in ECEF frame points to the Prime Meridian, and again the y-axis completes the right-hand rule. The ECEF frame is typically used for geolocation and determining the magnetic field vector at the satellite's position.

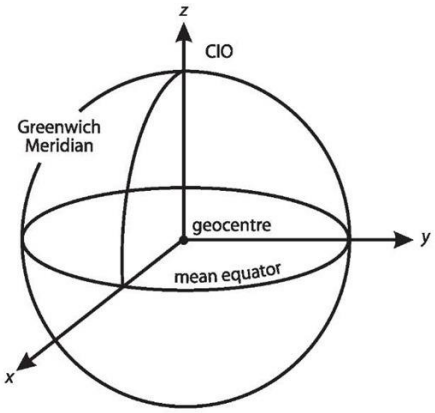


Figure 2.2: Earth Centered Earth Fixed Frame

2.1.3. Perifocal coordinate system

Another useful frame is the P,Q,W frame where $\hat{P}$ and $\hat{Q}$ are unit vectors in the orbit plane, with $\hat{P}$ directed to perigee, $\hat{W}$ along the angular momentum vector and $\hat{Q}$ completing a right hand triad, i.e., $\hat{P} \times \hat{Q} = \hat{W}$. This reference frame is useful to define the initial condition and the position of the spacecraft along its orbit and then to identify its position and velocity with respect to the inertial frame.

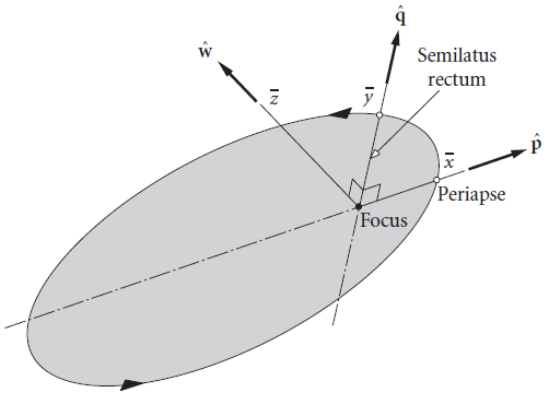


Figure 2.3: Perifocal Coordinate System

2.1.4. NED

NED: North-East-Down (NED) is an inertial frame defined relative to Earth's surface with the x-axis pointing towards north, z-axis downwards perpendicular to Earth's

reference ellipsoid. The y-axis completes a right-handed orthogonal coordinate system, with the positive y-axis pointing towards East. Earth's reference ellipsoid is a mathematically defined surface fitted to approximate the shape of the Earth.

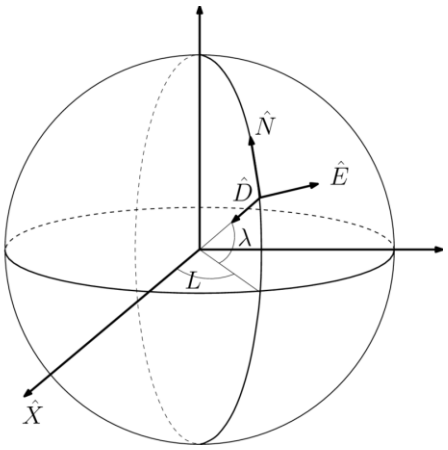


Figure 2.4: North-East-Down (NED) Frame

2.1.5. Roll, Pitch, Yaw Frame (Orbital frame)

The roll pitch yaw (RPY) frame is a common frame of reference for nadir pointing spacecraft as one axis is always pointing towards the center of the Earth. The RPY frame is an orbital frame where the roll axis (x-direction) is along the velocity tangent of the orbit, the yaw axis (z-direction) points towards the center of the earth, and the pitch axis (y-direction) points normal to the orbital plane.

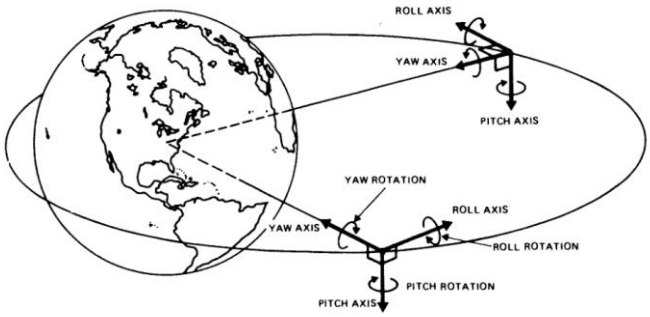


Figure 2.5: Roll Pitch Yaw Frame

2.1.6. Body coordinate system

The body-fixed or spacecraft-fixed coordinate system is a user defined frame that is fixed on the spacecraft. This coordinate system places its origin at the spacecraft's center of mass (COM) and follows the right-hand rule for the user's defined orientation. The coordinate axes can align with the spacecraft's principal axes using a coordinate

transformation which uncouples the equations of motion. The body-fixed frame used has the x-axis normal to the 3U side, y-axis normal to the 6U side, and the z-axis normal to the 2U side as.

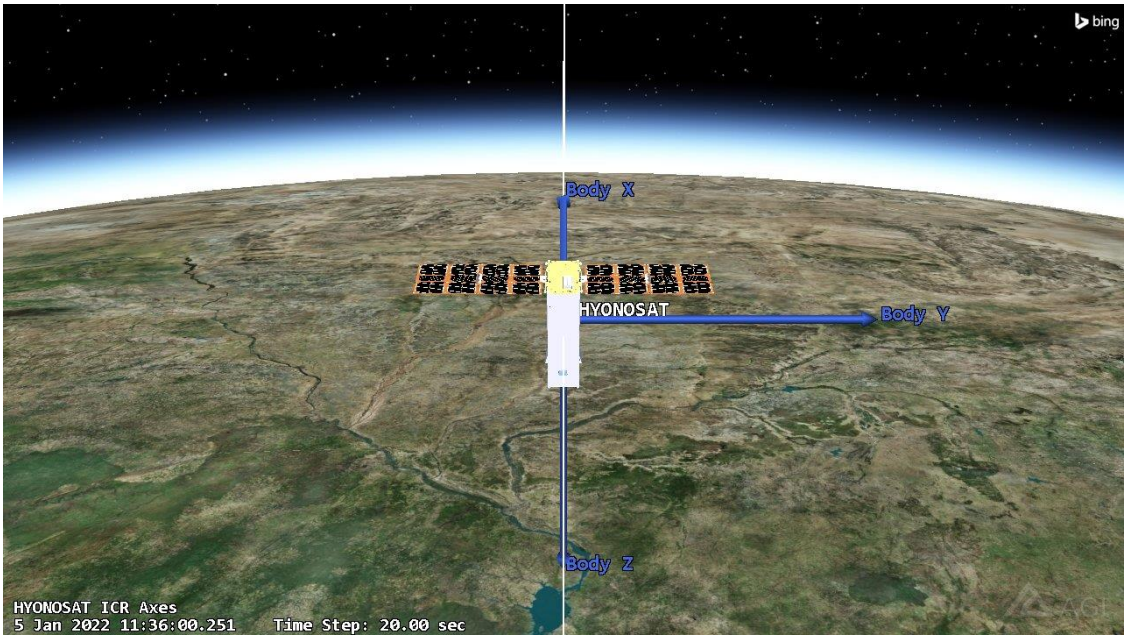


Figure 2.6: Body coordinate frame

2.2. Attitude kinematics

Kinematics is the study of a rigid-body motion without considering the forces that cause them to move. Rotational kinematics describes the orientation of a body that is in rotational motion. Let the two reference frames N and B each be defined through sets of orthonormal right-hand sets of vectors $\{\hat{n}\}$ and $\{\hat{b}\}$.

$$\{\hat{n}\} = \begin{Bmatrix} \hat{n}_1 \\ \hat{n}_2 \\ \hat{n}_3 \end{Bmatrix} \qquad \{\hat{b}\} = \begin{Bmatrix} \hat{b}_1 \\ \hat{b}_2 \\ \hat{b}_3 \end{Bmatrix} \tag{2.2.1}$$

The reference frame B can be thought of as being a generic rigid body, and the reference frame N could be associated with some inertial coordinate system.

2.2.1. Direction Cosine Matrix

Let the three angles α_{1i} be the angles formed between the first body vector $\hat{b}_1$ and the three inertial axes $\hat{n}_1$, $\hat{n}_2$, and $\hat{n}_3$. The cosines of these angles are called the direction cosines of $\hat{b}_1$ relative to the N frame.

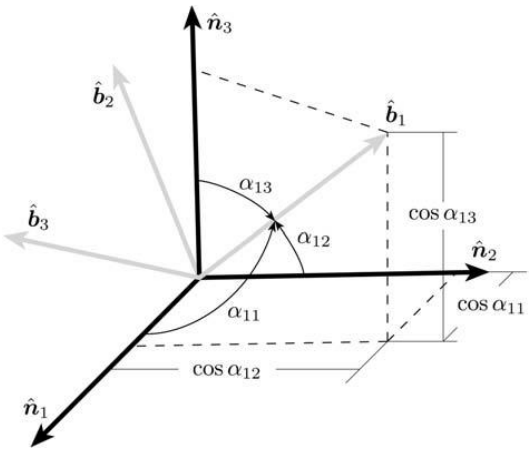


Figure 2.7: Direction cosines

The set of orthonormal base vectors $\{\hat{b}\}$ can be compactly expressed in terms of the base vectors $\{\hat{n}\}$ as

$$\{\hat{b}\} = \begin{bmatrix} \cos\alpha_{11} & \cos\alpha_{12} & \cos\alpha_{13} \\ \cos\alpha_{21} & \cos\alpha_{22} & \cos\alpha_{23} \\ \cos\alpha_{31} & \cos\alpha_{32} & \cos\alpha_{33} \end{bmatrix} \{\hat{n}\} = [C]\{\hat{n}\} \tag{2.2.2}$$

Where the matrix $[C]$ is called direction cosine matrix that is orthogonal. Note that each entry of $[C]$ can be computed through:

$$C_{ij} = \hat{b}_i \cdot \hat{n}_j = cos\,\alpha_{ij} \tag{2.2.3}$$

To find the kinematic differential equation in terms of the direction cosine matrix $[C]$, let us write the instantaneous angular velocity vector ω of the B frame relative to the N frame in B frame orthogonal components as

$$\omega = \omega_1\hat{b}_1 + \omega_2\hat{b}_2 + \omega_3\hat{b}_3 \tag{2.2.4}$$

After introducing the skew-symmetric tilde matrix operator

$$[\tilde{x}] = \begin{bmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{bmatrix} \tag{2.2.5}$$

The kinematic equation is

$$[\dot{C}] = -[\tilde{\omega}][C] \tag{2.2.6}$$

Using explicit frame labelling

$$\left[\dot{BN}\right] = -\left[\widetilde{\omega}_{B|N}\right][BN] \tag{2.2.7}$$

Given an arbitrary time history of $\omega(t)$, this equation represents a rigorously linear differential equation that can be integrated to yield the instantaneous direction cosine matrix $[C]$. A major advantage of the kinematic differential equation for $[C]$ is that it is linear and universally applicable. There are no geometric singularities present in the attitude description or its kinematic differential equations. However, this advantage comes at the cost of having a highly redundant formulation. Several other attitude parameters will be presented in the following sections that include a minimal number (3) of attitude parameters. However, all minimal sets of attitude coordinates have kinematic differential equations that contain some degree of nonlinearity and also embody geometric and/or mathematical singularities. Only the once-redundant Euler parameters (quaternions) will be found to retain a singularity-free description and possess linear kinematic differential equations analogous to the direction cosine matrix.

2.2.2. Euler Angles

The most commonly used sets of attitude parameters are the Euler angles. They describe the attitude of a reference frame B relative to the frame N through three successive rotation angles $(\theta_1, \theta_2, \theta_3)$ about the sequentially displaced body-fixed axes $\{\hat{b}\}$. There are 12 possible combinations of rotations which can be divided into two types. The first type incorporates rotations about each axis and has a singularity when $\theta = \pm 90\, deg$. The second type repeats a rotation about an axis but not in successive rotations and has a singularity at $\theta = 0\, deg$ and $180\, deg$. A common choice in rotation sequence is 3-2-1 which is a rotation about the z-axis then a rotation about the intermediate y-axis and finally a rotation about the transformed x-axis which the direction cosines matrix is given by:

$$C_{321}(\phi, \theta, \psi) = C_x(\phi)C_y(\theta)C_z(\psi) =$$

$$= \begin{bmatrix} c(\theta)c(\psi) & c(\theta)s(\psi) & -s(\theta) \\ s(\phi)s(\theta)c(\psi) - c(\phi)s(\psi) & s(\phi)s(\theta)s(\psi) + c(\phi)c(\psi) & s(\phi)c(\theta) \\ c(\phi)s(\theta)c(\psi) + s(\phi)s(\psi) & c(\phi)s(\theta)s(\psi) - s(\phi)c(\psi) & c(\phi)c(\theta) \end{bmatrix} \tag{2.2.8}$$

Where $c(a) = \cos a$ and $s(a) = \sin(a)$. For the kinematic equation in Euler angles for the rigid-body spacecraft's attitude is given by:

$$\begin{pmatrix} \dot{\psi} \\ \dot{\theta} \\ \dot{\phi} \end{pmatrix} = \frac{1}{\cos\theta} \begin{bmatrix} 0 & \sin\phi & \cos\phi \\ 0 & \cos\phi\cos\theta & -\sin\phi\cos\theta \\ \cos\theta & \sin\phi\sin\theta & \cos\phi\sin\theta \end{bmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix}^B = [B(\psi, \theta, \phi)]\omega^B \tag{2.2.9}$$

The Euler angles provide a compact, three-parameter attitude description whose coordinates are easy to visualize. One major drawback of these angles is that a rigid body or reference frame is never further than a 90-deg rotation away from a singular orientation. Therefore, their use in describing large, and in particular arbitrary, rotations is limited. Also, their kinematic differential equations are fairly nonlinear, containing computationally intensive trigonometric functions. The linearized Euler angle kinematic differential equations are valid only for a relatively small domain of rotations.

2.2.3. Quaternions

Another popular set of attitude coordinate are the four Euler parameters (quaternions). They provide a redundant, non-singular attitude description and are well suited to describe arbitrary, large rotations. Quaternions can be used as an alternative to Euler angles. They have the advantage of avoiding the singularities associated with Euler angle sequences. Quaternions add an additional parameter, a scalar, to the vector. Another advantage of using quaternions is computational efficiency since the equations of motion are then purely algebraic and do not contain any trigonometric functions. The quaternion vector q is defined in terms of the principal rotation elements as

$$\begin{aligned} q_1 &= e_1 \sin\left(\frac{\Phi}{2}\right) \\ q_2 &= e_2 \sin\left(\frac{\Phi}{2}\right) \\ q_3 &= e_3 \sin\left(\frac{\Phi}{2}\right) \\ q_4 &= \cos\left(\frac{\Phi}{2}\right) \end{aligned} \tag{2.2.10}$$

The quaternion satisfies the holonomic constraint:

$$q_1^2 + q_2^2 + q_3^2 + q_4^2 = 1 \tag{2.2.11}$$

The direction cosine matrix can be written in terms of quaternion as

$$A(q) = \begin{bmatrix} q_1^2 - q_2^2 - q_3^2 + q_4^2 & 2(q_1q_2 + q_3q_4) & 2(q_1q_3 - q_2q_3) \\ 2(q_1q_2 - q_3q_4) & -q_1^2 + q_2^2 - q_3^2 + q_4^2 & 2(q_2q_3 + q_1q_4) \\ 2(q_1q_3 + q_2q_4) & 2(q_2q_3 - q_1q_4) & -q_1^2 - q_2^2 + q_3^2 + q_4^2 \end{bmatrix} \tag{2.2.12}$$

From which we can obtain the quaternion as:

$$\begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix} = \begin{bmatrix} \frac{1}{4q_4}(C_{23} - C_{32}) \\ \frac{1}{4q_0}(C_{31} - C_{13}) \\ \frac{1}{4q_0}(C_{12} - C_{21}) \\ \pm\frac{1}{2}(C_{11} + C_{22} + C_{33} + 1)^{\frac{1}{2}} \end{bmatrix} \tag{2.2.13}$$

For the kinematic equation let us indicate with $q(t)$ the quaternion at a time t. The quaternion at time $t + \Delta t$ is represented as a quaternion composition:

$$q(t + \Delta t) = q(\Delta t) \otimes q(t) \tag{2.2.14}$$

where $q(\Delta t)$ is the quaternion representing the rotation taking place between time t and $t + \Delta t$.

$$\frac{dq}{dt} = \frac{1}{2}\Omega(\omega)q(t) = \frac{1}{2}\begin{bmatrix} 0 & \omega_z & -\omega_y & \omega_x \\ -\omega_z & 0 & \omega_x & \omega_y \\ \omega_y & -\omega_x & 0 & \omega_z \\ -\omega_x & -\omega_y & -\omega_z & 0 \end{bmatrix} q(t) \tag{2.2.15}$$

This is the fundamental kinematical equation for the spacecraft attitude. Remarkably, if we have a way to determine angular velocities onboard (e.g., by gyros), then we need not to integrate the dynamical attitude equations. It suffices to integrate the kinematical equation above. The Eq. 2.2.15 can also be written as

$$\begin{bmatrix} \dot{q}_0 \\ \dot{q}_1 \\ \dot{q}_2 \\ \dot{q}_3 \end{bmatrix} = \frac{1}{2}\begin{bmatrix} q_0 & -q_1 & -q_2 & -q_3 \\ q_1 & q_0 & -q_3 & q_2 \\ q_2 & q_3 & q_0 & -q_1 \\ q_3 & -q_2 & q_1 & q_0 \end{bmatrix} \begin{bmatrix} 0 \\ \omega_1 \\ \omega_2 \\ \omega_3 \end{bmatrix} \tag{2.2.16}$$

2.3. Rigid body dynamics

Dynamics is the study of motion and the forces affecting motion. In this study, the spacecraft is assumed to be a rigid body meaning that there is no deformation of any particle of the spacecraft with respect to any other particle. Therefore, the motion of the spacecraft is a superposition of a translation (orbital dynamics) and a rotation (attitude dynamics).

The equation of motion for the spacecraft attitude is based on Euler's moment equation: the change of angular momentum of the satellite is equal to the external torque applied about the center of mass of the rigid body:

$$\dot{\boldsymbol{h}} = \boldsymbol{T_{ext}} \tag{2.3.1}$$

where the angular momentum vector of the rigid body is given by

$$\boldsymbol{h} = I\boldsymbol{\omega} \tag{2.3.2}$$

Where I is the inertia matrix and $\boldsymbol{\omega}$ is the angular velocity vector.

$$\begin{bmatrix} h_x \\ h_y \\ h_z \end{bmatrix} = \begin{bmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{yx} & I_{yy} & -I_{yz} \\ -I_{zx} & -I_{zy} & I_{zz} \end{bmatrix} \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} \tag{2.3.3}$$

Combining Eq. 2.3.1 with Eq. 2.3.2 yields the equations of angular motion of the spacecraft in body-fixed frame with respect to the inertial frame:

$$I\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times I\boldsymbol{\omega} = \boldsymbol{T_{ext}} \tag{2.3.4}$$

This is the well-known Euler's moment equation. By choosing a body-fixed coordinate system that is aligned with the principal body axes, the inertia matrix $[I]$ will be diagonal and Eq. 2.3.4 reduces to

$$\begin{cases} I_x\dot{\omega}_x + (I_z - I_y)\omega_y\omega_z = T_x \\ I_y\dot{\omega}_y + (I_x - I_z)\omega_x\omega_z = T_y \\ I_z\dot{\omega}_z + (I_z - I_y)\omega_x\omega_y = T_z \end{cases} \tag{2.3.5}$$

Since the HyonoSat satellite is a 6-unit (6U) CubeSat, one can use the following inertia tensor for a cuboid (with the body frame located at the center) of height h, width w, and depth d, and mass m:

$$I_{CG} = \begin{bmatrix} \frac{1}{12}m(h^2 + d^2) & 0 & 0 \\ 0 & \frac{1}{12}m(w^2 + d^2) & 0 \\ 0 & 0 & \frac{1}{12}m(w^2 + h^2) \end{bmatrix} \tag{2.3.6}$$

It is important that $I_y > I_x > I_z$ in order to maintain the equilibriums of the satellite in the magnetic field (it is the result of how the gravity will affect the moment of inertia). Eq. 2.3.5 are nonlinear, so they don't have analytical closed-form solution. We shall break down the external torque *T* into two principal parts: $\boldsymbol{T_c}$, the control moments to be used for controlling the attitude motion of the satellite; and $\boldsymbol{T_d}$, those moments due to different disturbing environmental phenomena. The total torque vector is thus:

$$\boldsymbol{T} = \boldsymbol{T_c} + \boldsymbol{T_d} \tag{2.3.7}$$

The momentum of the entire system will be divided between the momentum of the rigid body $\boldsymbol{h_B} = \left[h_x, h_y, h_z\right]^T$ and the momentum of the moment exchange devices $\boldsymbol{h_w} = \left[h_{wx}, h_{wy}, h_{wz}\right]^T$. Finally, $\boldsymbol{h} = \boldsymbol{h_B} + \boldsymbol{h_w}$. With this definition, the general equations in the body axis frame become:

$$\begin{pmatrix} T_x \\ T_y \\ T_z \end{pmatrix} = \begin{pmatrix} \dot{h}_x + \dot{h}_{wx} + \left(\omega_y h_z - \omega_z h_y\right) + \left(\omega_y h_{wz} - \omega_z h_{wy}\right) \\ \dot{h}_y + \dot{h}_{wy} + \left(\omega_z h_x - \omega_x h_z\right) + \left(\omega_z h_{wx} - \omega_x h_{wz}\right) \\ \dot{h}_z + \dot{h}_{wz} + \left(\omega_x h_y - \omega_y h_x\right) + \left(\omega_x h_{wy} - \omega_y h_{wx}\right) \end{pmatrix} \tag{2.3.8}$$

In coordination with the group of Structures and Mechanisms, we have obtained the inertia matrix in the body reference for the stowed and open configuration with which we have refined our attitude control simulations that will follow.

2.3.1. Linearized Attitude Dynamics Equations of Motion

In the first phase of the design stage, it is important to transform Eq. 2.3.5 into a more easily treatable form. The angular motion can be approximated by infinitesimal angular motion, which means small Euler angles and angle derivatives. With these assumptions, the dynamics equations can be Laplace transformed, thus gaining the important advantage of using linear control theory. If the body coordinate axes are principal axes and since reaction wheels inside the CubeSat don't change the overall inertial angular momentum of the system but instead merely transfers the momentum change to the spacecraft, the linearised attitude dynamics equation expressed with the Euler angles are:

$$\begin{cases} T_{dx} + T_{cx} = I_x\ddot{\phi} + 4\omega_0^2\left(I_y - I_z\right)\phi + \omega_0\left(I_x - I_z - I_y\right)\dot{\psi} + \dot{h}_{wx} - \omega_0 h_{wz} \\ T_{dy} + T_{cy} = I_y\ddot{\theta} + 3\omega_0^2(I_x - I_z)\theta + \dot{h}_{wy} \\ T_{dz} + T_{cz} = I_z\ddot{\psi} + \omega_0^2\left(I_y - I_x\right)\psi + \omega_0\left(I_z + I_x - I_y\right)\dot{\phi} + \dot{h}_{wz} + \omega_0 h_{wx} \end{cases} \tag{2.3.9}$$

In Eq. 2.3.9, h_{wx}, h_{wy}, h_{wz} are the momentum components of the wheels with axes of rotation along the $\boldsymbol{X_B}, \boldsymbol{Y_B}$ and $\boldsymbol{Z_B}$ body axes of the satellite; $h_{wx} = I_{wx}\omega_{wx}, h_{wy} = I_{wy}\omega_{wy}$, and $h_{wz} = I_{wz}\omega_{wz}$, where I_{wx}, I_{wy}, I_{wz} are the moments of inertia of the individual wheels and $\omega_{wx}, \omega_{wy}, \omega_{wz}$ are the angular velocities of the wheels. The terms $\dot{h}_{wx}, \dot{h}_{wy}, \dot{h}_{wz}$ are the angular moments that the wheels exert on the s/c along the body axes. If $\dot{\omega}_{wx}$ is the angular acceleration of the $\boldsymbol{X_B}$ axis wheel, then $\dot{h}_{wx} = I_{wx}\dot{\omega}_{wx}$ is the negative of the angular momentum that the $\boldsymbol{X_B}$ wheel exerts on the satellite about its $\boldsymbol{X_B}$ axis. The same applies for the $\boldsymbol{Y_B}$ and $\boldsymbol{Z_B}$ axes wheel components. Attitude control of a s/c can be achieved by controlling these angular accelerations, which are internal torques exerted on the satellite. If in addition, external (inertial) torques such as magnetic or reaction torques are applied to the satellite, they are incorporated in $\boldsymbol{T_c}$, the vector of control torques.

In general, the rotation axes of the wheels are not necessarily aligned along the satellite's body axes. Moreover, there may be more (or less) than three wheels in the satellite. In such cases, the momentum and the angular acceleration of the wheels will be transformed to the body axes, so that they comply with Eq. (2.3.9).

2.4. Environment Disturbance Torque

Before a satellite decays, the attitude of the latter gets continually disturbed by different forces, called disturbance torques. These forces need to be taken into account when designing the AODCS, to make sure the spacecraft operates as it should. This section will discuss how to calculate these external disturbances and estimate their magnitude over the CubeSat (worst case values).

The external torques acting on the satellite include the gravity gradient, solar radiation pressure, aerodynamic drag, and magnetic disturbance.

In Table 2.1 below are reported the worst-case value for each of these:

Gravity gradient torque T_g	4.3242e-09 Nm
Solar radiation pressure torque T_{sp}	8.1655e-08 Nm
Magnetic disturbance torque T_m	4.9487e-07 Nm
Aerodynamic drag torque T_a	7.7698e-07 Nm
TOTAL DISTURBANCE TORQUE	1.3578e-06 Nm

Table 2.1: Disturbance Torque

2.4.1. Aerodynamic Disturbance Torque

Even in low orbit the spacecraft will still experience drag due to Earth's atmosphere. The drag is directed in the opposite direction of the orbital velocity vector. It is a

function of the area that the velocity vector passes through, the coefficient of drag, speed of the satellite, and atmospheric density. The aerodynamic drag force is given by:

$$T_a = \frac{1}{2}\rho A C_d v^2 (c_{pa} - c_g) \quad (2.4.1)$$

Where C_d is the drag coefficient (usually between 2 and 2.5); ρ the atmospheric density; A the surface area; v the spacecraft velocity; c_{pa} the center of aerodynamic pressure and c_g the center of gravity.

2.4.2. Gravity Gradient Torque

Because of the variation in the Earth's gravitational field, any non-symmetrical object in orbit is subject to a gravitational torque. A useful method to calculate this torque is to use the mass moment of inertia of the satellite in body coordinates. The gravity gradient torque is normal to the local vertical coordinate and is inversely proportional to the cube of the distance to the center of Earth. So, it is influenced mainly by orbit altitude and spacecraft inertias:

$$T_g = \frac{3\,\mu}{2R^3} \left|I_z - I_y\right| \sin(2\theta) \quad (2.4.2)$$

where T_g is the maximum gravity torque, μ is the Earth's gravity constant, R is orbit radius, θ is the maximum deviation of the Z-axis from local vertical in radians, and I_z and I_y are moments of inertia about z and y (or x, if smaller) axes in [kgm^2].

2.4.3. Solar Radiation Pressure

A torque can be produced by solar radiation pressure when the center of pressure from the incident radiation is not aligned with the center of mass. The solar radiation pressure is caused by incident solar radiation on the surface of the satellite. The intensity of the pressure is determined by the satellite's radiative properties, geometry, and the position and orientation in relation to the Sun.

The solar radiation varies as the inverse square of the distance from the sun, and therefore the solar radiation pressure is altitude independent. Major sources of electromagnetic radiation pressure are solar illumination, solar radiation reflected from the Earth and radiation emitted from the Earth and its atmosphere. Since the radiation from the Earth is small compared to the solar radiation, the radiation emitted from the Earth will not be considered.

Solar radiation pressure is highly dependent on the type of surface being illuminated. A surface is either transparent, absorbent, or a reflector, but most surfaces are a combination of the three. In general, solar arrays are absorbers and the spacecraft body is a reflector. The worst-case solar radiation torque is:

$$T_{sp} = \frac{F_s}{c} A_s(1+q)\cos(I)\,(c_{ps} - c_g) \tag{2.4.3}$$

Where F_s is the solar constant, 1367 W/m^2, c is the speed of fight, A_s is the surface area, c_{ps} is the location of the center of solar pressure, c_g is the center of gravity, q is the reflectance factor (ranging from 0 to 1, we use 0.6), and I is the angle of incidence of the Sun (used equal to 0 to evaluate worst case).

2.4.4. Magnetic disturbance torque

The magnetic disturbance stems from the interaction between the spacecraft's residual magnetic field and the Earth's magnetic field. It depends mainly on orbit altitude and inclination and residual spacecraft magnetic dipole:

$$T_m = DB$$
$$with \quad B = \frac{2M}{R^3} \tag{2.4.4}$$

where D is the residual dipole of the vehicle in Am^2, and B is the Earth's magnetic field in Tesla, approximated as $2M/R^3$ for a polar orbit to half that at the equator. M is the magnetic moment of the Earth, $7.96 \times 10^{15}\ Tesla \cdot m^3$, and R is the radius from Earth center to spacecraft in [m].

2.5. Orbit Propagator

An orbit propagator makes it possible to simulate realistic disturbances. This is obvious when looking at the Earth's magnetic field, which varies in strength and direction depending on the location of the satellite, thus giving a changing disturbance torque from the satellite's magnetic residual. Additionally, having the satellites position relative to the Earth and knowing where the Sun is, makes it possible to calculate an estimate of the disturbance torque caused by solar radiation and makes it possible to emulate sun sensors. When evaluating the performance of the pointing controller it is relevant to know the satellite's orbital position. An orbit propagator implemented in the software on board the satellite is also important because the attitude estimator needs to know the expected measurements from the sensors.

For all of these reasons, STK is used in order to compute an accurate orbit propagator, by taking into account a sophisticated model. The main characteristics of the analysis and propagator are reported below.

- Scenario

Analysis period	Start: 1 Jan 2022 10:00:00.000 UTCG Stop: 2 Jan 2022 10:00:00.000 UTCG

Table 2.2: Period of scenario with STK software

- Basic Orbit (noon/midnight Sun Synchronous orbit)

Propagator	J2 Perturbation
Step Size	0.1 [s]
Semimajor Axis	6878.14 km
Eccentricity	0
Inclination	97.4065 [deg]
Argument of Perigee	0 [deg]
RAAN	281.037 [deg]
True Anomaly	0 [deg]

Table 2.3: Orbit parameters chosen in STK software.

The J2 Perturbation (first order) propagator accounts for secular variations in the orbit elements due to Earth oblateness. This propagator does not model atmospheric drag or solar or lunar gravitational forces.

J2 is a zonal harmonic coefficient in an infinite series representation of the Earth's gravity field. It represents the dominant effects of Earth oblateness. The even zonal harmonic coefficients of the gravity field are the only coefficients that result in secular changes in satellite orbital elements. The J2 propagator includes the first-order secular effects of the J2 coefficient.

The 3D orbit view and the ground track of the orbit are represented below.

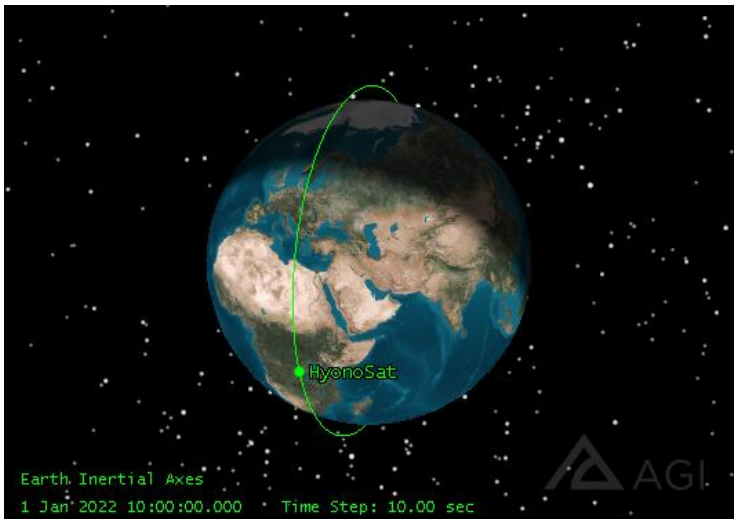


Figure 2.8: 3D orbit view with STK software.



Figure 2.9: Ground track with STK software.

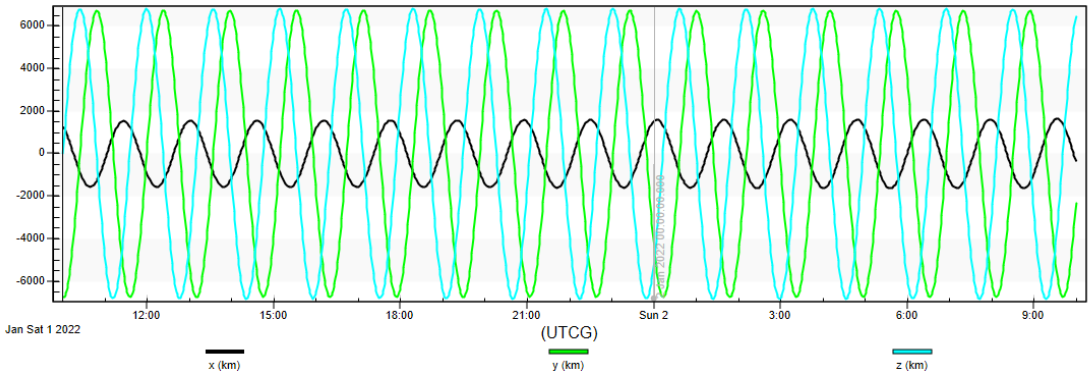


Figure 2.10: ECI coordinates during time with STK software.

2.6. Magnetic field model

In order to simulate the behaviour of magnetometers and magneto torquers a model of Earth's magnetic field is needed. Here we present two models for the Earth's magnetic field: a simplistic dipole geomagnetic model and a more sophisticated and accurate International Geomagnetic Reference Field (IGRF) model. The geomagnetic field model to be implemented on the satellite is obviously of central importance for the attitude control. The better the model, taking into account the computational time and required power, the better the control ultimately will be. Nevertheless, we start with a simple dipole model, and then move on to the IGRF model.

2.6.1. Dipole model

The satellite will be falling around the Earth's magnetic field at an altitude of around 500 km. At a larger scale it is possible to look at this magnetic field as a dipole. This means that one basically considers the Earth as a large dipole magnet, with the magnetic vector field exiting the south pole and entering the north pole (Figure 2.11).

Dipole model is a simple symmetrical model of the geomagnetic field, that completely ignores the longitudinal components of the magnetic field vector and can thus simply be illustrated two-dimensionally, as in Figure 2.11. We note that the geographic and magnetic poles do not (necessarily) overlap, in fact the magnetic equator has an inclination $i_m = 11.5°$ and the longitude of the ascending node is equal to $\Omega_m = 20.3°$.

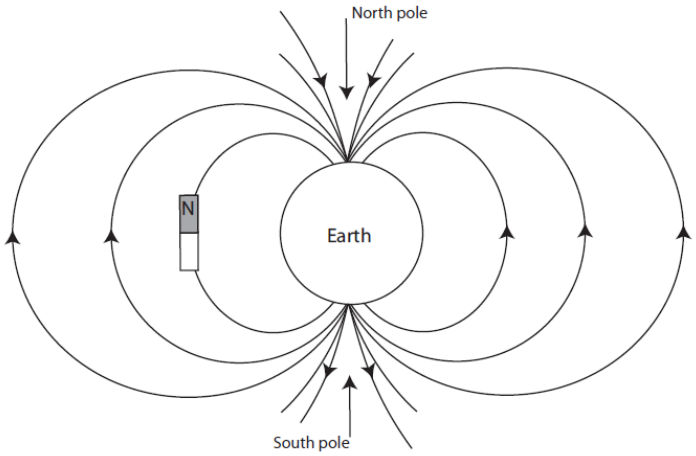


Figure 2.11: Dipole Model

In particular, it is possible to obtain the components of the magnetic field along the NED frame (North, East and Down), by taking advantage of this simple formulation:

$$[\vec{B}]_{NED} = \left(\frac{R_{\otimes}}{R}\right)^3 \begin{bmatrix} cos\,L\,a & -sin\,L\,a\,cos\,L\,o & -sin\,L\,a\,sin\,L\,o \\ 0 & -sin\,L\,o & cos\,L\,o \\ 2\,sin\,L\,a & 2\,cos\,L\,a\,cos\,L\,o & 2\,cos\,L\,a\,sin\,L\,o \end{bmatrix} \begin{bmatrix} 30215\ \text{nT} \\ 2133\ \text{nT} \\ -5766\ \text{nT} \end{bmatrix} \qquad (2.6.1)$$

In order to obtain the evolution of the magnetic field in this reference frame, the orbit propagation is necessary and also the computation of the latitude and longitude at all times. For this purpose, the MATLAB's function *ecef2lla* is used.

2.6.2. International Geomagnetic Reference Field

The IGRF model estimated every 5 years by the International Association of Geomagnetism and Aeronomy (IAGA), is essentially a set of Gaussian coefficients g_n^m and h_n^m, that can be used in a spherical harmonic model to approximate the Earth's magnetic field. Because the geomagnetic field is time-varying, the IGRF model also includes the so-called secular variation (SV) coefficients such that for the 5 years after the most recent epoch, the SV can be used for forward linear extrapolation.

Here MATLAB's built-in function *igrfmagm* from the Aerospace Toolbox is used. Given the latitude, longitude, altitude and the time of the analysis, this function provides the values of the magnetic field in the NED frame, according to the IGRF model.

2.6.3. IGRF vs Dipole models

Two different time evolution of the magnetic field are obtained in this way in the NED frame. From this computation, through a series of rotations, it is possible to compute the magnetic field evolution in different reference frames. In particular, it is obtained the evolution in the ECEF, ECI and orbital frames.

In order to pass from the NED to the ECEF frame, the following rotation is considered:

$$\left[\vec{B}\right]_{ECEF} = \begin{bmatrix} -\sin Lat \cos Long & -\sin Long & -\cos Lat \cos Long \\ -\sin Lat \sin Long & \cos Long & -\cos Lat \sin Long \\ \cos Lat & 0 & -\sin Lat \end{bmatrix} \left[\vec{B}\right]_{NED} \qquad (2.6.2)$$

The time evolution of the magnetic field in the ECEF frame is reported here and the two models can be easily compared.

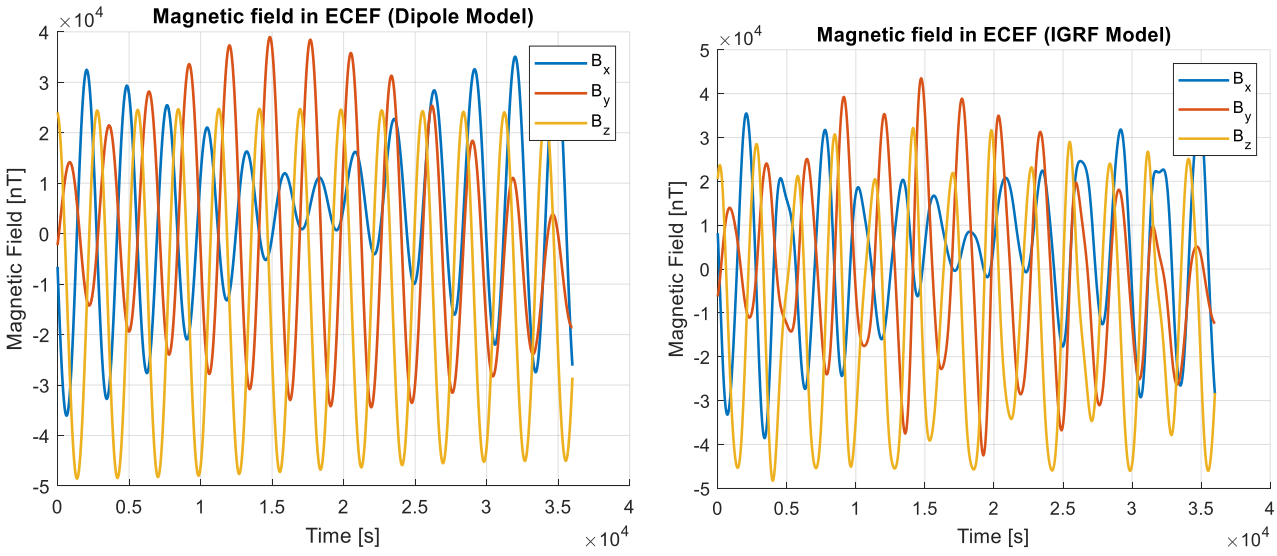


Figure 2.12: Earth magnetic field in ECEF using Dipole Model (left) and IGRF Model (right)

From the ECEF frame it is possible to pass to the ECI frame, through the following relation.

$$[\vec{B}]_{ECI} = \begin{bmatrix} \sqrt{B_{ECEF}(1)^2 + B_{ECEF}(2)^2}\cos(\gamma + \alpha) \\ \sqrt{B_{ECEF}(1)^2 + B_{ECEF}(2)^2}\sin(\gamma + \alpha) \\ B_{ECEF}(3) \end{bmatrix} \tag{2.6.3}$$

Where α is the angle between the initial epoch and the current time, while γ is the angle formed by the vector of the magnetic field projected in the (x, y) plane with the x axis.

Here the plots for the evolution of the magnetic field in ECI frame are presented for the two models analysed.

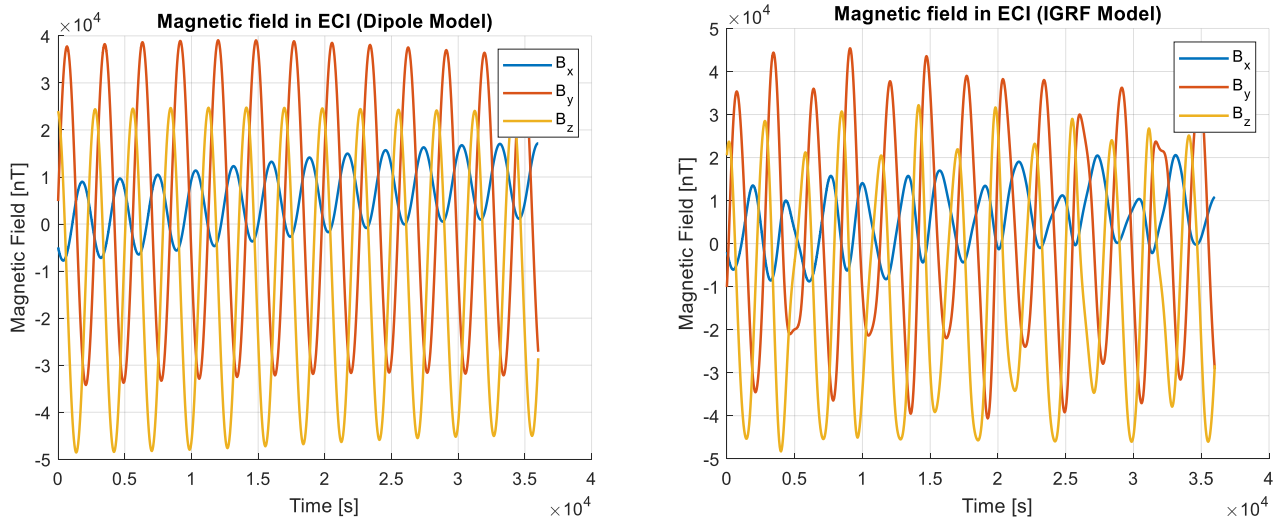


Figure 2.13: Earth magnetic field in ECI using Dipole Model (left) and IGRF Model (right)

In conclusion, the transformation from the ECI to the orbital frame is reported here.

$$[\vec{B}]_{orb} = \begin{bmatrix} \cos\Omega\cos\omega - \sin\Omega\sin\omega\cos i & \sin\Omega\cos\omega + \cos\Omega\sin\omega\cos i & \sin\omega\sin i \\ -\cos\Omega\sin\omega - \sin\Omega\cos\omega\cos i & -\sin\Omega\sin\omega + \cos\Omega\cos\omega\cos i & \cos\omega\sin i \\ \sin\Omega\sin i & -\cos\Omega\sin i & \cos i \end{bmatrix} [\vec{B}]_{ECI} \quad (2.6.4)$$

In this case the following time evolution of the magnetic field is obtained for the two models.

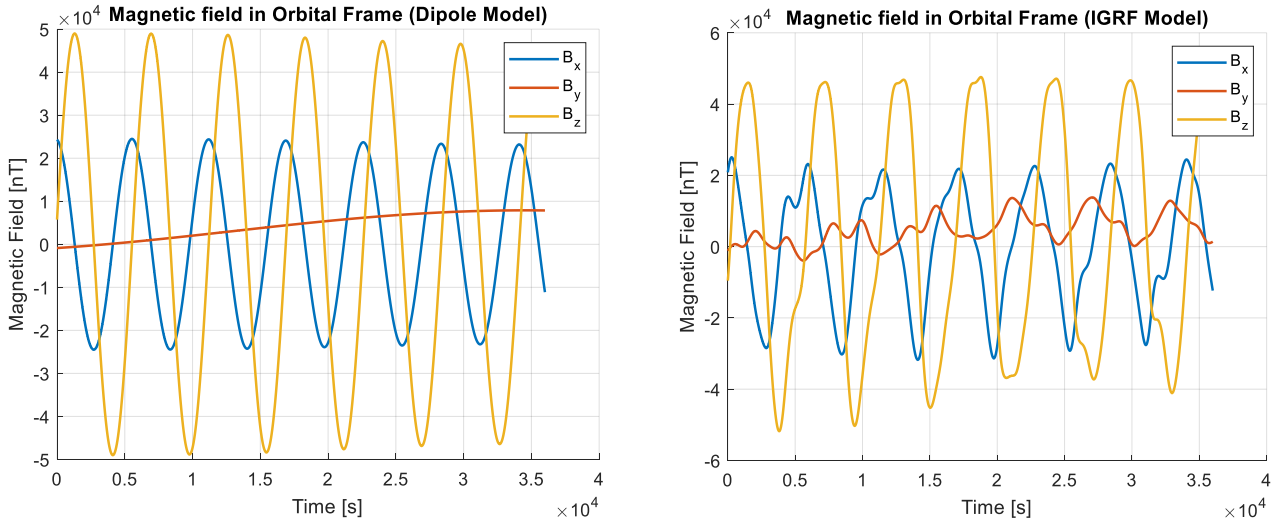


Figure 2.14: Earth magnetic field in Orbital Frame using Dipole Model (left) and IGRF Model (right)

From the graphs obtained above we can show that between the simple magnetic dipole model and the more accurate IGRF model, we have a perfect agreement between the two. For the simulations that follow we will use the IGRF model to have more accurate results.

3. AODCS Design

Attitude Orbit Determination Control System (AODCS) is the onboard system responsible for determining and controlling the orbit and satellite attitude. AODCS is vital when speaking of satellite orbit and attitude. It provides the following major functions:

- Sensor data acquisition and pre-processing
- Measurement processing
- Attitude and orbit estimation
- Actuator commanding

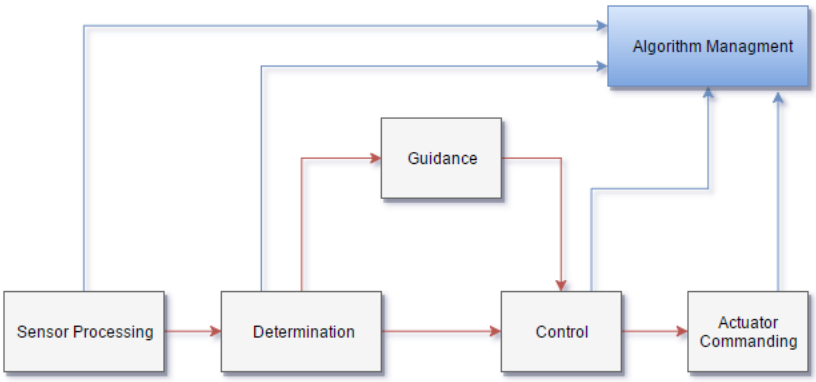


Figure 3.1. AODCS high level architecture.

After measuring and process the data collection from the sensors, AODCS also commands the actuators to employ torques needed in order to re-orient the satellite to a desired attitude and maintain the orbit. This is visible on Figure 3.1, which illustrates the high-level architecture and how the data flows.

3.1. Sensors

Sensors are instruments used for determining the angular deviations of the axes of a spacecraft from pre-set directions. Attitude sensors are distinguished according to the principle of acquisition and processing of information. In fulfilling their functions, attitude sensors may use any celestial bodies (the sun or planets), or the force fields generated by the bodies. Attitude sensors also exist which do not require external sources of information, and which are fully isolated from space, for example, gyroscopes.

A set of sensors present on the satellite gather information of the surrounding environment and on the satellite itself. They allow the satellite OBC to process and analysis their data to produce the satellite's attitude and orbit determination. Reference

sensors are also needed, and Table 3.1 gives a quick comparison between the sensor types.

Sensor type	**Advantages**	**Disadvantages**
Star tracker	- Very high accuracy. - Lost in space function.	- Dependent on star identification. - Expensive. - Heavy and large. - High power consumption. - Sensitive to spin rates. - No output when pointing towards the Sun.
Sun sensor	- Simple sensor. - Coarse accuracy. - Low power consumption. - Analog sun sensors are cheap.	- Coarse accuracy. - Disturbed by Earth Albedo. - No output during eclipse. - Digital 2-axis sun sensors are expensive.
Magnetometer	- Simple sensor. - Low power consumption. - Very cheap. - Very small. - Always available.	- Coarse accuracy. - Magnetic field not completely known. - Affected by on-board electronics. - Only applicable in LEO.
GPS antenna array	- Resistant against spin rates. - Also provides position, velocity, and time.	- Expensive. - Large antenna array setup. - High power consumption.
Horizon sensor	- Earth always available. - Simple sensor.	- Coarse accuracy. - Accuracy affected by the Sun and Moon. - Not as widely used as sun sensors.

Table 3.1: Advantages and disadvantages of different reference sensor types, if used in a CubeSat.

Star trackers and GPS antenna arrays are both very expensive and hard to fit within the size and the power constraints of a CubeSat but the accuracy requirement for the mission does furthermore require the use of e.g., a high precision star tracker. Since our orbit is Sun synchronous were chosen sun sensors that will have an eclipse period of about 35 minutes, furthermore, Sun sensors can also be very small and low cost. Moreover, for the detumbling strategy, as will be seen below, the B-dot control was chosen and therefore the magnetometers were adopted. IMU are also a part of the AODCS useful for determining the attitude. The following criteria were used during the sensor component selection process for HyonoSat:

- Resilience towards the harsh space environment.
- Low power consumption.
- Low weight and size.
- Ease of implementation.
- Accuracy.
- Cost.

It is also important to analyse the accommodation chosen for the sensors:

- The sun sensor has been placed in the + x face of the CubeSat so that the sun is in visibility for a part of the orbit, for the remaining part the sensors that are present and available on the orientable solar panels are used.
- The star tracker has been placed in the -y direction so that during the synchronous orbit it can look into deep space and thus correctly detect the stars in its field of view and at the same time is not disturbed by sunlight and therefore the solar vector is perpendicular to the sensor boresight.
- The magnetometer is oriented in accordance with the datasheet of the instrument to be aligned with the body axes of the satellite.
- The IMU is oriented in accordance with the datasheet of the instrument to be aligned with the body axes of the satellite.

The previous assumptions lead to the following configuration obtained with STK shown in the Figure 3.2.

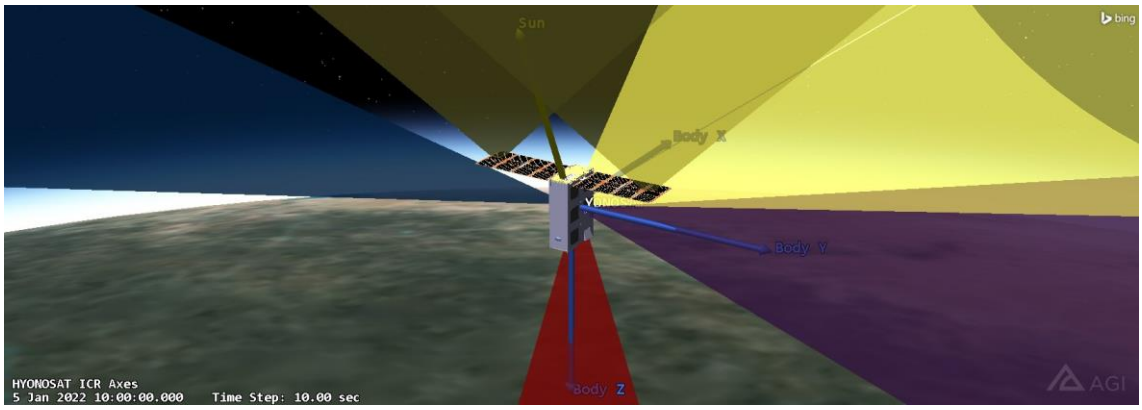


Figure 3.2: Sensors positions with STK software.

3.1.1. Sun sensors

The most common sensor for attitude determination in space missions is the sun sensor. The sun sensor is a device that can provide two axis attitude knowledge of a satellite by using the sun as a reference through the incident ray angles. Multiple sun sensors are often mounted on the satellites outer panels reducing the number of solar cells, thus

having a large impact on the power generated on smaller satellites. There are three main types of sun sensors: the analog sensors, digital sensors, and the sun presence sensors. While digital sensors are the most accurate type of sun sensor, they are also the most expensive. Analog sensors are the most common as they provide moderate quality detection for their relatively lower monetary cost.

Our choice is the **nanoSSOC-A60**. Sun Sensor on a Chip (SSOC) is a two-axes and low-cost sun sensor for high accurate sun-tracking, pointing and attitude determination. The device measures the incident angle of sun ray in two orthogonal axes, providing a high sensitivity based on the geometrical dimensions of the design. NanoSSOC sun sensor is based on MEMS fabrication processes to achieve high integrated sensing structures. NanoSSOC-A60 has tiny size, low weight, and low power consumption to be the perfect ADCS solution for nanosatellite platforms like CubeSats.



Figure 3.3: NanoSSOC-A60

Type	2 orthogonal axes
Field of View	±60º
Accuracy	< 0.5 º (3sigma) < 0.1 º (precision)
Electrical interface	4 voltage outputs 10-pin micro-connector
Power supply	3.3V / 5V < 2mA consumption
Mechanical interface	27.4 x 14 x 5.9 mm 4 g
Housing	Aluminium 6082 Black anodizing

Table 3.2: NanoSSOC-A60 Specifications

https://www.solar-mems.com/wp-content/uploads/2020/04/Solar-MEMS-nanoSSOC-A60.pdf

3.1.2. Star trackers

Star sensors measure star coordinates in the spacecraft frame and provide attitude information when these observed coordinates are compared with known star directions obtained from a star catalogue. In general, star sensors are the most accurate of attitude sensors, achieving accuracies to the arc-second range. This impressive capability is not

provided without considerable cost, however. Star sensors are heavy, expensive, and require more power than most other attitude sensors. In addition, computer software requirements are extensive, because measurements must be pre-processed and identified before attitudes can be calculated. Star sensors also suffer from both occultation and interference from the Sun, the Earth, and other bright sources. Despite these disadvantages, the accuracy and versatility of star sensors have led to applications in a variety of different spacecraft attitude environments.

Star sensing and tracking devices can be divided into three major classes:

- Star scanners, which use the spacecraft rotation to provide the searching and sensing function.
- Gimbaled star trackers, which search out and acquire stars using mechanical action.
- Fixed head star trackers, which have electronic searching and tracking capabilities over a limited field of view.

Our choice is the **ST200** from Hyperion Technologies. The ST200 is one of the world's smallest and lightest fully autonomous, low power star trackers, aimed at applications in pico- and nano-satellite platforms. With an accuracy of 30 arcseconds and its extremely small size the ST200 can be used in practically every platform that requires attitude determination. The ST200 is a co-development with partner Berlin Space Technologies.



Figure 3.4: ST200 Hyperion Technologies

Accuracy	30 arcseconds (3-sigma)
Update rate	5 Hz
Interface	TTL UART
Power consumption	650 mW (nominal)
Mass	42 g
Supply voltage	3.65V (3.6 – 5.0V)
Others	Various suitable baffle options are available on demand

Table 3.3: ST200 Hyperion Technologies Specifications

https://hyperiontechnologies.nl/wp-content/uploads/2015/07/HT_ST200_v2.1-flyer.pdf

3.1.3. Magnetometers

Magnetometers are widely used as spacecraft attitude sensors for a variety of reasons: they are vector sensors, providing both the direction and magnitude of the magnetic field; they are reliable, lightweight, and have low power requirements, they operate over a wide temperature range, and they have no moving parts. Magnetometers measure the sum of the ambient field that is of interest and any local fields produced by the spacecraft. Magnetometers are not accurate inertial attitude sensors because the magnetic field is not completely known, and the models used to predict the magnetic field direction and magnitude at the spacecraft's position are subject to relatively substantial errors. A total of three units, with one magnetorquer per axis are needed to allow full control of the spacecraft. Magnetorquers are very attractive as they offer a cheap attitude control solution for satellite being constrained in weight and volume.

Our choice is **NanoSense-M315** from GOMspace that is a flight proven compact and low noise magnetometer with an I2C interface designed especially for nanosatellites with high ADCS requirements. The compact design allows flexible mounting which in turns allows the magnetometer to be placed far away from magnetic disturbance sources.

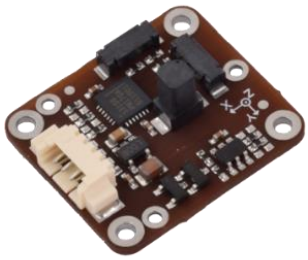


Figure 3.5: NanoSense-M315 GOMspace

Mass	8 g
Dimensions	23 x 20 x 8 mm (0.004U)
Operational temperature	-40 – 85 °C
Supply voltage	3.3 V
Supply current (Not sampling)	100 µA
Supply current (Sampling)	2.5 mA
Noise (1-sigma)	15 nT
Sample frequency (max)	140 Hz
Measurement range	± 800 µT

Table 3.4: NanoSense-M315 GOMspace Specifications

https://gomspace.com/UserFiles/Subsystems/datasheet/gs-ds-nanosense-m315-12.pdf

3.1.4. IMU

An inertial measurement unit as the name states, is an instrument that measures the components of angular and linear velocities on each axis of the satellite. To do so, it uses gyroscopes and accelerometers. As the IMU is usually not aligned with the satellite body frame, its data must be rotated from IMU referential to the satellite body frame.

Our choice is **STIM377H** by Sensonor. It is a small, tactical grade, low weight, high performance non-GPS aided Inertial Measurement Unit (IMU) in a hermetic package. STIM377H contains 3 highly accurate MEMS gyros, 3 high stability accelerometers and 3 inclinometers. The IMU is factory calibrated and compensated over its entire operating temperature range.



Figure 3.6: STIM377H Sensonor

GENERAL	
Weight	55 g
Supply Voltage	5 V
Power Consumption	1.5 W
Sample Rate	2000 SPS
GYRO	
Resolution	0.22 °/h
Bias instability	0.3 °/h
Angular random walk	0.15 °/√h
ACCELEROMETER	
Resolution	1.9 µg
Bias instability	0.05 mg
Velocity random walk	0.07 m/s/√h

Table 3.5: STIM377H Sensonor Specifications

https://sensonor.azurewebsites.net/media/1465/stim377h-product-brief.pdf

3.2. Actuators

The stability or alignment of a satellite is very important for the effective operation of various components. Communication antennae must be pointed in the correct direction to function properly while cameras usually have to provide photographs of specific objects or areas. As such, a variety of attitude control devices have been developed and tested to successfully stabilize a satellite in orbit. These devices produce the torque that is necessary for the attitude control of a spacecraft. Magnetic control has been used over many years for small spacecraft attitude control. The main drawback of magnetic control is that magnetic torque is two-dimensional, and it is only present in the plane perpendicular to the magnetic field vector. The satellite cannot be controlled precisely in three-dimensional space using only magnetorquers, but the combination of magnetorquers with one reaction wheel expands the two-dimensional control torque possibilities to be three-dimensional.

Taking into account these considerations the following design and accommodation has been chosen for the actuators:

- The magnetorquers are 3 and each of these is placed on an axis of the CubeSat.
- The reaction wheels are 4 and are placed in a pyramid array configuration, the center of mass is positioned as close as possible to that of the CubeSat.
- The thrusters for orbit maintenance are placed in the opposite direction to the direction of motion and aligned to the CoM of the CubeSat.

3.2.1. Magnetorquers

Magnetorquers are used to attain control of a satellite under the influence of small torques relies on magnetic actuators (magnetic torquers or torqrods in short). A torqrod is a thin, long coil where a variable, controlled current can flow. The current endows the coil with a magnetic moment M, which interacts with the ambient magnetic field generating a torque. Magnetic torques can be used either directly for attitude control or to unload momentum accumulated by reaction wheels or CMGs. The simplest torquers are made of N turns of wire in a loop of area A; sending a current I through such a coil produces a magnetic dipole moment of magnitude $m = NIA$ in a direction perpendicular to the plane of the coil. This relation has the advantage of being a perfectly linear function of the current, but large areas or many turns are required to produce the dipole moments needed by many spacecrafts. The dipole interacts with the magnetic field generating a torque $\vec{\tau} = \vec{m} \times \vec{B}$.

Our choice is **MTQ400** by Hyperion Technologies. The MTQ400 magnetorquers are aimed at applications in 6-12U CubeSats. Offering one of the highest energy efficiencies in the industry, combined with fine-grained dipole-moment strength control, it allows precise pointing and rapid de-saturation of any reaction wheels in the satellite. With a unique boost-mode, these magnetorquers can be pushed up to higher magnetic dipole moments for short duration maneuvers, through sacrificing some energy efficiency. In return however, these magnetorquers will offer the fastest detumbling of any satellite in this mass class, allowing the users to start their operations in the shortest amount of time. They furthermore feature an inherently safe, passive detumbling mode, which works without any external controller.



Figure 3.7: MTQ400 Hyperion Technologies

Nominal Dipole Moment	0.5 Am^2
Boost Mode Dipole Moment	2 Am^2
Dimensions	11.5 x 80 mm
Mass	44.1
Power Consumption	270 mW

Table 3.6: MTQ400 Hyperion Technologies Specifications

https://hyperiontechnologies.nl/products/mtq400/

3.2.2. Reaction Wheels

Momentum wheel is a flywheel designed to operate at a biased, or nonzero, momentum. It provides a variable-momentum storage capability about its rotation axis, which is usually fixed in the vehicle. Reaction wheel is a flywheel with a vehicle-fixed axis designed to operate at zero bias. The gimbal rings allow control of the le direction of the flywheel momentum vector in the spacecraft body. Reaction wheels are used as the primary attitude control actuators on most spacecraft. Momentum-bias spacecraft may use one or two reaction wheels, but full three-axis attitude control requires three or more wheels. Providing extra reaction wheels for redundancy gives some protection, but failure of one reaction wheel is often followed by failure of other wheels of the same design on the same spacecraft.

Our choice is **RW210** by Hyperion Technologies. The RW210 series reaction wheels are low mass, low power reaction control wheels, which allow CubeSats and other pico- or nanosatellites to control their attitude.

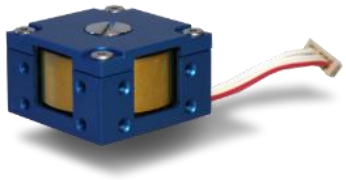


Figure 3.8: RW210 Hyperion Technologies

Total momentum storage	+/- 6.0 mN.m.s
Maximum torque	+/- 0.1 mN.m
Maximum rotation rate	15000 rpm
Outer dimensions	25 x 25 x15 mm
Mass	48 g
Power consumption	<800 mW peak

Table 3.8: RW210 Hyperion Technologies Specifications

https://hyperiontechnologies.nl/wp-content/uploads/2019/11/HT_RW210.pdf

3.2.3. Thrusters

Thrusters generate both forces and torques, so they can be used for both trajectory control and attitude control. Thrusters have the advantage of not being dependent on an ambient magnetic or gravitational field, so they can be used for control or momentum unloading in any orbit.

To maintain the orbit, our choice is to use an electric thruster that is the Field-emission electric propulsion (FEEP). It is an advanced electrostatic space propulsion concept, a form of ion thruster, that uses a liquid metal as a propellant – usually either cesium, indium, or mercury. A FEEP device consists of an emitter and an accelerator electrode. A potential difference of the order of 10 kV is applied between the two, which generates a strong electric field at the tip of the metal surface. The interplay of electric force and the liquid metal's surface tension generates surface instabilities, which give rise to Taylor cones on the liquid surface. At sufficiently high values of the applied field, ions are extracted from the cone tip by field evaporation or similar mechanisms, which then are electrically accelerated to high velocities – typically 100 km/s or more.

The device that has been chosen is the multiFEEP by Morpheus Space. The MultiFEEP design lies at the forefront of micro propulsion technology. It is a unique

system for satellite missions, which have challenging requirements with regard to total delta-V, maximum thrust levels and precise thrust vector control.



Figure 3.10: multiFEEP Morpheus Space

Dynamic Thrust Range	1- 120 µN
Maximum thrust	140 µN
Specific Impulse	2600 to 8500 s
Propellant Mass Range	33 - 156 g
Total Impulse Range	up to 13000 Ns
Total System Power	0.4 W - 19 W
Total System Mass (dry	280 g
Total System Size (LxWxH)	90 x 45 x 45 mm

Table 3.9: multiFEEP Morpheus Space

https://satcatalog.com/datasheet/Morpheus%20Space%20-%20multiFEEP.pdf

4. Orbit Determination

OD is the method of estimating the state (position and velocity, or other describing parameters) of an orbiting object. This is done by developing a set of equations describing a variety of forces and using measurements to determine initial and refine later states. The complex nature of these equations leads to highly non-linear equations. This, combined with our lack of fully grasping the physics behind perturbing forces and noisy measurements, limit the accuracy of predicting a satellite's state.

In the astrodynamics world, there are three playing fields of propagation: analytical methods (also referred to as general perturbation techniques), numerical integration procedures (also referred to as special perturbation techniques), and a middle ground called semi-analytical methods. General perturbation methods superimpose the approximated effects of the most prominent perturbations on the Keplerian equations of motion. Special perturbation methods numerically integrate orbit dynamics equations, which are derived from a more comprehensive force model. Semi-analytical methods solve the 2-body problem analytically and then numerically integrate the perturbations.

The currently most popular propagator, SGP4, is an analytical method. The earliest satellite orbit observations were performed with cameras. Since then, the typical OD methodologies involved the use of radar or laser ranging devices from Earth stations. These were used to derive orbit describing parameters, which were uploaded to the satellite for on-board propagation. Global navigation satellite systems (GNSS) revolutionized OD industry. Satellites equipped with GPS receivers are able to accurately determine their orbit state without the aid of any ground segment. The precision of this technique is superior to propagation systems and is thus a natural choice for missions requiring more accurate navigation solutions.

4.1. SGP4

The most widely used propagation technique is SGP4, a semi-analytical method originally developed in the 1960s. Its popularity is due to it being open source, the regular public release of new TLEs, the relatively small code size and light computational load. It uses a set of mean Kepler elements and a B* drag term to approximate the average orbit of a satellite. To obtain these, the periodic variations in true satellite states need to be eliminated. SGP4 then needs to try to reconstruct these periodic variations for its output.

TLEs (Two Line Elements) are mean Kepler element sets compatible with SGP4 and regularly published by NORAD (North American Aerospace Defense Command) for

almost every artificial satellite that is still in orbit. These are widely used for smaller satellites. TLEs have several sources of error, and no covariance estimates (quantity that rates the confidence of a TLE being accurate) are published.

The classical elements vary due to perturbations. Earth's full gravity model explains small oscillations that occur, while aerodynamic drag has secular effects. The SGP4 propagation technique is initialized with a set of mean classical elements, free of these perturbation's short-term effects. This mean set is referred to as the TLE set. NORAD produces TLEs of almost all satellites on a regular basis. A new TLE set is published when the previous set's propagated trajectory differs more than 5 km (90% confidence) from the current measurements, but empirical evidence shows otherwise. TLEs can be found on the CelesTrak website, which also offers the option to request a satellite's TLE history for a given date interval. Figure 4.1 shows the format of a typical TLE that SGP4 requires.

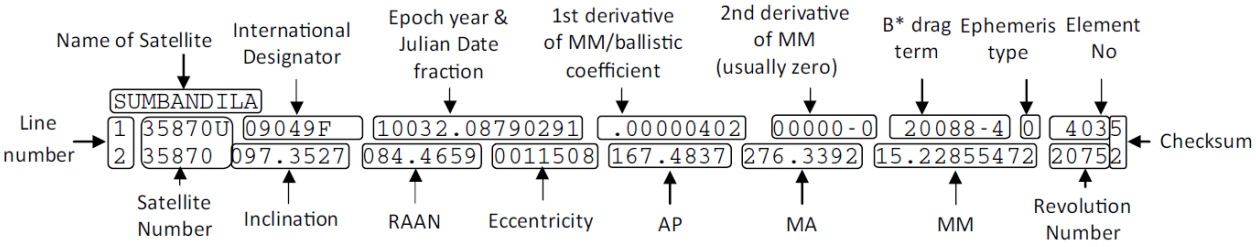


Figure 4.1: Format of a TLE set.

4.1.1. Generation of TLE with STK

- TLE – Trajectory Sampling Methods

Trajectory Sampling is used to generate a TLE based on a batch least squares algorithm. Samples used to compute element set data are based on the Start and Stop Times and Time Step entered. Using this algorithm, STK will generate TLEs by progressively minimizing the difference between the TLE-generated ephemeris and the original ephemeris by comparing the two at each step. The iterative process is completed when either convergence of the algorithm is achieved, or the maximum number of iterations is reached. Convergence is detected based on the relative error in the predicted sum of squares of the residuals. The atmospheric drag effects are modelled through the BSTAR element of the TLE when the propagation of the orbit is performed using SGP4. This is the drag parameter that is solved for by STK.

- TEME Residuals

Provides the residual (that is, position difference) between the satellite's ephemeris and the ephemeris generated using the solved-for TLE over the time span.

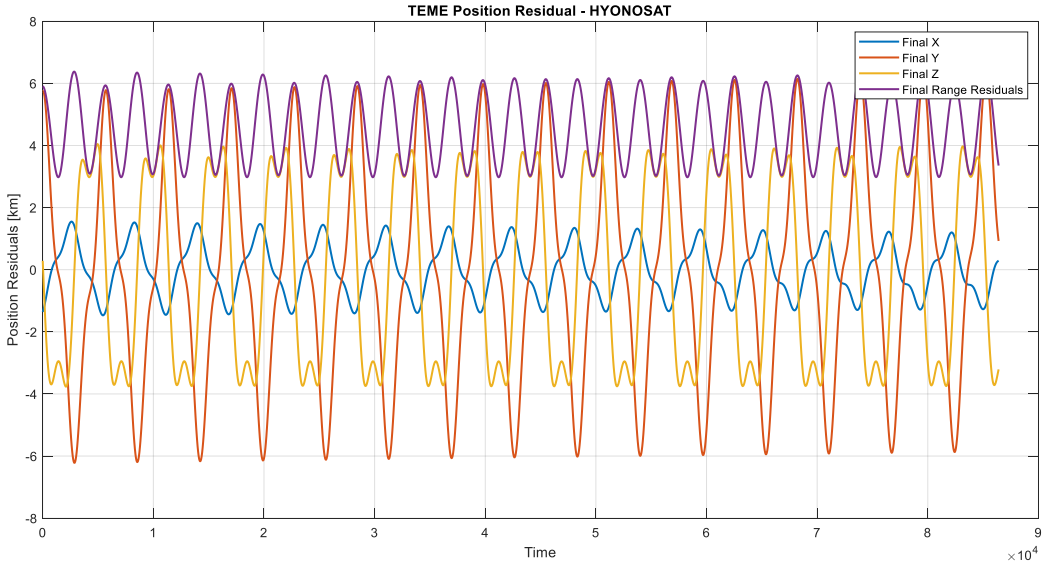


Figure 4.2: TEME position residual

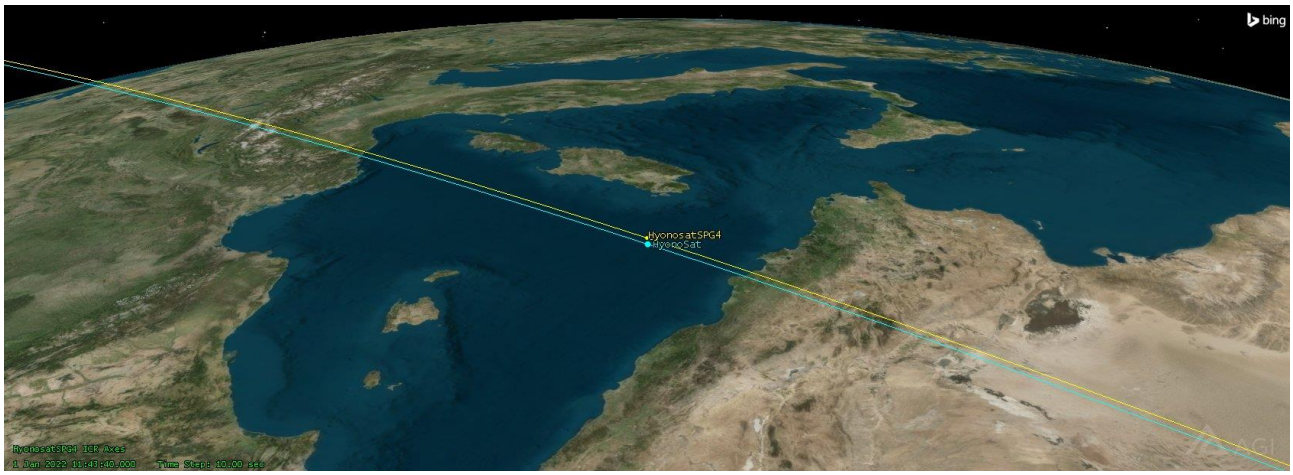


Figure 4.3: HyonoSat true orbit and SPG4 orbit with STK software

4.2. Range and Range Rate Data

In this simulation, we have used GMAT to read in simulated range and range rate measurement data for the HyonoSat to determine its orbit. The basic steps of this tutorial are:

- **Create and configure the spacecraft, spacecraft transponder, and related parameters**

In this section, we choose to represent the orbit in a Earth-centered coordinate frame. We create a new spacecraft, HyonoSat, and set its epoch and Cartesian coordinates. For the simulation, the Cartesian state represented the “true” state. Here, the Cartesian state represents the spacecraft operator’s best “estimate” of the state, the so-called a priori estimate. Because, one never has exact knowledge of the true state, we have

perturbed the Cartesian state above by a few km in each component as compared to the simulated true state shown in the comment field, for this reason there are two definitions of satellite reported as “Sim” and “Est”.

```
Create Spacecraft SimSat;

SimSat.DateFormat           = UTCGregorian;
SimSat.Epoch                = '01 Jun 2021 00:00:00.000';
SimSat.CoordinateSystem     = EarthMJ2000Eq;
SimSat.DisplayStateType     = Cartesian;
SimSat.X                    = 1316.744009807273
SimSat.Y                    = -6750.782859242079
SimSat.Z                    = 0
SimSat.VX                   = -0.9752757780489929
SimSat.VY                   = -0.1902280913832143
SimSat.VZ                   = 7.547556419338797
SimSat.DryMass              = 12;
SimSat.Cd                   = 2.2;
SimSat.Cr                   = 1.8;
SimSat.DragArea             = 0.06;
SimSat.SRPArea              = 0.06;
SimSat.Id                   = 'LEOSat';
SimSat.AddHardware          = {Transponder1, SpacecraftAntenna};

Create Spacecraft EstSat;

EstSat.DateFormat           = UTCGregorian;
EstSat.Epoch                = '01 Jun 2021 00:00:00.000';
EstSat.CoordinateSystem     = EarthMJ2000Eq;
EstSat.DisplayStateType     = Cartesian;
EstSat.X                    = 1316.744009807273
EstSat.Y                    = -6750.782859242079
EstSat.Z                    = 0
EstSat.VX                   = -0.9752757780489929
EstSat.VY                   = -0.1902280913832143
EstSat.VZ                   = 7.547556419338797
EstSat.DryMass              = 12;
EstSat.Cd                   = 2.2;
EstSat.Cr                   = 1.8;
EstSat.DragArea             = 0.06;
EstSat.SRPArea              = 0.06;
EstSat.Id                   = 'LEOSat';
EstSat.AddHardware          = {Transponder1, SpacecraftAntenna};
EstSat.SolveFors            = {CartesianState};
```

To estimate an orbit state for a given spacecraft, GMAT requires that a Transponder object, which receives the ground station uplink signal and re-transmits it, typically, to a ground station, be attached to the spacecraft. Below, we create the Transponder object and attach it to our spacecraft. Note that after we create the Transponder object, there are three fields, PrimaryAntenna, HardwareDelay, and TurnAroundRatio that must be set. The PrimaryAntenna is the antenna that the spacecraft transponder, SatTransponder, uses to receive and retransmit RF signals. HardwareDelay is the transponder signal delay in seconds. TurnAroundRatio is the ratio of the retransmitted to the input signal.

```
%
%   Spacecraft hardware
%

Create Antenna SpacecraftAntenna;
Create Transponder Transponder1;

Transponder1.PrimaryAntenna  = SpacecraftAntenna;
Transponder1.HardwareDelay   = 0.00005;
Transponder1.TurnAroundRatio = '240/221'

```

- **Create and configure the Ground Station and related parameters**

Then we create and configure our CAN GroundStation object. The script segment is broken into five sections. In the first section, we create our GroundStation object, and we set our Earth-Centered Fixed Cartesian coordinates. In the second section, we set the ID of the ground station so that GMAT will be able to identify data from this ground station contained in the GMD file. In the third section, we set the minimum elevation angle. Below this ground station to spacecraft elevation angle, no measurement data will be used to form an orbit estimate. In the fourth section, we specify which troposphere and ionosphere model we wish to use to model RF signal atmospheric refraction effects. Finally, in the fifth section, we attach three pieces of previously created required hardware to our ground station, a transmitter, a receiver, and an antenna.

```
%
%   GroundStation hardware
%

Create Transmitter Transmitter1;
Create Antenna Antenna1;
Create Receiver Receiver1;

Transmitter1.PrimaryAntenna = Antenna1;
Transmitter1.Frequency      = 2067.5;
Receiver1.PrimaryAntenna    = Antenna1;

```

Next, we create and configure the GDS GroundStation resource, and associated Transmitter resource.

```
Create GroundStation GDS;

GDS.CentralBody           = Earth;
GDS.StateType             = Cartesian;
GDS.HorizonReference      = Ellipsoid;
GDS.Location1             = 4637.874118023063;
GDS.Location2             = 1028.582597899062;
GDS.Location3             =  4241.640535865513;
GDS.Id                    = 'GDS';
GDS.AddHardware           = {Transmitter1, Receiver1, Antenna1};
GDS.MinimumElevationAngle = 10;
GDS.ErrorModels           = {RangeModel, RangeRateModel};

```

Next, we create and configure the CAN and MAD GroundStation resource, and associated Transmitter resource.

```
Create GroundStation CAN;

CAN.CentralBody           = Earth;
CAN.StateType             = Cartesian;
CAN.HorizonReference      = Ellipsoid;
CAN.Location1             = 1258.168895420548;
CAN.Location2             =  346.1275104213947;
CAN.Location3             = 6222.291027834557;
CAN.Id                    = 'CAN';
CAN.AddHardware           = {Transmitter1, Receiver1, Antenna1};
CAN.MinimumElevationAngle = 10.;
CAN.ErrorModels           = {RangeModel, RangeRateModel};

Create GroundStation MAD;

MAD.CentralBody           = Earth;
MAD.StateType             = Cartesian;
MAD.HorizonReference      = Ellipsoid;
MAD.Location1             = 1258.168895420548;
MAD.Location2             = 346.1275104213947;
MAD.Location3             = 6222.291027834557;
MAD.Id                    = 'MAD';
MAD.AddHardware           = {Transmitter1, Receiver1, Antenna1};
MAD.MinimumElevationAngle = 10.;
MAD.ErrorModels           = {RangeModel, RangeRateModel};
```

It is well known that all measurement types have random noise and/or biases associated with them. For GMAT, these effects are modelled using ground station error models. Since we have already created the GroundStation object and its related hardware, we now create the ground station error models. Since we wish to form an orbit estimate using both range and range rate data, we need to create two error models as shown below, one for range measurements and one for range rate measurements.

```
%
%   Error models
%

Create ErrorModel RangeModel;

RangeModel.Type           = 'Range';
RangeModel.NoiseSigma     = 0.010;
RangeModel.Bias           = 0.0;
RangeModel.SolveFors      = {};

Create ErrorModel RangeRateModel;

RangeRateModel.Type       = 'RangeRate';
RangeRateModel.NoiseSigma = 0.00001;
RangeRateModel.Bias       = 0.0;
RangeRateModel.SolveFors  = {};
```

The script segment above is broken into three sections. The first section defines an ErrorModel named RangeModel. The error model Type is Range which indicates that it is an error model for range measurements. The 1 sigma standard deviation of the

Gaussian white noise is set to 0.010 Range Units (RU) and the measurement bias is set to 0 RU.

The second section above defines an ErrorModel named RangeRateModel. The error model Type is RangeRate which indicates that it is an error model for range rate measurrments. The 1 sigma standard deviation of the Gaussian white noise is set to 0.00001Hz and the measurement bias is set to 0 Hz. The range and range rate NoiseSigma values above will be used to form measurement weighting matrices used by the estimator algorithm.

The third section above attaches the two ErrorModel resources we have just created to the CAN, GDS, and MAD GroundStation resources. Note that in GMAT, the measurement noise or bias is defined on a per ground station basis. Thus, any range measurement error involving the CAN, GDS, and MAD GroundStation is defined by the Range ErrorModel and any RangeRate measurement error involving the CAN, GDS, and MAD GroundStation is defined by the RangeRate ErrorModel.

- **Define the types of measurements to be processed**

Now we have created and configured a TrackingFileSet resource. This resource defines the type of data to be processed, the ground stations that will be used, and the file name of the input GMD file which will contain the measurement data. Note that in order to just cut and paste from our simulation, we name our resource simData. But, since, in this script, we are estimating, perhaps a better name would have been estData.

```
%
%   Tracking file sets
%

Create TrackingFileSet simData;

simData.AddTrackingConfig       = {{GDS, SimSat, GDS}, 'Range', 'RangeRate'};
simData.AddTrackingConfig       = {{CAN, SimSat, CAN}, 'Range', 'RangeRate'};
simData.AddTrackingConfig       = {{MAD, SimSat, MAD}, 'Range', 'RangeRate'};
simData.FileName                = {'SpacecraftNavigation_SolveFors_CartesianState_Range_RangeRate.gmd'};
simData.RampTable               = {};
simData.UseLightTime            = True;
simData.UseRelativityCorrection = False;
simData.UseETminusTAI           = False;
simData.SimRangeModuloConstant  = 67108864;
simData.SimDopplerCountInterval = 10.;
simData.DataFilters             = {};

Create TrackingFileSet estData;

estData.AddTrackingConfig       = {{GDS, EstSat, GDS}, 'Range', 'RangeRate'};
estData.AddTrackingConfig       = {{CAN, EstSat, CAN}, 'Range', 'RangeRate'};
estData.AddTrackingConfig       = {{MAD, EstSat, MAD}, 'Range', 'RangeRate'};
estData.FileName                = {'SpacecraftNavigation_SolveFors_CartesianState_Range_RangeRate.gmd'};
estData.RampTable               = {};
estData.UseLightTime            = True;
estData.UseRelativityCorrection = False;
estData.UseETminusTAI           = False;
estData.SimRangeModuloConstant  = 67108864;
estData.SimDopplerCountInterval = 10.;
estData.DataFilters             = {};

```

The script lines above are broken into three sections. In the first section, AddTrackingConfig is the field that is used to define the data types. The AddTrackingConfig lines tell GMAT to process range and range rate, two way

measurements for the GDS to Sat to GDS, CAN to Sat to CAN and MAD to Sat to MAD measurement strand.

The second section above sets some processing parameters that apply to both the range and range rate measurements. We set UseLightTime to True in order to generate realistic computed, C, measurements that take into account the finite speed of light. The other two parameters in this section, UseRelativityCorrection and UseETminusTAI, are set to True or False so that general relativistic corrections are applied or not to the light time equations. We set two other DSNsimData fields, SimDoppler-CountInterval and SimRangeModuloConstant.

- **Create and configure Force model and propagator**

We now create and configure the force model and propagator that will be used for the simulation. The script segment accomplishing this is shown below.

```
%
%   Propagators
%

Create ForceModel ODProp_ForceModel;

ODProp_ForceModel.CentralBody  = Earth;
ODProp_ForceModel.PointMasses  = {Earth};
ODProp_ForceModel.Drag         = None;
ODProp_ForceModel.SRP          = Off;
ODProp_ForceModel.ErrorControl = None;

Create Propagator ODProp;

ODProp.FM                      = ODProp_ForceModel;
ODProp.Type                    = 'RungeKutta56';
ODProp.InitialStepSize         = 60;
ODProp.Accuracy                = 1e-13;
ODProp.MinStep                 = 0;
ODProp.MaxStep                 = 60;
ODProp.MaxStepAttempts         = 50;

```

We set ErrorControl = None because for the current release of GMAT, batch estimation requires fixed step numerical integration. The fixed step size is given by Prop.InitialStepSize which has a default value of 60 seconds.

- **Create and configure Batch Estimator object**

As shown below, we create and configure the BatchEstimator object used to define our estimation process.

```
Create Simulator sim;

sim.AddData                    = {simData};
sim.EpochFormat                = 'UTCGregorian';
sim.InitialEpoch               = '01 Jun 2021 00:00:00.000';
sim.FinalEpoch                 = '04 Jun 2021 00:00:00.000';
sim.MeasurementTimeStep        = 60;
sim.Propagator                 = ODProp;
sim.AddNoise                   = On;

%
%   Estimator
%

Create BatchEstimator bat

bat.ShowProgress               = True;
bat.Measurements               = {estData}
bat.AbsoluteTol                = 0.0001;
bat.RelativeTol                = 0.001;
bat.MaximumIterations          = 10;
bat.MaxConsecutiveDivergences  = 3;
bat.Propagator                 = ODProp;
bat.ShowAllResiduals           = On;
bat.OLSEInitialRMSSigma        = 1000;
bat.OLSEMultiplicativeConstant = 3;
bat.OLSEAdditiveConstant       = 0;
bat.InversionAlgorithm         = 'Internal';
bat.EstimationEpoch            = 'FromParticipants';
bat.ReportStyle                = 'Normal';
bat.ReportFile                 = 'SpacecraftNavigation_SolveFors_CartesianState_Range_RangeRate.txt';

```

All of the fields above are described in BatchEstimator Help but we describe them briefly here as well. In the first script line above, we create a BatchEstimator object, bat. In the next line, we set the ShowProgress field to true so that detailed output of the batch estimator will be shown in the message window. Then we set the ReportStyle to Normal. And in the last line, we set the ReportFile field to the name of our desired output file which by default is written to GMAT's 'output' directory.

We set the Measurements field to the name of the TrackingFileSet resource we wish to use. Recall that the TrackingFileSet, estData, that we created in the Define the types of measurements that will be processed section defines the type of measurements that we wish to process. In our case, we wish to process range and range rate data associated with the CAN, GDS, and MAD ground stations.

The next four fields, AbsoluteTol, RelativeTol, MaximumIterations, and MaxConsecutiveDivergences define the batch estimator convergence criteria. The next script line sets the Propagator field which specifies which Propagator object should be used during estimation. We set this field to the ODPropagator object which we created in the Define the types of measurements that will be processed section. In the 11th script line, we set the ShowAllResiduals field to On show that the observation residuals plots, associated with the various ground stations, will be displayed.

The next three script lines set fields, OLSEInitialRMSSigma, OLSEMultiplicativeConstant, and OLSEAdditiveConstant, that are associated with GMAT's Outer Loop Sigma Editing (OLSE) capability that is used to edit, i.e., remove, certain measurements so that they are not used to calculate the orbit estimate. See the

“Behavior of Outer Loop Sigma Editing (OLSE)” discussion in the BatchEstimator Help for complete details.

Next, we set the EstimationEpochFormat field to 'FromParticipants’ which tells GMAT that the epoch associated with the solve-for variables, in this case the Cartesian State of Sat, comes from the value of Sat.Epoch.

Next, we set the InversionAlgorithm field to 'Internal' which specifies which algorithm GMAT should use to invert the normal equations. There are two other inversion algorithms, 'Cholesky' or 'Schur' that we could optionally use.

- **Run the mission and analyze the results**

The script segment used to run the mission is shown below.

```
219
220 BeginMissionSequence
221
222 RunSimulator sim;
223 RunEstimator bat;
224
```

The first script line, BeginMissionSequence, is a required command which indicates that the “Command” section of the GMAT script has begun. The second line of the script issues the RunEstimator command with the bat BatchEstimator resource, defined in the Create and configure Batch Estimator object section, as an argument. This tells GMAT to perform the estimation using parameters specified by the bat resource.

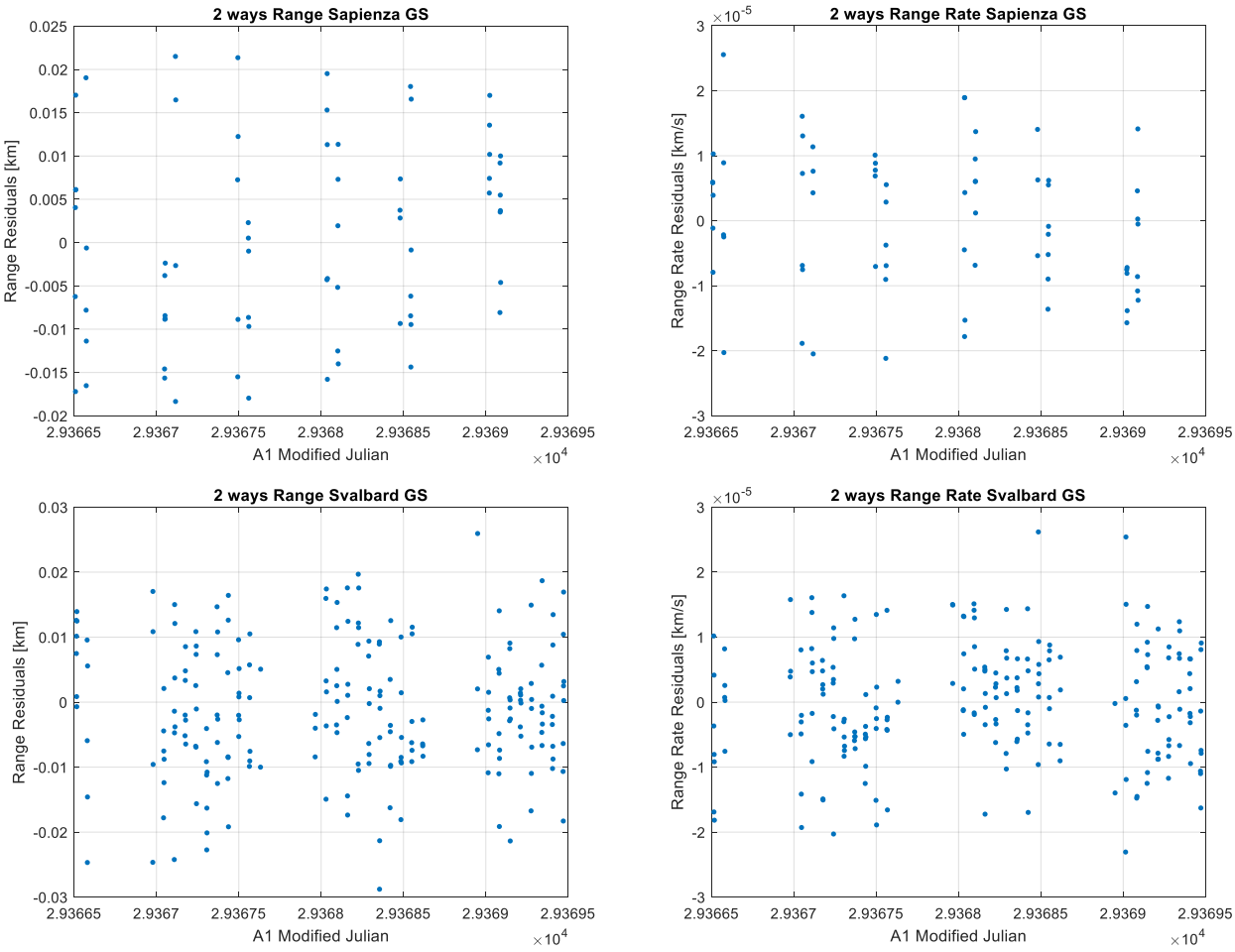


Figure 4.4: Range and Range Rate Residuals for the Sapienza GS (upper left and right), and for Svalbard GS (bottom left right)

4.3. GPS

Using the GMAT software again, a batch filter was always implemented, but this time using simulated observables for the on-board GNSS receiver. The GPS_PosVec observation type models position estimates provided by an on-board GPS receiver. Since this type of data is not derived from ground station measurement modeling, the error model for GPS_PosVec data is specified on the ErrorModels field of a Receiver resource instead. The receiver must be attached to the corresponding Spacecraft object.

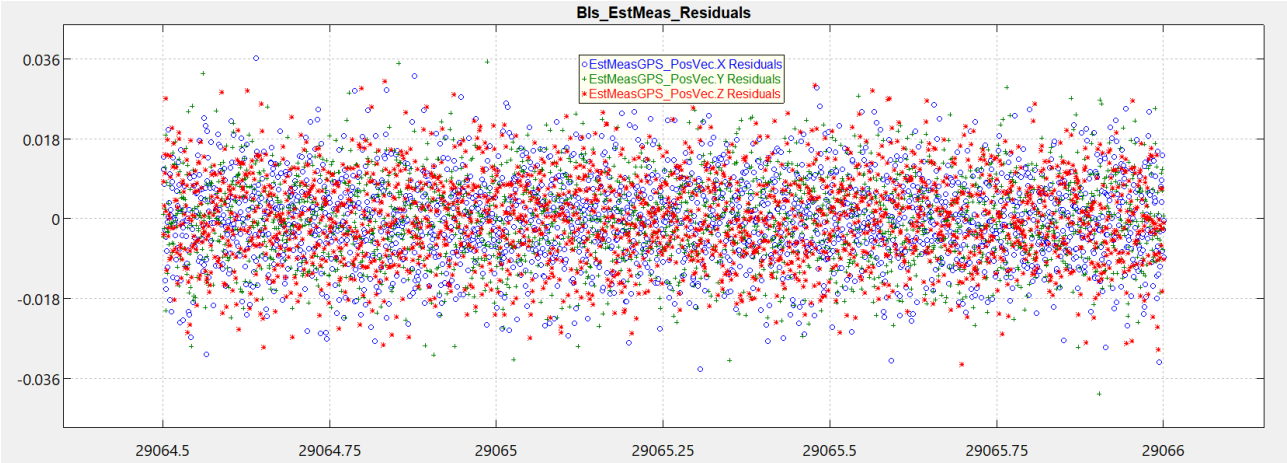


Figure 4.5: Position Vector residuals for the GNSS receiver with GMAT software.

In the simulation carried out with GMAT, after defining the SimSat and EstSat satellite, we implemented the on-board hardware and in particular the GNSS receiver with characteristics similar to those of the datasheet of the receiver chosen by the Payload group, SGR LIGO. Then we set up the force model and orbital propagator using a J2 propagator based on Runge Kutta's method. Finally, we set the error model and the selection of tracking data. The solver to analyze the simulated data is the batch filter with a timestep is 60 sec. So, in conclusion, the data is processed every 60 seconds and the GNSS is always on, with a duty cycle equal to 1.

The precision of this technique is superior to propagation systems and so it is a natural choice for our mission (also considering the fact that we already have a GPS on board).

5. Orbit Control

5.1. Orbit maintenance

Based on the requirement of maintaining orbit position in a range of 20 km, the satellite is maneuvered to correct its orbital elements. This correction is due to the perturbing forces which are acting on them, especially by the gravitational perturbation J2 and by atmospheric drag, particularly intense in the chosen LEO orbit. To achieve this maintenance, electrical thrusters are used, basing the control action on the following mathematical basis. The state vector is:

$$\begin{aligned} \underline{x} &\coloneqq [\underline{z}^T \quad x_6 \quad x_7] \\ &= [x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5 \quad x_6 \quad x_7]^T \\ &= \left[p \quad l \quad m \quad n \quad s \quad q \quad \frac{m}{m_0}\right]^T \end{aligned} \tag{5.1.1}$$

where the first six components are the nonsingular equinoctial elements, given by the following relations:

$$\begin{cases} l = e\cos(\Omega+\omega) \\ m = e\sin(\Omega+\omega) \\ n = \tan\dfrac{i}{2}\cos\Omega \\ s = \tan\dfrac{i}{2}\sin\Omega \\ q = \Omega+\omega+\theta_* \end{cases} \tag{5.1.2}$$

and the seventh component is the so-called mass ratio.

The governing equations for the nonsingular equinoctial elements are:

$$\dot{z} = \boldsymbol{G}(\underline{z}, x_6)\underline{a} \tag{5.1.3}$$

$$\dot{x}_6 = \sqrt{\frac{\mu_E}{x_1^3}}(1 + x_2\cos x_6 + x_3\sin x_6) + \sqrt{\frac{x_1}{\mu_E}}\frac{x_4\sin x_6 - x_5\cos x_6}{1 + x_2\cos x_6 + x_3\sin x_6}a_h \tag{5.1.4}$$

$$\begin{aligned} \dot{x}_7 &= \frac{\dot{m}}{m_0} = -\frac{u_T}{c} \\ 0 &\le u_T \le u_T^{(max)} \end{aligned} \tag{5.1.5}$$

$$\left(u_T \coloneqq \frac{T}{m_0} \quad and \quad u_T^{(max)} \coloneqq \frac{T_{max}}{m_0}\right)$$

where $\underline{a} = \underline{a_T} + \underline{a_P}$ is the non-Keplerian acceleration 3x1 vector which components are expressed with respect to local vertical local horizontal (LVLH) frame $(\hat{r}, \hat{\theta}, \hat{h})$ (see figure 5.1); a_h is the third component of $\underline{a}$. Moreover, $\underline{a_T} = \frac{\underline{u_T}}{x_7}$ and $\underline{a_P}$ contains the perturbations that act on the spacecraft, which main contribute is given by the harmonic J_2 and by atmospheric drag force.

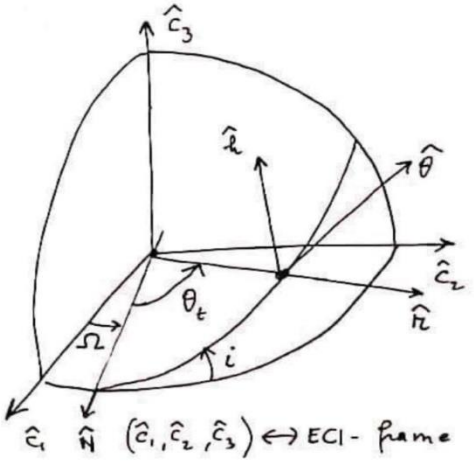


Figure 5.1: LVLH frame

The matrix $\boldsymbol{G}(\boldsymbol{z}, x_6)$ is written in the following form:

$$\boldsymbol{G} = \sqrt{\frac{x_1}{\mu_E}} \begin{bmatrix} 0 & \frac{2x_1}{\eta} & 0 \\ \sin x_6 & \frac{(\eta+1)\cos x_6 + x_2}{\eta} & -\frac{x_4 \sin x_6 - x_5 \cos x_6}{\eta} x_3 \\ -\cos x_6 & \frac{(\eta+1)\sin x_6 + x_3}{\eta} & \frac{x_4 \sin x_6 - x_5 \cos x_6}{\eta} x_2 \\ 0 & 0 & \frac{1 + x_4^2 + x_5^2}{2\eta} \cos x_6 \\ 0 & 0 & \frac{1 + x_4^2 + x_5^2}{2\eta} \sin x_6 \end{bmatrix} \quad (5.1.6)$$

The control vector is $\underline{u_T} = \frac{\underline{T}}{m_0}$, that is directly related to the thrust acceleration through the previous relation. In order to find the feedback control law that provides convergence toward the desired state, the Lyapunov function is defined as

$$V = \frac{1}{2}\underline{\Psi}^T \boldsymbol{K}\, \underline{\Psi} \tag{5.1.7}$$

where $\underline{\Psi}$ is a vector that becomes null when the actual state is equal to the target state:

$$\underline{\Psi} = \begin{bmatrix} x_1 - p_d \\ x_2^2 + x_3^2 - e_d \\ x_4^2 + x_5^2 - \tan^2\frac{i_d}{2} \end{bmatrix} \tag{5.1.8}$$

and $\boldsymbol{K}$ is the gain matrix, and it is diagonal and positive definite. It is important to remind that, for the Lyapunov theorem, $\underline{\Psi} = 0$ is an asymptotically stable solution if $V(\underline{\Psi}) > 0$ and $\dot{V}(\underline{\Psi}) < 0$ when $\underline{\Psi} \neq 0$, i.e., if $V(\underline{\Psi})$ positive definite with $\dot{V}(\underline{\Psi})$ negative definite exists. Since the time derivative of V is

$$\dot{V} = \underline{\Psi}^T \boldsymbol{K}\, \frac{\partial \Psi}{\partial \underline{z}}\, \underline{\dot{z}} = \underline{\Psi}^T \boldsymbol{K}\, \frac{\partial \Psi}{\partial \underline{z}}\, \boldsymbol{G}\left(\frac{\underline{u_T}}{x_7} + \underline{a_P}\right) \tag{5.1.9}$$

an appropriate choice of $\underline{u_T}$ that makes $\dot{V} < 0$ is

$$\begin{aligned} \underline{u_T} &= -x_7\left(\underline{b} + \underline{a_P}\right) \\ with \quad \underline{b} &= \boldsymbol{G}^T\left(\frac{\partial \Psi}{\partial \underline{z}}\right)^T \boldsymbol{K}^T \underline{\Psi} \end{aligned} \tag{5.1.10}$$

In fact, using this control law the time derivative of the Lyapunov function is

$$\dot{V} = -\underline{\Psi}^T \boldsymbol{K}\, \frac{\partial \Psi}{\partial \underline{z}}\, \boldsymbol{G}\boldsymbol{G}^T\left(\frac{\partial \Psi}{\partial \underline{z}}\right)^T \boldsymbol{K}^T \underline{\Psi} = -\underline{b}^T \underline{b} \leq 0 \tag{5.1.11}$$

If $\underline{u_T}$ obtained by the previous formula has magnitude greater than $u_T^{(\max)}$, the following law con be used:

$$\underline{u_T} = -u_T^{(max)} \frac{\underline{b} + \underline{a_P}}{\left|\underline{b} + \underline{a_P}\right|} \tag{5.1.12}$$

These two previous laws can be written together in the following form:

$$\underline{u_T} = -u_T^{(max)} \frac{x_7 \left(\underline{b} + \underline{a_P}\right)}{\max\left\{u_T^{(max)}, \left|x_7 \left(\underline{b} + \underline{a_P}\right)\right|\right\}} \tag{5.1.13}$$

It can be demonstrated that this control law makes $\dot{V} \leq 0 \ \ \forall t$ only if

$$\frac{u_T^{(max)}}{x_7} = \left|\underline{a_T^{(max)}}\right| > \left|\underline{a_P}\right| \ \ \forall t. \tag{5.1.14}$$

However, $\dot{V}$ can be greater than zero in limited time intervals without compromising necessarily the asymptotic convergence.

- **Analysis and results**

In the studied mission (LEO orbit with altitude of 474 km), the perturbations are always greater in magnitude than thrust acceleration available because the harmonic J_2 and atmospheric drag strongly affects the motion at low altitudes. Because of that, oscillations around the desired orbital position cannot be avoided.

In order to fulfil the 20 km requirement, the multi-FEEP thrusters by Morpheus Space have been chosen.

From computation, 280g of propellent are required for a 5-years mission if one uses the maximum thrust provided by one multi-FEEP system in calculations. However, since the propellent capacity of one system is not sufficient, it is necessary to employ two multi-FEEP systems and make them work at half of their maximum thrust.

Although increasing thrust would have positive effect in controlling the orbital position, this would result in a direct increase of needed power. But power that can be used is strongly limited by the necessity to allow more than one subsystem to work at the same time. In according to Power Subsystem Group, the solution founded is based on the use of the two thrusters simultaneously at half of their maximum thrust, so that total power is equal to the power needed by just one of these in the needed working condition but no unbalance torque is generated, firing for a time equal to the time of exposure to Sun reduced of 10 minutes (in which also the payload has to work and so, to limit consumed power, multiFEEP have to be switched off). Obviously, also during eclipse time thrusters are switched-off.

This strategy allows to maintain orbit parameters inside the established ranges, at the cost of an increase in propellant consumption: for this reason, 2 systems are needed to control spacecraft for the whole mission time, assuring also some extra grams of propellant for emergency firing.

Assuming a small injection error, the following results are obtained:

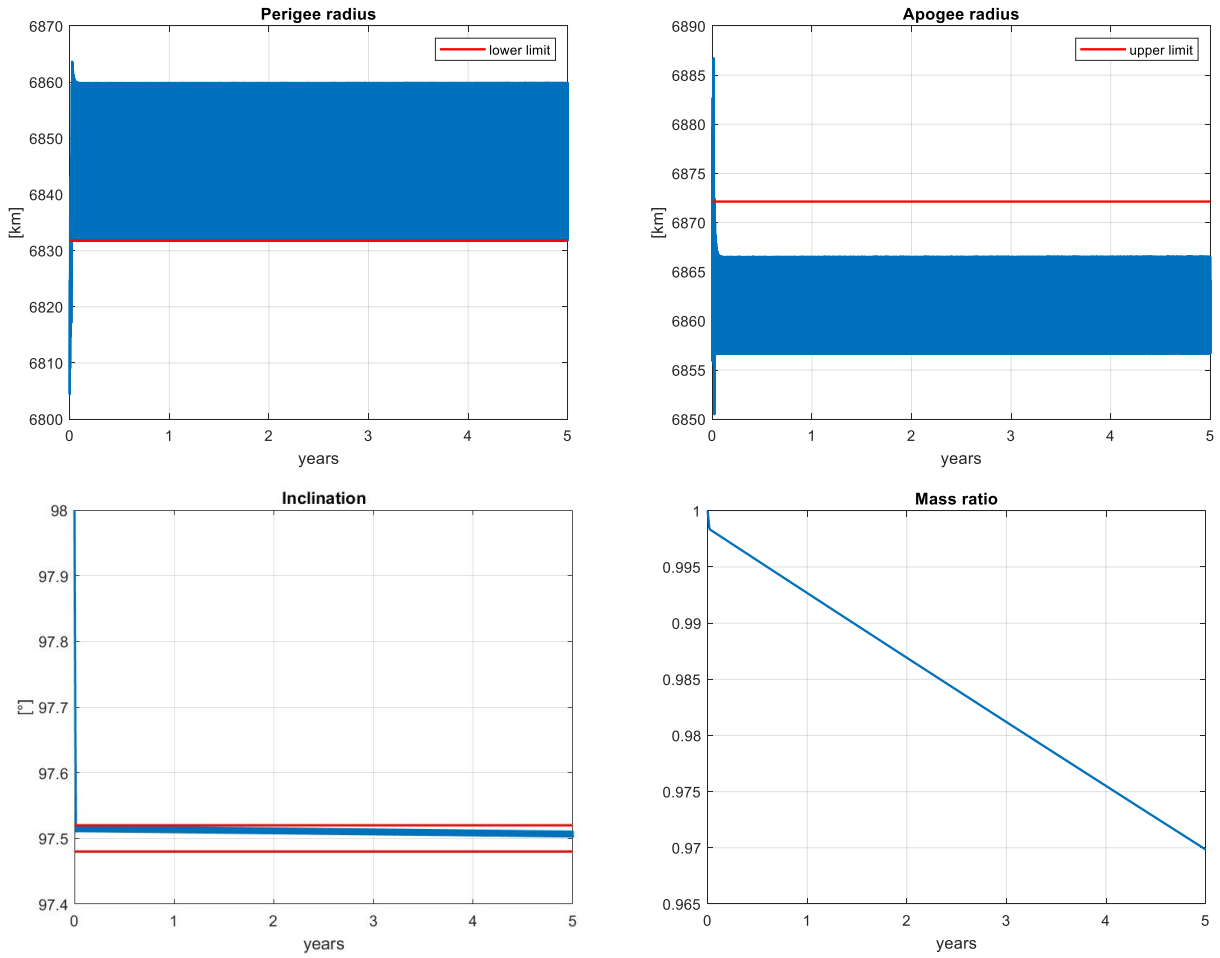


Figure 5.2: Perigee radius (upper left), apogee radius (upper right), inclination (bottom left), mass ratio (bottom right)

It's evident that requirements for apogee and perigee radius are satisfied, and also that orbit inclination is perfectly maintained around the desired value. To consider a more realistic situation, also a small acquisition period has been simulated at the beginning of the mission: supposing to start from an inclination of 98°, in very small-time system takes the spacecraft to desired value and maintain it in that configuration. Although this acquisition determines higher propellant consumption at the beginning, chosen MultiFEEP systems are perfectly able to fulfil the whole mission also in this "worst" scenario.

5.2. Deorbiting

The threats of space debris are increasing due to the launch of several multi-satellite constellations, particularly in low-Earth orbit. The international interest in the problem is reflected in the Inter-Agency Space Debris Coordination Committee (IADC) mitigation guidelines, which indicate the removal of space systems that interfere with the Low Earth Orbit (LEO) region no later than 25 years after the end of the mission.

They have mandated that all spacecraft must either deorbit within a given amount of time or move into a graveyard orbit for safe storage.

Most of the satellite will burn up in the atmosphere. In case one could expect heavy and heat resistant parts to impact the Earth' surface, one first may have to wait till the satellite is in a proper position, so the impact would be on a safe location (e.g., ocean), but for a very small satellite like a 6U CubeSat no part will survive during re-entry phase.

The majority of launched small spacecraft do not carry on-board propulsion, making them unable to achieve graveyard orbits for decommissioning. Therefore, they need to rely on deorbit techniques such as increasing the drag area. Otherwise, most efficient strategy is using a separate propulsion system to deorbit it in smaller time.

Low Earth orbiting satellites experience orbital decay and have physical lifetimes determined almost entirely by their interaction with the atmosphere. Prediction of such lifetimes or of a re-entry date is of great interest to satellite planners, users, trackers, and frequently to the general public.

The prediction of satellite lifetimes depends upon a knowledge of the initial satellite orbital parameters, the satellite mass to cross-sectional area (in the direction of travel), and a knowledge of the upper atmospheric density and how this respond to space environmental parameters which must also be predicted: the drag force that such a satellite experiences is due to its interaction with the few air molecules that are present at these altitudes. The density of the atmosphere at LEO heights is controlled by solar X-ray flux and particle precipitation from the magnetosphere and so varies with the current space weather conditions.

Even with a complete atmospheric model describing variations with time, season, latitude and altitude, complete specification of orbital decay is not possible because of uncertainties in the prediction of satellite attitude (which affects the relevant cross-sectional area), and solar and geomagnetic indices (which substantially modify the atmospheric model). Even when most of the quantities are known there appears to be an irreducible level below which it is not possible to predict. This level appears to be around 10% of the remaining satellite lifetime, irrespective of what that lifetime is.

The main parameter to be defined is the atmospheric density. Since mission height is below 500 km, orbits can be regarded as essentially circular, with the use of the semimajor axis in place of the orbital radius. The atmospheric density ρ is specified by a simple exponential with variable scale height H. For a fixed exospheric temperature T, H is made to vary with altitude h through the use of an effective atmospheric molecular mass m. This m includes both the actual variation in molecular mass with height and a compensation term for the variation in temperature over the considered range from 180 to 500 km. The variation in density due to the space environment is

introduced through T which is specified as a function of the solar radio flux F10.7 and the geomagnetic index Ap.

The two terms which are used in the model describe the effects of different agencies, both of which originate from the Sun. The solar X-ray output incident upon the Earth is generally absorbed at the base of the thermosphere (around 120 km) and this gives rise to a direct heating effect which propagates itself upward from this level. The solar 10 cm radio flux is used as a surrogate for the total solar X-ray flux which produces this effect. This flux can vary from a low of about 65 to over 300 Solar Flux Units ($1\ SFU\ =\ 10-22\ W/m^2/Hz$). The other agency is the precipitation of particles (mainly electrons and protons) from the magnetosphere down into the lower thermosphere. The energy dumped by this precipitation again acts to heat the atmosphere, which subsequently changes the atmospheric density. Most of these particles originate from the Sun. They are expelled in Coronal Mass Ejections, travel through the interplanetary medium, and eventually arrive at the Earth. Precipitation of such particles is well correlated with large variations in the geomagnetic field as measured at ground level, and quantified by a number of geomagnetic indices, including the planetary Ap index used here. This index, computed every 24 hours, hovers just above zero in quiet times, but may rise to above 400 (no units) at times of major geomagnetic storms.

The set of defining equations for the model are given by:

$$T = 900 + \ 2.5\ (\ F10.7\ -70\ ) + 1.5Ap \rightarrow [Kelvin] \tag{5.2.1}$$

$$m = 27 - 0.012\ (\ h - 200\ ) \rightarrow 180 < h[km] < 500 \tag{5.2.2}$$

$$H = \frac{T}{m} \rightarrow [km] \tag{5.2.3}$$

$$\rho = 6 * 10^{-10} * exp\left(-\frac{h-175}{H}\right) \rightarrow \left[\frac{kg}{m^3}\right] \tag{5.2.4}$$

All constants were empirically derived to give an appropriate fit to the standard models. It should be noted that the only really valid output of this model is the density. The intermediate variables used in deriving this density in general do not correspond to true atmospheric values at any height within the considered range. The temperature may be regarded as the mean asymptotic value for the exosphere at large altitudes. The molecular mass might be regarded as an integrated mean value from the base of the thermosphere up to the specified height. In our model these parameters are averaged as $Ap = 15$ and $F10.7 = 150$.

A more complete analysis is then carried out by using GMAT software, in which dynamical model is defined considering not only drag force (using MSISE90 model for atmosphere but averaging parameters as $Ap = 15$ and $F10.7 = 150$), but considering also solar radiation pressure effect and, above all, the effect of J2 in the gravitational field that is one of the main contributors for the spacecraft capture. To stop propagation process, a final altitude of 50 km has been fixed (at this altitude, in facts, CubeSat can be considered already burnt).

In GMAT computation all orbital effects are taken into account, also the very small eccentricity introduced in the orbit during non-controlled deorbiting. For this reason, with respect to the simple model based on the integration of the orbit period, in all the figure some oscillation can be detected. Despite this simplicity, results obtained by the two models are very similar and close each other, meaning that also simplified model gives very reliable results.

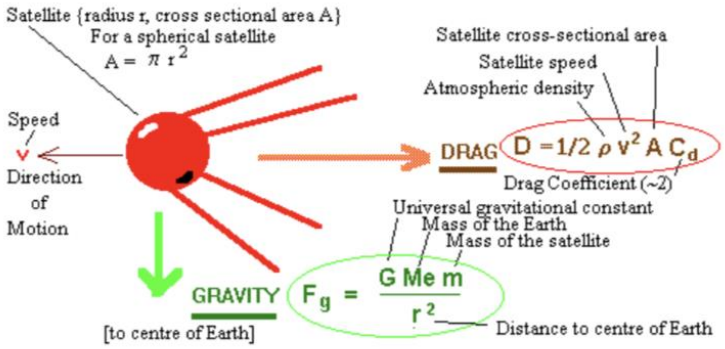


Figure 5.3: Basic forces acting on the spacecraft

```
%----------------------------------------
%---------- ForceModels
%----------------------------------------

Create ForceModel DefaultProp_ForceModel;
GMAT DefaultProp_ForceModel.CentralBody = Earth;
GMAT DefaultProp_ForceModel.PrimaryBodies = {Earth};
GMAT DefaultProp_ForceModel.SRP = On;
GMAT DefaultProp_ForceModel.RelativisticCorrection = Off;
GMAT DefaultProp_ForceModel.ErrorControl = RSSStep;
GMAT DefaultProp_ForceModel.GravityField.Earth.Degree = 4;
GMAT DefaultProp_ForceModel.GravityField.Earth.Order = 4;
GMAT DefaultProp_ForceModel.GravityField.Earth.StmLimit = 100;
GMAT DefaultProp_ForceModel.GravityField.Earth.PotentialFile = 'JGM2.cof';
GMAT DefaultProp_ForceModel.GravityField.Earth.TideModel = 'None';
GMAT DefaultProp_ForceModel.SRP.Flux = 1367;
GMAT DefaultProp_ForceModel.SRP.SRPModel = Spherical;
GMAT DefaultProp_ForceModel.SRP.Nominal_Sun = 149597870.691;
GMAT DefaultProp_ForceModel.Drag.AtmosphereModel = MSISE90;
GMAT DefaultProp_ForceModel.Drag.HistoricWeatherSource = 'ConstantFluxAndGeoMag';
GMAT DefaultProp_ForceModel.Drag.PredictedWeatherSource = 'ConstantFluxAndGeoMag';
GMAT DefaultProp_ForceModel.Drag.CSSISpaceWeatherFile = 'SpaceWeather-All-v1.2.txt';
GMAT DefaultProp_ForceModel.Drag.SchattenFile = 'SchattenPredict.txt';
GMAT DefaultProp_ForceModel.Drag.F107 = 150;
GMAT DefaultProp_ForceModel.Drag.F107A = 150;
GMAT DefaultProp_ForceModel.Drag.MagneticIndex = 3;
GMAT DefaultProp_ForceModel.Drag.SchattenErrorModel = 'Nominal';
GMAT DefaultProp_ForceModel.Drag.SchattenTimingModel = 'NominalCycle';
GMAT DefaultProp_ForceModel.Drag.DragModel = 'Spherical';
```

Figure 5.4: GMAT code to implement complete force model.

For this project different solution are analysed, before choosing the most suitable one for the mission.

5.2.1. Uncontrolled deorbiting

This strategy relies on the aerodynamic drag in order to reduce orbit dimension and achieve dismission of the satellite. When a spacecraft travels through an atmosphere it experiences a drag force in a direction opposite to the direction of its motion. This drag force is given by the expression:

$$D = \frac{1}{2}\rho v^2 A C_d \tag{5.2.5}$$

where D is the drag force, ρ is the atmospheric density, v is the speed of the satellite, A is its cross-sectional area perpendicular to the direction of motion, and C_d is the drag coefficient. At the altitudes at which satellites orbit, C_d is generally assumed to be equal to 2.2, although experiments have shown that this can vary widely. Because it is usually difficult to separate out independent variations in the cross-sectional area from the variations in the drag coefficient, we shall henceforth use an effective cross-sectional area $A_e = AC_d$ for the rest of the model.

We can use this drag force in Newton's second law together with energy considerations of a circular orbit to derive an expression for the change in the orbital radius and period of the satellite with time. For a circular orbit we have the following relation between period P and semimajor axis a:

$$P = 2\pi\sqrt{\frac{a^3}{\mu}} \tag{5.2.6}$$

The reduction in the period due to atmospheric drag is given by:

$$\frac{dP}{dt} = -3\pi a\rho\left[\frac{AC_d}{m}\right] \tag{5.2.7}$$

The last two equations, together with the equations modelling the atmospheric density, can be iterated from the starting satellite altitude and time. In other words, the satellite is flown around its orbit using appropriate past or forecast values for the space environment variables (F10.7 and Ap). Re-entry is assumed to occur when the satellite has descended to an altitude of 180 km.

Conceptual meaning of this orbit radius dimension is very clear in the following picture:

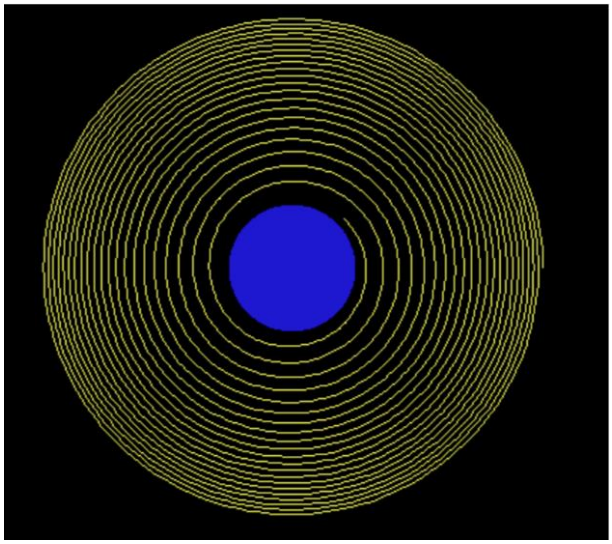


Figure 5.5: Scheme of the orbit radius reduction

In this diagrammatic view of how a satellite decays in low Earth orbit (LEO), many more orbits occur than shown here. Also, the radial scale has been drawn very much enlarged to show how the spiral opens out toward the end of the satellite's life, when it undergoes a fiery re-entry into the Earth's lower atmosphere. Computing it for our purpose, the following results are obtained:

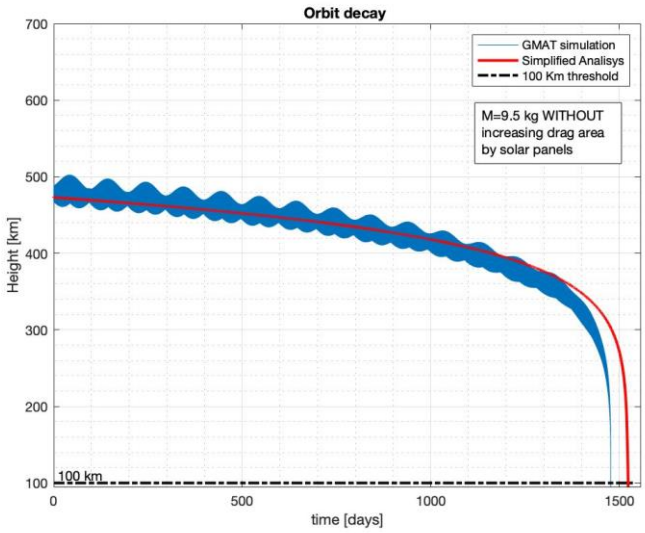


Figure 5.6: Orbit decay without using solar panels to increase drag area.

As clearly seen in picture, results obtained by the simplified model described above fit well with complete GMAT one.

Since orbit is quite low, deorbit time is around 4 years, perfectly inside the 25 years range imposed.

Oscillation in GMAT plot, as explained above, are due to the small eccentricity of the free-spacecraft new orbits: thanks to atmospheric drag, Earth shape and solar radiation pressure, dimension of the orbit is decreased passage by passage, resulting in not perfect circular orbits.

Up to 350 km, height reduction is slow because atmospheric drag and J2 term are not so influent. But then, starting from 350 km, these terms are quite effective and so orbit size decrease faster. Reached 180 km, slope of the curve is almost vertical, meaning that orbit fall is very fast and so spacecraft is rapidly captured by atmosphere.

To reduce deorbit time, a clever solution can be to use solar panels to increase surface area and generate drag: since in this mission orientable solar panels have been chosen, during deorbiting they can be oriented such that all the area is in the same direction of motion. Using this tricky strategy, deorbit time can be heavily reduced. Appling this strategy, the following results are obtained:

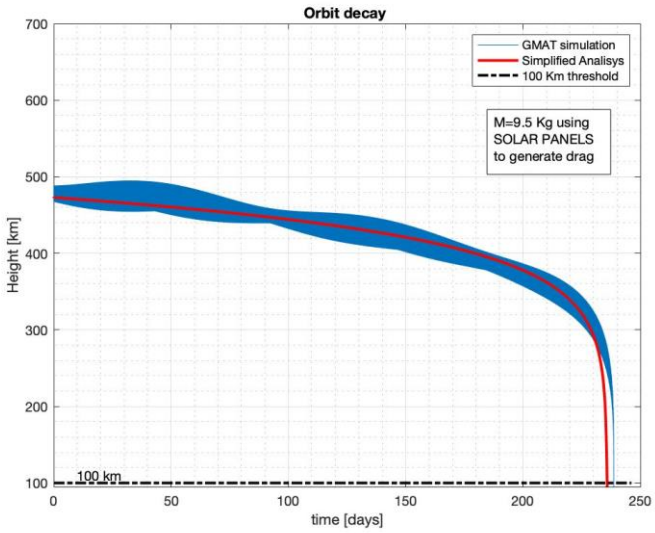


Figure 5.7: Orbit decay using solar panels to increase drag area.

As evident in picture, increasing area using the orientable solar panels allow to take around 235 days with respect to 4 years of before, enhancing a very good deorbit time.

A complete reference for the altitude with respect to time can be found in the output table reported, also giving information on the reduction of the orbit period and number of orbits in one day.

5.2.2. Passive-Controlled strategy: Drag sail.

Drag devices are particularly attractive because they are independent of spacecraft propulsion capabilities. They can be deployed to increase the drag area for faster deorbiting in compliance with the requirements. Other notable advantages are simplicity and small volumes occupied while stowed. Moreover, this requires no further active control after deployment.

Recently, this technology has been implemented in several small spacecraft missions, and several companies and institutions are developing prototypes that are increasingly more mature, providing solutions to the space debris problem for missions that do not have resources for an active system.

Most of the drag device designs consist of a deployable “drag sail” made of a thin film (the sail) that is deployed via extension mechanism. The feasibility of drag sails for small spacecraft has now been demonstrated on several missions. Examples can be found in CanX-7 (University of Toronto Institute for Aerospace Studies Space Flight Laboratory) and DragSail (Surrey Space Centre) spacecrafts. However, all of these missions were demonstrative, and payload was uniquely constituted of the drag sail.

So, there are not any examples of missions with other purposes that used these devices for deorbiting strategy.

5.2.3. Active-Controlled deorbiting: Thrusters

De-orbiting spacecraft requires velocity increment at a certain point in the spacecraft's orbit where the velocity increment or burns of the thrusters shall take place. The velocity increments apogee, ΔV_a, follow from orbital mechanics calculations. The amount of required propellant follows from Tsiolkowski's equation. The deorbit system must be as independent as possible from the satellite system itself, i.e., even if the satellite fails, the system must have power for its operation and be able to have simple communications with the satellite operating centre. This centre can then uplink commands that allows the satellite to further carry out an autonomous deorbit manoeuvre. The performance of the propellant shall be good. In general, the propellant mass for deorbiting will be small in comparison to the satellite mass at the end of mission. For these cases, it is not the specific impulse, but the density impulse that determines the performance.

To deorbit a satellite, one may consider a circular orbit, giving a ΔV to reduce orbit dimension (typically, to deorbit in fast way, final height of the orbit after the impulse must be around 80 km). The velocity in the apogee of an elliptical orbit is always lower than that of a circular orbit at the apogee altitude of the elliptical orbit. Therefore, the values presented here are upper limits.

$$\Delta V = \sqrt{\frac{\mu}{R_i}} - \sqrt{\frac{2\mu}{R_i + R_f}\frac{R_f}{R_i}} = 113\,\frac{\mathrm{m}}{\mathrm{s}} \tag{5.2.8}$$

with respective time equal to half Hohmann transfer period

$$t = \pi\sqrt{\frac{a^3}{\mu}} = 0.75 \text{ hours} \tag{5.2.9}$$

The strategy is to decrease the apogee velocity such that an orbit results with its perigee at 80 km altitude. Atmospheric drag then will rapidly decelerate and heat the satellite. A schematic of this manoeuvre is reported in following picture.

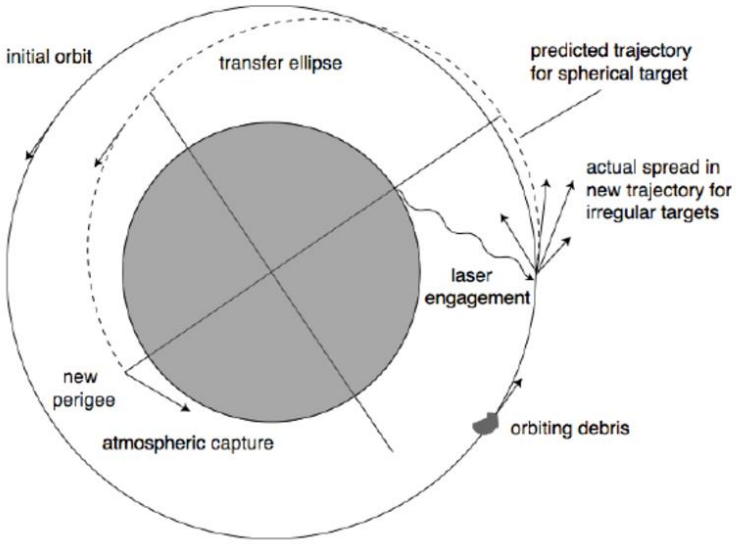


Figure 5.8: Basic scheme of a thruster-controlled deorbiting

In order to evaluate this strategy, basing on propulsive system found in the market analysis, best option to deorbit the satellite would be Dawn Aerospace Propulsive 1.5U unit. Using this engine to deorbit the satellite, this would be the result:

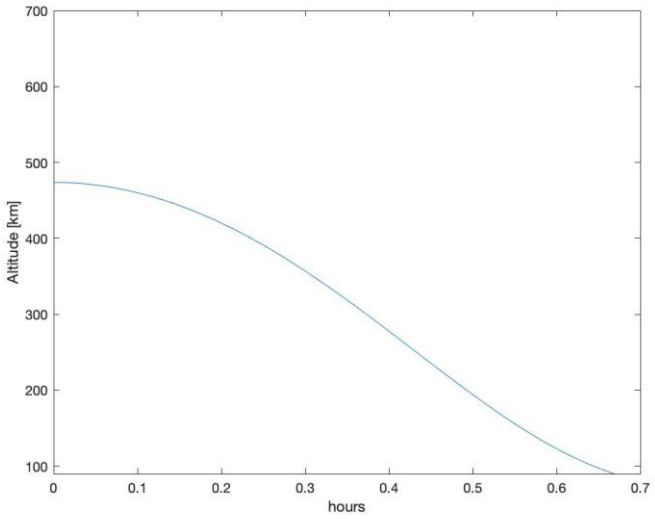


Figure 5.9: Active deorbiting using 1.5U motor.

As clearly seen in picture, giving the complete needed ΔV, complete deorbit time is very low (about 0.75 hours as computed before).

This is just an "ideal solution", since this motor must be discarded because of the weight (2.25 kg) and, above all, because can't fit well inside the structure due to dimension not compatible with the other components being present inside the spacecraft itself.

Making a trade-off with the remaining motors suitable for this CubeSat, the unique solution that can fit well inside the spacecraft, giving a ΔV as high as possible is the Dawn Aerospace Propulsive 1U unit.

With respect to the 1.5U configuration, this motor is lighter (1.4 kg) but contains only 310 grams of propellant, that means a possible $\Delta V = 82 \frac{m}{s}$.

Using this solution spacecraft can be taken to an altitude of about 200 km, putting it on ellipses (since only one impulsive burn is given and so it continues to travel on the "transfer ellipses", not being orbit re-circularized). As consequence, deorbiting is not immediate but takes some time more than expected, as seen in the following figure.

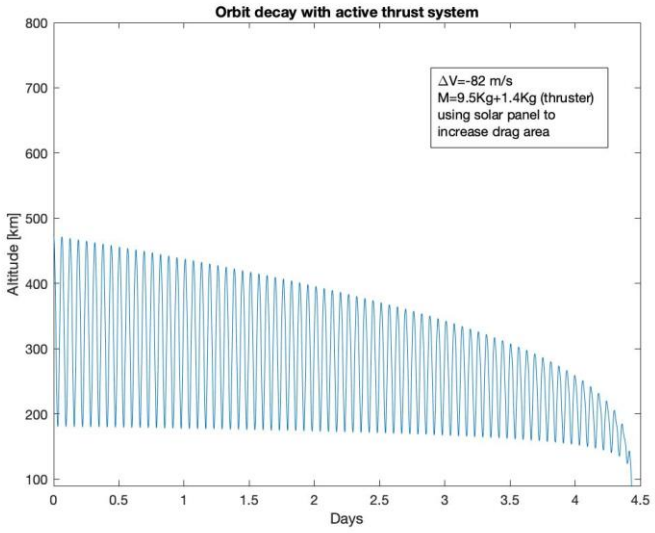


Figure 5.10: Active deorbiting using 1U motor.

Permanence on ellipses determines an unwanted oscillation of perigee and apogee, but, since the perigee is around 200 km, also in this case spacecraft atmospheric capture takes less than 5 days.

Due to this unwanted strong oscillation and making a trade-off between benefits (fast deorbit) and disadvantages (strong altitude oscillation, higher mass of the CubeSat, higher price for the launch, difficulties in integrating it inside the spacecraft), this solution has been discarded in according with System Engineering Subsystem Group.

5.2.4. Final deorbiting choice

In order to achieve a fast and safe deorbiting a trade-off between the various possibilities is carried out. Thruster controlled strategy permits to obtain deorbiting in faster way, but the benefits introduced by this short deorbit time are not balanced by the disadvantages introduced, as explained above. For this reason, this solution is

discarded. Drag sail represents an innovative possibility but flight heritage is quite low and for this reason is not chosen.

Finally, uncontrolled deorbiting permits to deorbit within the limit of 25 years without introducing additional masses or components, limiting so the risk of failure of one of the systems. By using orientable solar panels, also in this uncontrolled strategy deorbit time can be strongly reduced using only components already present on-board, limiting so also the mass of the spacecraft and thus the launch cost. For this reason, uncontrolled deorbiting with solar panels oriented in motion direction is the final choice. A representation of last orbits is plotted in next figure.

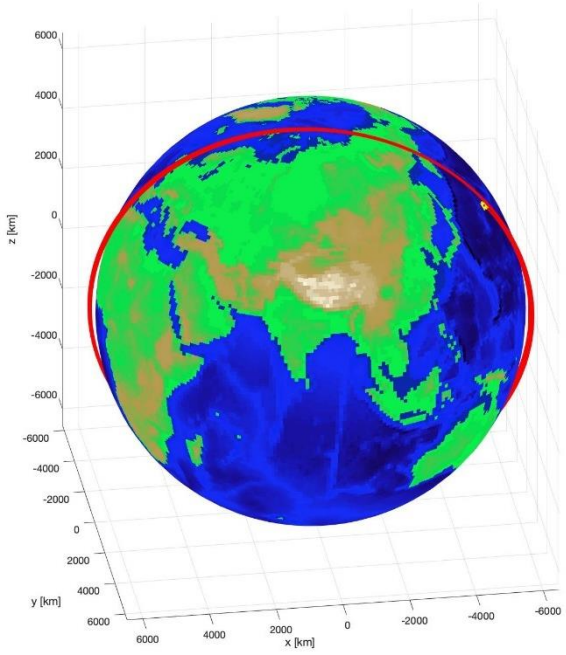


Figure 5.11: Orbit decay representation (yellow point is the final one reached)

6. Attitude Determination

Attitude estimation refers to the process of estimating the current attitude state of a system from a set of measured observations. The state estimation problem involves finding the best estimate of the true system state using a dynamic model and measurements that are both corrupted with random noise of known statistics. It is useful to divide attitude determination approaches into two categories. The first category comprises static determination approaches that depend on measurements taken at the same time, or close enough in time that spacecraft motion between the measurements can be ignored or easily compensated for. The second category comprises filtering approaches that make explicit use of knowledge of the motion of the spacecraft to accumulate a "memory" of past measurements. Static approaches have the requirement that at each time there are enough observations available to fully compute the attitude, but they typically require no a priori attitude estimate. A purely deterministic approach incorporates just enough observation information to uniquely determine the attitude. Filtering can generally provide a more accurate attitude estimate than static methods because it incorporates memory of past observations.

6.1. Simulation of observables

The task of this simulation is to compute the HyonoSat attitude by using the data reported in the matrices OBS, which include ST and SS measurements, and OMEGA_MEAS, which stores the spacecraft angular velocity measured by the gyroscopes. Regarding the gyroscopes error model, we have considered the values of $\sigma_v = 10^{-7}\, rad/s$ and $\sigma_u = 5 * 10^{-9}\, rad/s^2$ (σ_v is the standard deviation of η_1, whereas σ_u is the standard deviation of η_2 that is the RRW, $\dot{b} = \eta_2$). For the first 600s, the ST and SS work properly, and all measurements are available. After 600s, the HyonoSat enters in the eclipse and only ST data are available.

The **`data.mat`** file contains the following input data.

Name	Size	Structure
Qtrue	9001x4	Simulated quaternions that represent the "true values". The first 3 columns are the vector part of the quaternion $\vec{q}$, the fourth column is the scalar part of the quaternion
OMEGA_TRUE	9001x3	Simulated angular rates that represent the "true values" without any gyroscope errors.
OMEGA_MEAS	9001x3	Measured angular rates that include the error of the gyros.

OBS	4x7x9001	A 3-D matrix that include all ST and SS observations. The first 3 rows store the three different measurements collected by the star tracker while the 4th row stores the measurements collected by the SS, at the same time t_i. The 7 columns are W_i, V_i, and σ_i for each measurement. The time evolution is stored in the 3rd dimension of the matrix.

Table 6.1: Observable's specifications

In order to test the algorithms of the attitude determination, it is necessary to generate the measurements contained in this file and, in particular, STK and MATLAB are used to generate them.

The simulation on STK starts the 1st of January 2022 at 10:05 and it ends at 10:20 of the same day and the satellite is assumed nadir pointed for all the simulation.

First of all, during this time interval, the attitude evolution is saved in terms of quaternions and angular velocity. The quaternions are reported in the matrix Qtrue previously introduced, while the angular velocity is contained inside OMEGA_TRUE. Through MATLAB also the matrix OMEGA_MEAS is obtained, by adding random values to the omega measured through the function *randn* of the order of 10^{-7}, according to the gyroscope error defined previously. This is important to distinguish the real angular velocity and the measured one, affected by errors.

Then, the measurements of the sun sensor are simulated. The time evolution of the vector position of the sun from the satellite, divided by its magnitude, is saved. In particular the position is saved along both the ICRF and the body frame. The sun sensor measurements are contained in the 4th row of the OBS matrix. The position along the ICRF forms the V_i vector, while the W_i is obtained by adding to the sun position in the body frame the random values with the function *randn* of the order of 0.1°, that is the accuracy of the sun sensor selected for our design.

Finally, the measurements of the star tracker are simulated. In particular, three stars in visibility in that time interval are selected and their position in the ICRF is saved from STK. These values form the V_i vectors, contained in the first three rows of the OBS matrix. Then, through the quaternions, the attitude matrix is defined, and it allows to obtain the position of the stars from the ICRF to the body frame, by considering the relation: $\overrightarrow{W_i} = R\,\overrightarrow{V_i}$. Also in this case, in order to take into account, the error provided by the device, the function *randn* of the order of 10 arcsec is used and it is added to these vectors, by taking into account the accuracy of the star tracker selected for our design.

6.2. The TRIAD algorithm

Some spacecraft attitude determination methods use exactly two vector measurements. The two vectors are typically the unit vector to the Sun and unit vectors to two stars tracked by star tracker for fine attitude determination. The name “TRIAD” can be considered either as the word “triad” or as an acronym for TRIaxial Attitude Determination.

Suppose that we have measured the components of two-unit vectors in the spacecraft body frame and also know these components in some reference frame. We consider only unit vectors because the length of the vector has no information relevant to attitude determination. We have seen that three parameters are required to specify the attitude matrix A, so two-unit vector measurements are required to determine the attitude matrix, in general. TRIAD is based on the following idea, if we have an orthonormal right-handed triad of vectors $\vec{V} = \{v_1; v_2; v_3\}$ in the reference frame, and a corresponding orthonormal right-handed triad $\vec{W} = \{w_1; w_2; w_3\}$ in the spacecraft body frame, then we build a triad of orthogonal unit-vectors measured by the sensors:

$$\hat{x}_1 = w_1 \tag{6.2.1}$$

$$\hat{x}_2 = \frac{w_1 \times w_2}{|w_1 \times w_2|} \tag{6.2.2}$$

$$\hat{x}_3 = \hat{x}_1 \times \hat{x}_2 \tag{6.2.3}$$

Then we build a triad of orthogonal unit-vectors from the knowledge of celestial object in the inertial reference frame

$$\hat{y}_1 = v_1 \tag{6.2.4}$$

$$\hat{y}_2 = \frac{v_1 \times v_2}{|v_1 \times v_2|} \tag{6.2.5}$$

$$\hat{y}_3 = \hat{y}_1 \times \hat{y}_2 \tag{6.2.6}$$

From these calculations we can build the attitude matrix A:

$$A = \begin{bmatrix} | & | & | \\ \hat{x}_1 & \hat{x}_2 & \hat{x}_3 \\ | & | & | \end{bmatrix} \begin{bmatrix} | & | & | \\ \hat{y}_1 & \hat{y}_2 & \hat{y}_3 \\ | & | & | \end{bmatrix}^T \tag{6.2.7}$$

From the attitude matrix we can perform the decoding for determining the quaternion

$$A(q) = \begin{bmatrix} q_1^2 - q_2^2 - q_3^2 + q_4^2 & 2(q_1 q_2 + q_3 q_4) & 2(q_1 q_3 - q_2 q_3) \\ 2(q_1 q_2 - q_3 q_4) & -q_1^2 + q_2^2 - q_3^2 + q_4^2 & 2(q_2 q_3 + q_1 q_4) \\ 2(q_1 q_3 + q_2 q_4) & 2(q_2 q_3 - q_1 q_4) & -q_1^2 - q_2^2 + q_3^2 + q_4^2 \end{bmatrix} \tag{6.2.8}$$

$$\begin{cases} q_1 = \dfrac{1}{4q_4}(A_{23} - A_{32}) \\ q_2 = \dfrac{1}{4q_4}(A_{31} - A_{13}) \\ q_3 = \dfrac{1}{4q_4}(A_{12} - A_{21}) \\ q_4 = \dfrac{1}{2}(A_{11} + A_{22} + A_{33} + 1)^{\frac{1}{2}} \end{cases} \tag{6.2.9}$$

Finally, we can evaluate the error quaternion for each step with the knowledge of the true quaternion.

$$\delta q = q_{true} \otimes q_{triad}^{-1}$$

Assuming the hypothesis of small rotations, we can represent the error quaternion in terms of Euler angle errors in the configuration 321 (roll, pitch, yaw) using the following expression:

$$\delta\vec{\theta} = \begin{bmatrix} \delta\theta_x \\ \delta\theta_y \\ \delta\theta_z \end{bmatrix} = 2 \begin{bmatrix} \delta q_1 \\ \delta q_2 \\ \delta q_3 \end{bmatrix} \tag{6.2.10}$$

For what concerning the covariance matrix associated to the estimate of the Euler angle errors we have used the following formula

$$P_{\theta\theta}^{-1} = \frac{1}{\sigma_1^2}\{I - \hat{w}_1 \hat{w}_1^T\} + \frac{1}{\sigma_2^2}\hat{s}_4 \hat{s}_4^T \tag{6.2.11}$$

$$\hat{s}_4 = \frac{[\hat{w}_2 \times (\hat{w}_1 \times \hat{w}_2)]}{|\hat{w}_1 \times \hat{w}_2|} \tag{6.2.12}$$

Which provides the inverse of the covariance matrix that is the information matrix, where the σ are the standard deviations of the measurements that consider the error made in the measurement of the vector $\vec{W}$ and the error on the knowledge of the vector $\vec{V}$. Below are the graphs relating to the two configurations that represent the estimated quaternions and the errors on the same (in terms of Euler angles 321) with the relative uncertainties.

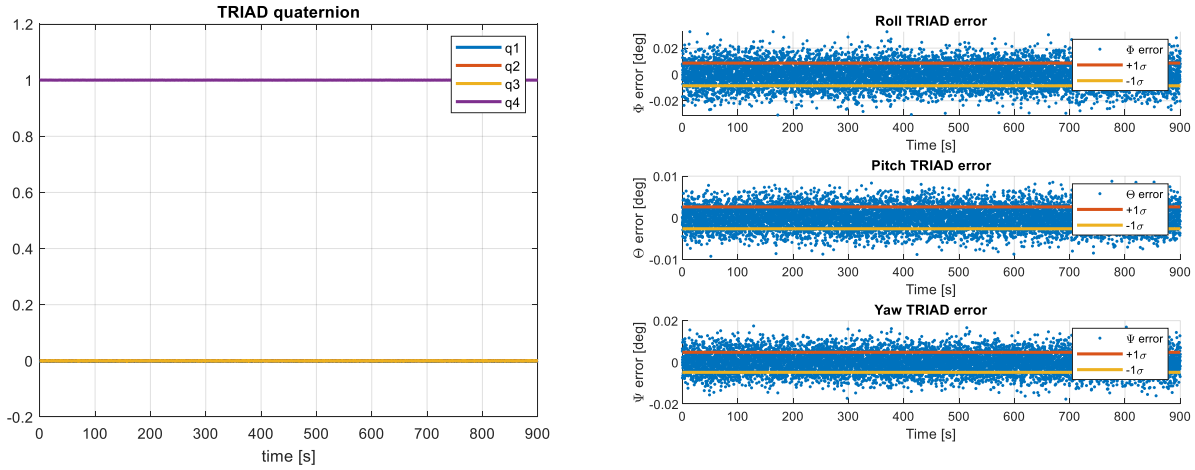


Figure 6.1: Estimated quaternions (left), Euler angles errors (right)

6.3. Quest Algorithm

We can improve on the TRIAD method in two ways: by allowing arbitrary weighting of the measurements and by allowing the use of more than two measurements. The latter is especially important for use with star trackers that can track many stars simultaneously. Almost all of these improvements are based on a problem posed in 1965 by Grace Wahba. Wahba's problem is to find the orthogonal matrix A with determinant $+1$ that minimizes the cost function:

$$J(A) = \frac{1}{2}\sum_{i=1}^{n} a_i |w_i - Av_i|^2 = 1 - g(A) \tag{6.3.1}$$

Where $g(A)$ is defined as

$$g(A) = \sum_{i=1}^{n} a_i w_i^T A v_i = tr(AB^T) \tag{6.3.2}$$

And B is the so called "attitude profile matrix".

$$B = \sum_{i=1}^{n} a_i w_i v_i^T \tag{6.3.3}$$

The Quest Algorithm transforms the maximization of the $g(A)$ function into an eigenvalue problem for a 4x4 matrix constructed with quaternions.

$$g(A) = [\vec{q} \quad q_4] \begin{bmatrix} S - \sigma I & Z \\ Z^T & \sigma \end{bmatrix} \begin{bmatrix} \vec{q} \\ q_4 \end{bmatrix} \tag{6.3.4}$$

$$S = B^T + B \tag{6.3.5}$$

$$Z = \sum_{i=1}^{n} a_i \hat{w}_i \times \hat{v}_i \tag{6.3.6}$$

$$\sigma = tr(B) \tag{6.3.7}$$

The maximization of $g(A)$ occurs when the quaternion q is the eigenvector of K to which corresponds the maximum eigenvalue which is shown to be close to 1. We therefore proceed to the solution of the characteristic equation with the iterative numerical method of Newton- Raphson starting from $\lambda = 1$ as the initial value. Finally, the optimal quaternion is calculated as follows:

$$q_{opt} = \frac{1}{\sqrt{\gamma^2 + |X|^2}} \begin{bmatrix} X \\ \gamma \end{bmatrix} \tag{6.3.8}$$

With

$$X = (\alpha I + BS + S^2)Z \tag{6.3.9}$$

$$\gamma = (\lambda + \sigma)\alpha - \det(S) \tag{6.3.10}$$

$$\alpha = \lambda^2 - \sigma^2 + tr(adjS) \tag{6.3.11}$$

We proceed with the calculation of the error on the quaternion according to the following formula:

$$\delta q = q_{true} \otimes q_{opt}^{-1} \tag{6.3.12}$$

Also, in this case the hypothesis of small rotations is assumed and therefore the errors on the quaternions are represented in terms of errors on the Euler angles with the same formula used for the TRIAD method. The following formulas are used to calculate the covariance matrix:

$$P_{qq} = \frac{1}{4}\sigma_{tot}^2 \left[I_{3\times 3} - \sum_{i=1}^{n} a_i \hat{w}_i \hat{w}_i^T \right]^{-1} \tag{6.3.13}$$

$$(\sigma_{tot}^2)^{-1} = \sum_{i=1}^{n} (\sigma_i^2)^{-1} \tag{6.3.14}$$

$$a_i = \left(\frac{\sigma_{tot}^2}{\sigma_i} \right)^2 \tag{6.3.15}$$

$$P_{\theta\theta} = 4P_{qq} \tag{6.3.16}$$

The graphs obtained from the algorithm are shown below.

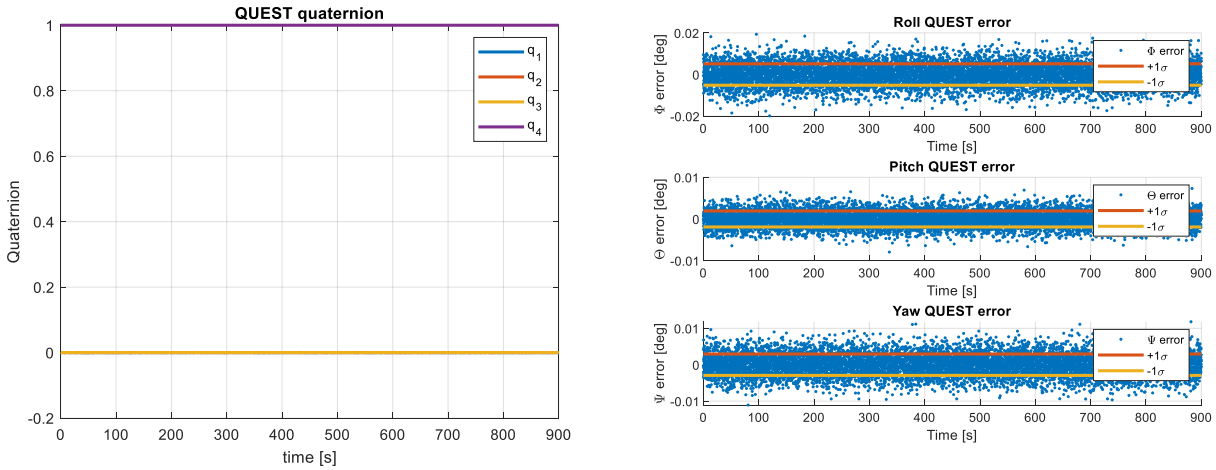


Figure 6.2: Estimated quaternions (left), Euler angles error (right)

6.4. Multiplicative Extended Kalman Filter

To initialize the Kalman Filter we use the results obtained with the Quest Algorithm in the first step. The state vector for this algorithm consists of 7 elements, 4 for the quaternion and 3 for the gyroscope bias components. At each step we integrate, through the kinematics equation, the quaternion estimated in the previous step, and we integrate the state transition matrix to also propagate the covariance matrix. With the propagated quaternion and with the quaternion calculated through the Quest method we obtain the innovation z with which we can then update the state vector (thus updating both the quaternion and the bias) thanks to the Kalman *K* matrix. Finally with the matrix of

propagated covariance we compute the new estimate of the covariance matrix. The formulas for this procedure are briefly given below.

Kinematic attitude equation
$$\frac{dq}{dt} = \frac{1}{2}\Omega(\omega)q(t) \tag{6.4.1}$$

State transition matrix equation
$$\frac{d}{dt}\Phi_{k|k-1} = F(t)\Phi_{k|k-1} \tag{6.4.2}$$

Initial condition
$$\Phi_{k-1|k-1} = I_{6\times 6} \tag{6.4.3}$$

State transition matrix
$$\Phi_{k|k-1} = \begin{bmatrix} \Theta_{k|k-1} & \Psi_{k|k-1} \\ 0_{3\times 3} & I_{3\times 3} \end{bmatrix} \tag{6.4.4}$$

$$F(t) = \begin{bmatrix} -[\omega_{k-1|k-1}\times] & -\frac{1}{2}I_{3\times 3} \\ 0_{3\times 3} & 0_{3\times 3} \end{bmatrix} \tag{6.4.5}$$

Propagated covariance matrix
$$\bar{P}_k = \Phi_{k|k-1}P_{k-1}\Phi_{k|k-1}^T + Q_{k-1} \tag{6.4.6}$$

Process noise covariance matrix
$$Q_{k-1} = \begin{bmatrix} \sigma_v^2 I_{3\times 3} & 0_{3\times 3} \\ 0_{3\times 3} & (\sigma_u dt)^2 I_{3\times 3} \end{bmatrix} \tag{6.4.7}$$

Predicted measurement
$$\hat{q}_{k|k-1} \tag{6.4.8}$$

Quest measurement
$$p_k \tag{6.4.9}$$

Innovation
$$z_k = p_k \otimes \hat{q}_{k|k-1}^{-1} \tag{6.4.10}$$

Sensitivity matrix
$$H_k = [I_{3\times 3} \quad 0_{3\times 3}] \tag{6.4.11}$$

Quest covariance matrix
$$R_k \tag{6.4.12}$$

Kalman matrix
$$K_k = \bar{P}_k H_k^T (H_k \bar{P}_k H_k^T + R_k)^{-1} \tag{6.4.13}$$

State update
$$\delta\hat{x}_k = \begin{bmatrix} \delta\hat{q}_k \\ \delta\hat{b} \end{bmatrix} = K_k z_k \tag{6.4.14}$$

Covariance matrix
$$P_k = (I - K_k H_k)\bar{P}_k (I - K_k H_k)^T + K_k R_k K^T \tag{6.4.15}$$

Update quaternion
$$\hat{q}_k = \begin{bmatrix} \delta\hat{q}_k \\ 1 \end{bmatrix} \otimes \hat{q}_{k|k-1} \tag{6.4.16}$$

Update bias $$\hat{b}_k = \hat{b}_{k|k-1} + \delta\hat{b}_k \quad (6.4.17)$$

The results of this method are reported below.

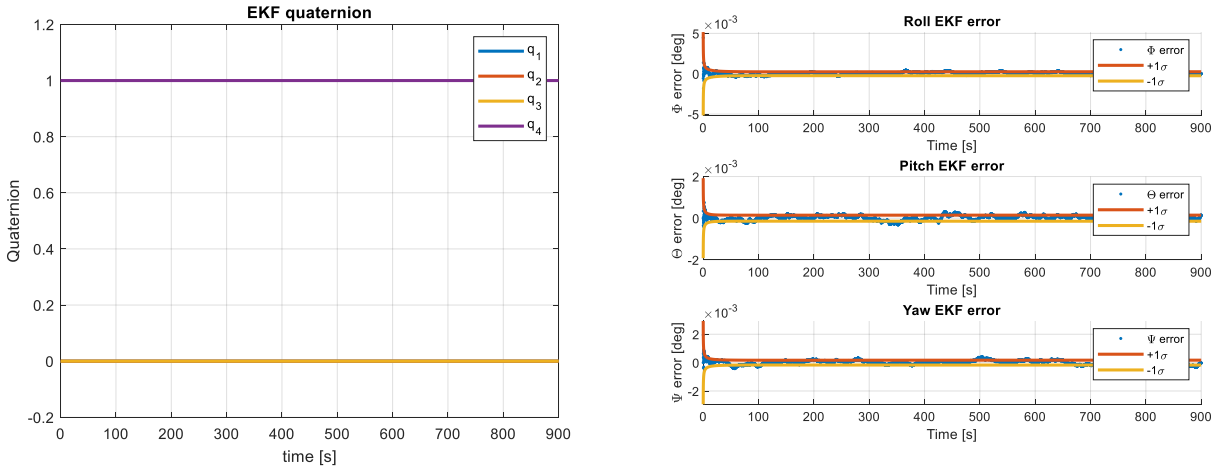


Figure 6.3: Estimated quaternion (left), Euler angles error (right)

6.5. Analysis of the results

From the results obtained with the three algorithms illustrated, one can first of all observe a full agreement regarding the estimation of quaternions. However, the errors made in determining the latter are decidedly different and this is demonstrated by the graphs of the errors on the Euler angles with the relative standard deviations. The TRIAD method is the one that has the greatest uncertainties and errors, and this is due to the fact that, compared to other algorithms, the TRIAD method can only process two measurements at a time collected at the same time. The QUEST method also considers measurements collected at the same time, however, unlike the TRIAD method, it processes more measurements at a time, and this generally involves minor uncertainties due to the fact that, in its formalism, weight factors are introduced that take into account the individual accuracy of measurements. The best results and therefore the best performances are obtained with the MEKF method which has the ability to process different observations at different instants of time. In the first two methods we make a static determination of the state and do not consider the "history" of the previous state, in other words, we process the observations and determine the state for each moment of observation. The MEKF, on the other hand, has the ability to propagate the information that it has collected in the previous step, thus filtering between the previous information propagated and the new processed observation.

An important observation is that in the interval between 0 and 600s the uncertainties of the QUEST method are greater due to the fact that, in this time interval there are

fewer measurements, the only ones available are those coming from the ST and not from the SS since the satellite is in eclipse.

6.6. Angular velocity and bias estimation

As regards the angular velocity, a simplified model of the Farrenkopf noise model for gyroscopes is adopted. In calculating the error on the angular velocity, the following relationships are used:

Angular velocity $$\vec{\omega} = \vec{g} - \vec{b} - \vec{\eta}_1 \quad (6.6.1)$$

ARW (angle random walk) $$\vec{\eta}_1 \quad (6.6.2)$$

Measured angular velocity $$\vec{g} \quad (6.6.3)$$

Estimated angular velocity $$\hat{\vec{\omega}} = \vec{g} - \hat{\vec{b}} \quad (6.6.4)$$

Angular velocity error $$\delta\vec{\omega} = \vec{\omega}_{TRUE} - \hat{\vec{\omega}} \quad (6.6.5)$$

To associate an uncertainty with the measurement of the error on the angular velocity, it can be considered that this is given by the following formula:

$$\delta\vec{\omega} = -\left(\delta\vec{b} + \vec{\eta}_1\right) \quad (6.6.6)$$

And since the two variables are independent of each other, the uncertainty associated with a function of two independent variables is given by the following relationship:

Variance $$\sigma^2_{\delta\vec{\omega}} = \sigma^2_{\delta\vec{b}} + \sigma^2_{\delta\vec{\eta}_1} \quad (6.6.7)$$

Standard deviation $$\sigma_{\delta\vec{\omega}} = \sqrt{\sigma^2_{\delta\vec{b}} + \sigma^2_{\delta\vec{\eta}_1}} \quad (6.6.8)$$

The standard deviation of η_1 is equal to $\sigma_v = 10^{-7}\ rad/s$, the standard deviation of the bias is calculated by taking the last three elements of the diagonal of the covariance matrix determined with the MEKF method and making the square root.

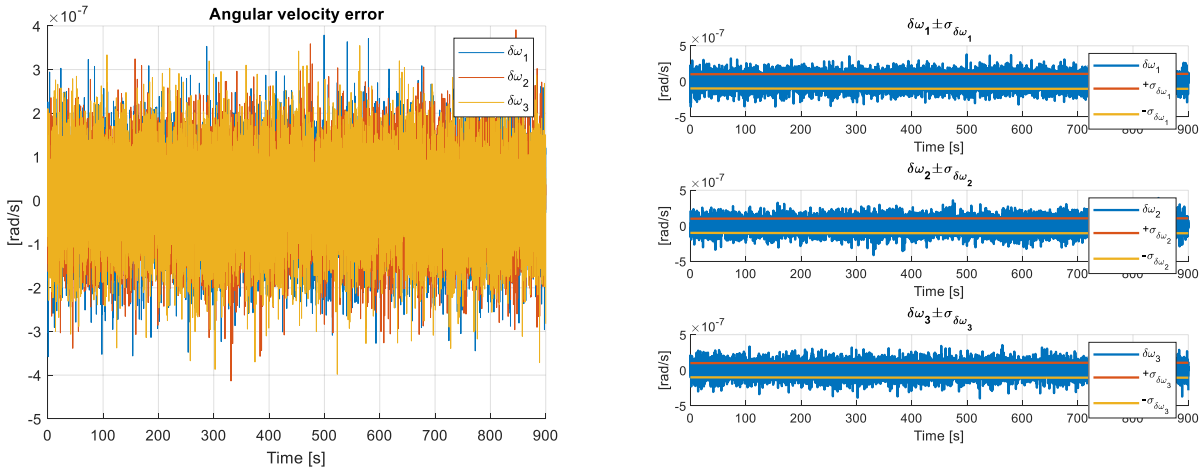


Figure 6.4: Angular velocity error (left), angular velocity error with uncertainties (right)

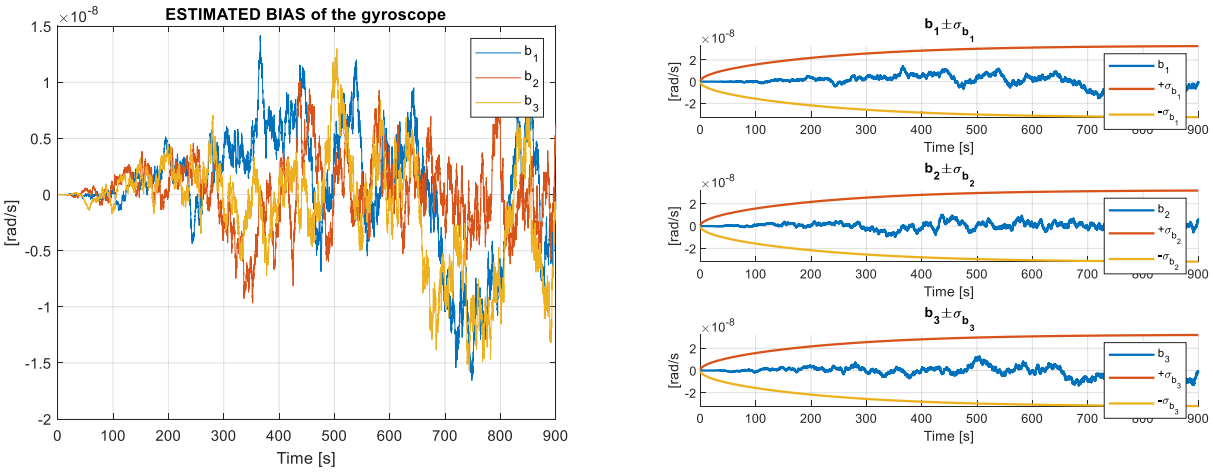


Figure 6.5: Estimated BIAS (left), estimated BIAS with uncertainties (right).

7. Attitude Control

The expression *attitude control* has the general meaning of controlling the attitude of the satellite. In practise, there exist a multitude of variations to this simple and apparently straightforward expression. For the HyonoSat mission, the functionality of the AODCS is broken down into a set of modes, used in different phases of the spacecraft's mission. The transitions between these modes will, to some extent, be autonomous. A subset of the AODCS mode transitions will be controllable manually from a ground station during a pass or using a flight planner inside the spacecraft. The modes are listed in Table 7.1.

Mode	Description
Detumble	Attenuate the change in spacecraft attitude
Nadir Pointing	Spacecraft should point at the Earth's centre
Safe Mode	A safe mode for when an emergency arise
Deorbiting	The act or process of going out of orbit

Table 7.1. AODCS modes

The attitude and orbit control of a satellite is performed with the aid of hardware that can be classified as either *attitude determination* or *control* hardware. The attitude determination hardware enables direct measurement (or estimation) of the *s/c* attitude with respect to some reference coordinate system in space. The attitude determination hardware used for the HyonoSat mission are sun and star sensors. Control hardware provides translational and angular accelerations so that the orbit and the location of the satellite within that orbit, as well as its angular attitude, can be varied at will. Control hardware includes reaction wheels, reaction thrusters, and magnetic torquers (also called *torqrods)*.

7.1. The basic attitude control equation

Generally speaking, the satellite attitude dynamics equations are three second-order nonlinear equations. Automatic control theory does not provide exact analytical solutions and design procedures for such dynamic plants, so linearization of these equations is necessary if the satellite control engineer wishes to use standard automatic control techniques. The attitude dynamics equations need not be based on the Euler angles only as in Eq. 2.3.9. In Eq. 2.3.9 it is clearly shown that (in automatic control terminology) the "plant," with respect to a single satellite-body axis, consists of two integrators. To control such a system, "control torques" are necessary. These torques can be produced actively with control hardware instruments, or by natural effects such as gravity gradient moments. The control torques to be activated are always a function

of the attitude errors. Since we are dealing with second-order systems, some damping control must also be provided for improved stability. This means that the control torques will have to include a term that is dependent on the attitude rates to be measured or estimated. If, in addition, the steady-state error is to be nulled, then an integral of the attitude error can be added to the control torque equation. Control torque equations can be written in the following form:

$$\begin{aligned} T_{cx} &= k_{px}\phi_{er} + k_{dx}\dot{\phi}_{er} + k_{ix}\int \phi_{er}dt \\ T_{cy} &= k_{py}\theta_{er} + k_{dy}\dot{\theta}_{er} + k_{iy}\int \theta_{er}dt \\ T_{cz} &= k_{pz}\psi_{er} + k_{dz}\dot{\psi}_{er} + k_{iz}\int \psi_{er}dt \end{aligned} \quad (7.1.1)$$

These are the well-known PID controller equations (for control gains that are Kp proportional and Kd derivative and Ki integral).

7.2. Detumbling

When a satellite separates from its launcher, it tumbles in space in an uncontrolled manner. Before the satellite can begin its nominal operation, the angular velocity has to be reduced. The purpose of the detumbling controller is to dampen the angular velocity vector of the satellite. A satellite in LEO with an altitude of 500 km is considered detumbled when its angular rate has decreased to 0.13°/s or less.

The stabilization of a spacecraft after deployment is never a trivial issue, and several satellites have been lost due to anomalies or dysfunctions occurred during this phase. Typically, the angular rates at the deployment are much higher than those desired for attitude manoeuvring and the satellite is said to be tumbling. Therefore, a specific detumbling control must be designed to stabilize the spacecraft within the minimum time compatible with the mission requirements. After the spin motion of the spacecraft has been damped to the desired level, the control policy can be switched to fine pointing or attitude manoeuvring and mission operations can start.

Focusing on the HyonoSat CubeSat, an algorithm producing the detumbling using only three orthogonal magnetorquers and the measurements by a three-axis magnetometer is developed. The algorithm is specifically designed to be implemented on a low computationally efficient CubeSat onboard computer. The B-dot controller is chosen in this project because it is the easiest and most common detumbling controller for small satellites.

7.2.1. The B-dot Controller

The torque generated by the magnetic torquers is given by

$$\boldsymbol{L} = \boldsymbol{m} \times \boldsymbol{B} \tag{7.2.1}$$

where **m** is the commanded magnetic dipole moment generated by the torquers and **B** is the local geomagnetic field expressed in body-frame coordinates. The magnetic moment is given in units of Am^2 and the magnetic flux density of the geomagnetic field is given in units of $1Wb/m^2 = 1$ Tesla (T) $=10^4$ Gauss (G).

A magnetic control law is developed that can be used to detumble a spacecraft (i.e. null its angular velocity vector). The control is effected by commanding a magnetic dipole moment:

$$\boldsymbol{m} = \frac{k}{\|\boldsymbol{B}\|}\boldsymbol{\omega} \times \boldsymbol{b} \tag{7.2.2}$$

Where:

$$\boldsymbol{b} = \frac{\boldsymbol{B}}{\|\boldsymbol{B}\|} \tag{7.2.3}$$

ω is the angular velocity and k is a positive scalar gain. Since no angular velocity information is available, a good approximation to $\boldsymbol{m}$ is given by

$$\boldsymbol{m} = -\frac{k}{\|\boldsymbol{B}\|}\dot{\boldsymbol{B}} \tag{7.2.4}$$

where **B** is the field sensed by onboard magnetometers. This is an alternative version of the well-known B-dot control. But the absolute angular velocity can be reduced down to a value of the same order of magnitude as the orbit rate (around 10^{-3} rad/s) as in the case when the standard B-dot command law is used. Note that in practical application, $\dot{\boldsymbol{B}}$ would be computed by a finite difference approach, which introduces more noise in the control signal.

$$\dot{\boldsymbol{B}} = \frac{\boldsymbol{B}(t+dt) - \boldsymbol{B}(t)}{dt} \tag{7.2.5}$$

A gain expression based on analysing the closed-loop dynamics of the component of ω perpendicular to the Earth's magnetic field:

$$k = \frac{4\pi}{T_{orb}}(1 + \sin\xi_m)J_{min} \tag{7.2.6}$$

Where T_{orb} is the orbital period in seconds, ξ_m is the inclination of the spacecraft orbit relative to the geomagnetic equatorial plane and J_{min} is the minimum principal moment of inertia. Note that k is always positive.

- **Simulations and Results**

The B-dot control law was simulated for the HyonoSat CubeSat in two different scenarios: the first is a situation in which the initial angular rate is set to $\omega_0 = [5, 5, 5]\frac{deg}{s}$, and the second refers to a more truthful situation in which we have an initial angular velocity of 30 degrees per second on each axis, $\omega_0 = [30, 30, 30]\frac{deg}{s}$. The initial Euler angle is set for both simulations to $[\phi, \theta, \psi] = [30\ 40\ 50]\ deg$. The following figures show that the control strategy is satisfactory and even in the case in which the initial angular velocities are very high, the B-dot strategy performs a correct detumbling and the magnetic moments of the torqrods are always within the limits dictated by the datasheets.

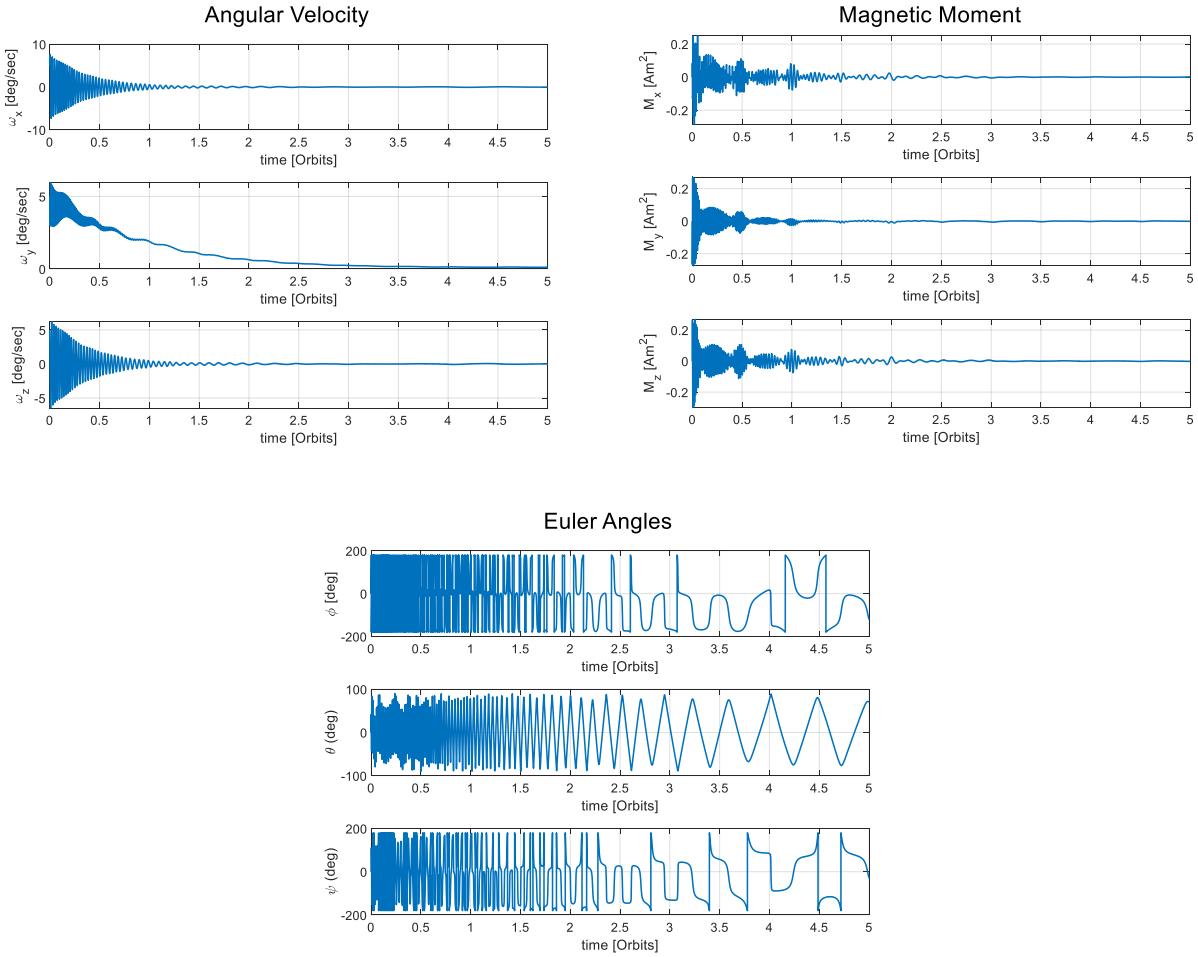


Figure 7.1: Angular velocity (upper left), magnetic moment (upper right), Euler angles (bottom) for the $\omega_0 = [5, 5, 5]\frac{deg}{s}$

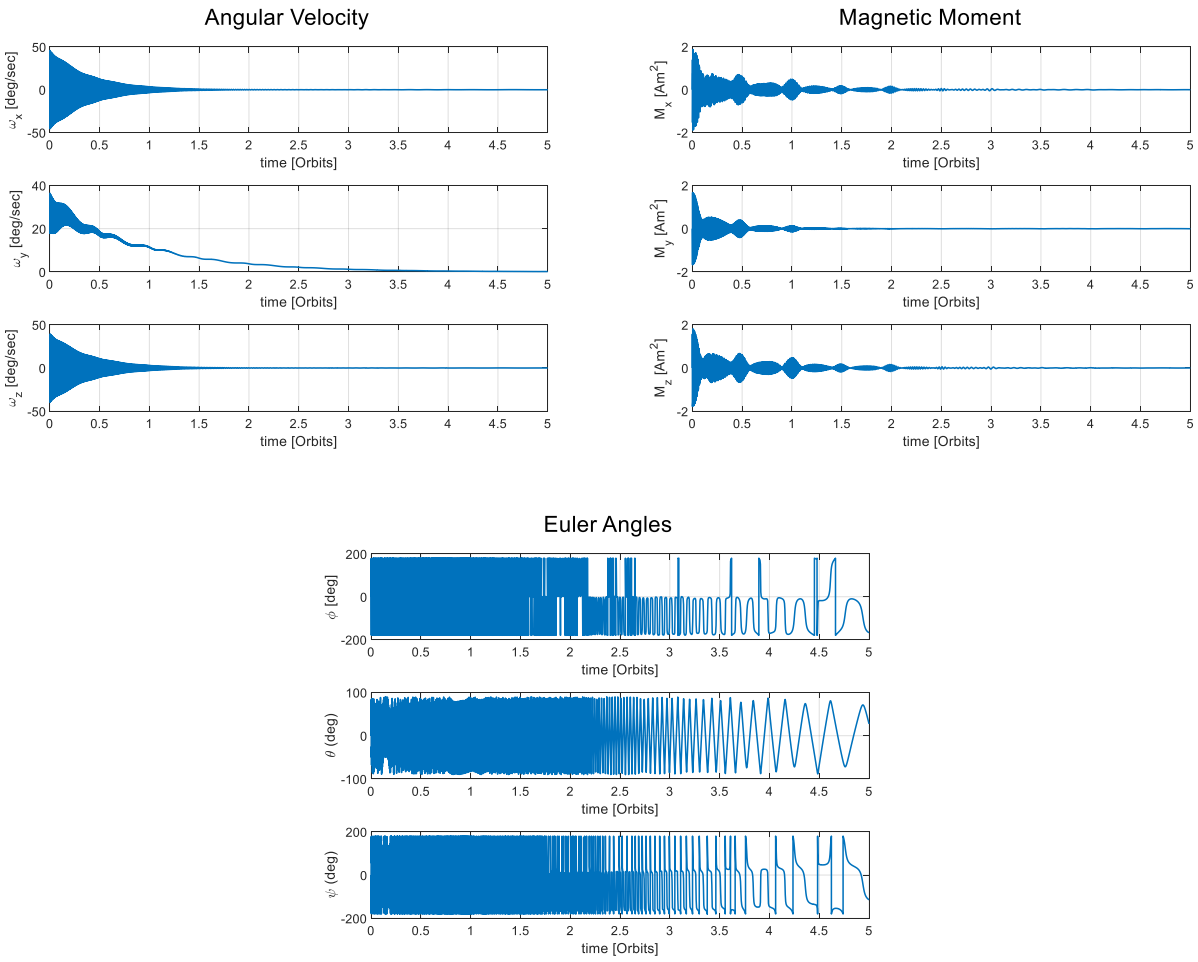


Figure 7.2: Angular velocity (upper left), magnetic moment (upper right), Euler angles (bottom) for the $\omega_0 = [30, 30, 30] \frac{deg}{s}$

7.3. Linear Nadir Pointing

The Nadir pointing mode is the mode used when the spacecraft is performing active pointing on the Earth surface. In Nadir pointing mode the spacecraft is pointing perpendicular to the Earth surface, in effect pointing at the Earth centre. In the Nadir mode, the payload and antennas are orientated in the z-axis. Three different attitude control strategies have been developed for the nadir pointing mode. The three strategies are reported below, underlining the differences and the accuracy obtained from each.

Two different simulations based on the two linear control strategies PD and PID have been conducted, the initial conditions for the Euler angles and the angular velocities are the following: $[\phi, \theta, \psi] = [0\ 0\ 0]\ deg$, $\omega_0 = [0, 0, 0] \frac{deg}{s}$. Moreover, it has been assumed that the disturbance torques are constant and are applied on all 3 axes and are equal to $\boldsymbol{T} = 1.36 * 10^{-6}\ N$. The main difference between the two methods is that with

the PID we can cancel the errors on the Euler angles while with the PD we have a constant error which is of the order of half degree.

7.3.1. Three axis magnetic ACS

An effective way to obtain the three-axis stabilization against external disturbances is to use magnetic torqrods, which - by interacting with the earth's magnetic field - can produce the needed moments to counteract the disturbances. If $\boldsymbol{B}$ is the earth's magnetic field vector and if $\boldsymbol{M}$ is the magnetic dipole value produced by energizing the current-carrying coils inside the satellite, then the mechanical moment exerted on the body will be:

$$\boldsymbol{T} = \boldsymbol{M} \times \boldsymbol{B} = \begin{bmatrix} i & j & k \\ M_x & M_y & M_z \\ B_x & B_y & B_z \end{bmatrix} \tag{7.3.1}$$

The vector product of Eq. 7.3.1 can be put in the form:

$$\begin{bmatrix} T_{cx} \\ T_{cy} \\ T_{cz} \end{bmatrix} = \begin{bmatrix} 0 & B_z & -B_y \\ -B_z & 0 & B_x \\ B_y & -B_x & 0 \end{bmatrix} \begin{bmatrix} M_x \\ M_y \\ M_z \end{bmatrix} \tag{7.3.2}$$

The value of $\boldsymbol{T_c}$ is computed according to Eq. 7.1.1, assuming that the attitude angles can be measured. The problem we are left with is finding the magnetic control dipole to be produced in the satellite with the three magnetic torqrods M_x, M_y, M_z. We need to find the inverse of Eq. 7.3.2, but the inverse does not exist because the matrix is singular. This means that it is impossible to achieve control of the three body axes by means of the created magnetic dipole vector in the satellite. The physical reason is that Eq. 7.3.1 is a vector product, so no moment can be achieved about the direction of $\boldsymbol{B}$. However, there is a possibility of damping all three body axes by active control with three magnetic torqrods aligned along the principal body axes. In this scenario, the principle of perpendicularity is used. As already mentioned, Eq. 7.3.2 has no inverse, and M_x, M_y, M_z cannot be found exactly. Let us return to the basic magnetic moment equation (Eq. 7.3.1), and multiply (using vector product) both sides by $\boldsymbol{B}$. Using the scalar identity for the vector product, we have

$$\boldsymbol{B} \times \boldsymbol{T_c} = \boldsymbol{B} \times (\boldsymbol{M} \times \boldsymbol{B}) = (\boldsymbol{B} \cdot \boldsymbol{B})\boldsymbol{M} - (\boldsymbol{B} \cdot \boldsymbol{M})\boldsymbol{B} \tag{7.3.3}$$

Suppose that the magnetic dipole vector $\boldsymbol{M}$, applied inside the satellite, it always perpendicular to the earth's magnetic vector $\boldsymbol{B}$ (this is not actually true). If this

condition is satisfied, then the scalar dot product $\boldsymbol{B} \cdot \boldsymbol{M}$ equals zero and Eq. 7.3.3 reduces to

$$\boldsymbol{M} = \frac{(\boldsymbol{B} \times \boldsymbol{T_c})}{\|\boldsymbol{B}\|^2} \tag{7.3.4}$$

Since this control law is not exact, any potential design must be carefully checked by numerous simulations under different disturbance conditions.

- **Simulation and results**

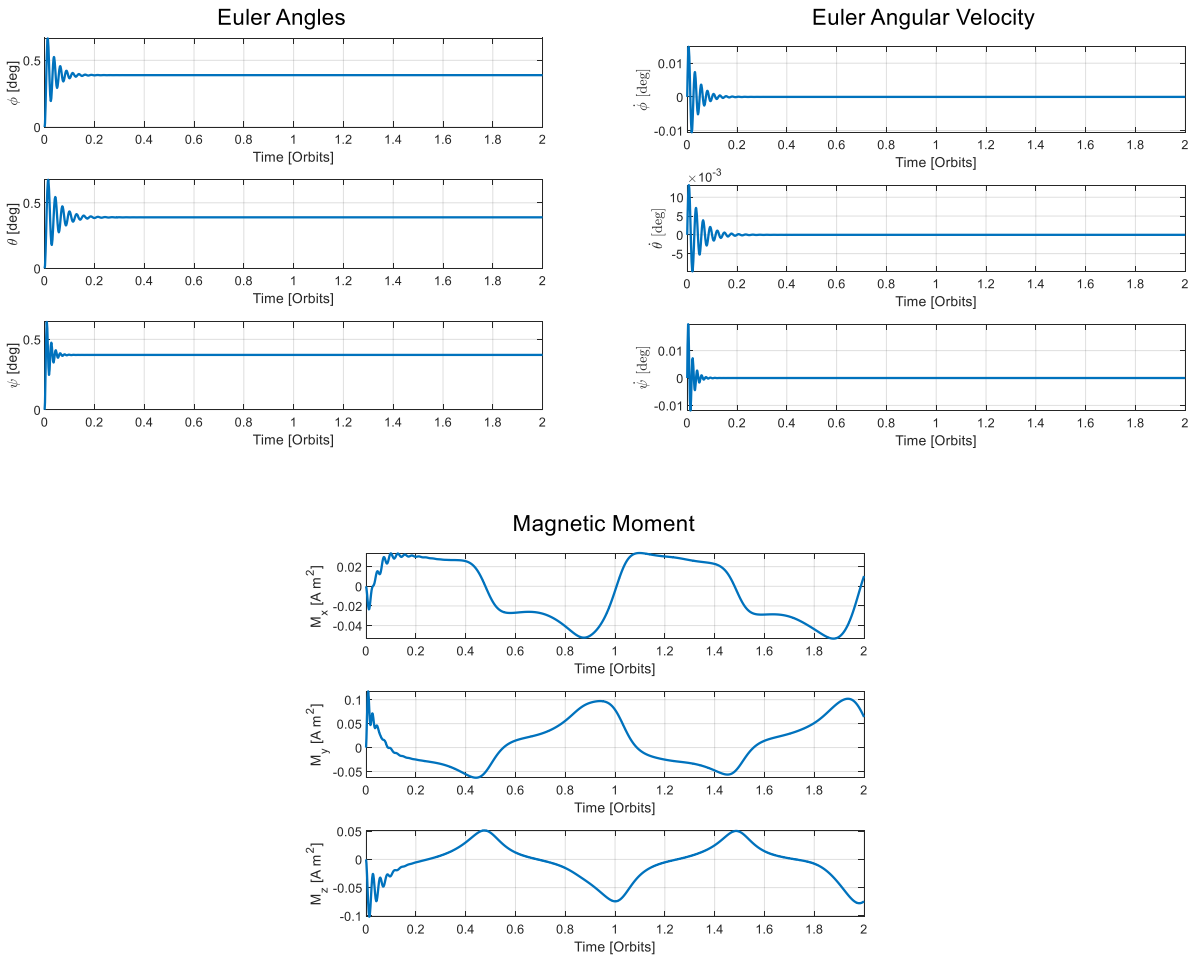


Figure 7.3: Euler angles (upper left), Euler angular velocity (upper right), magnetic moment (bottom) for the **PD controller**.

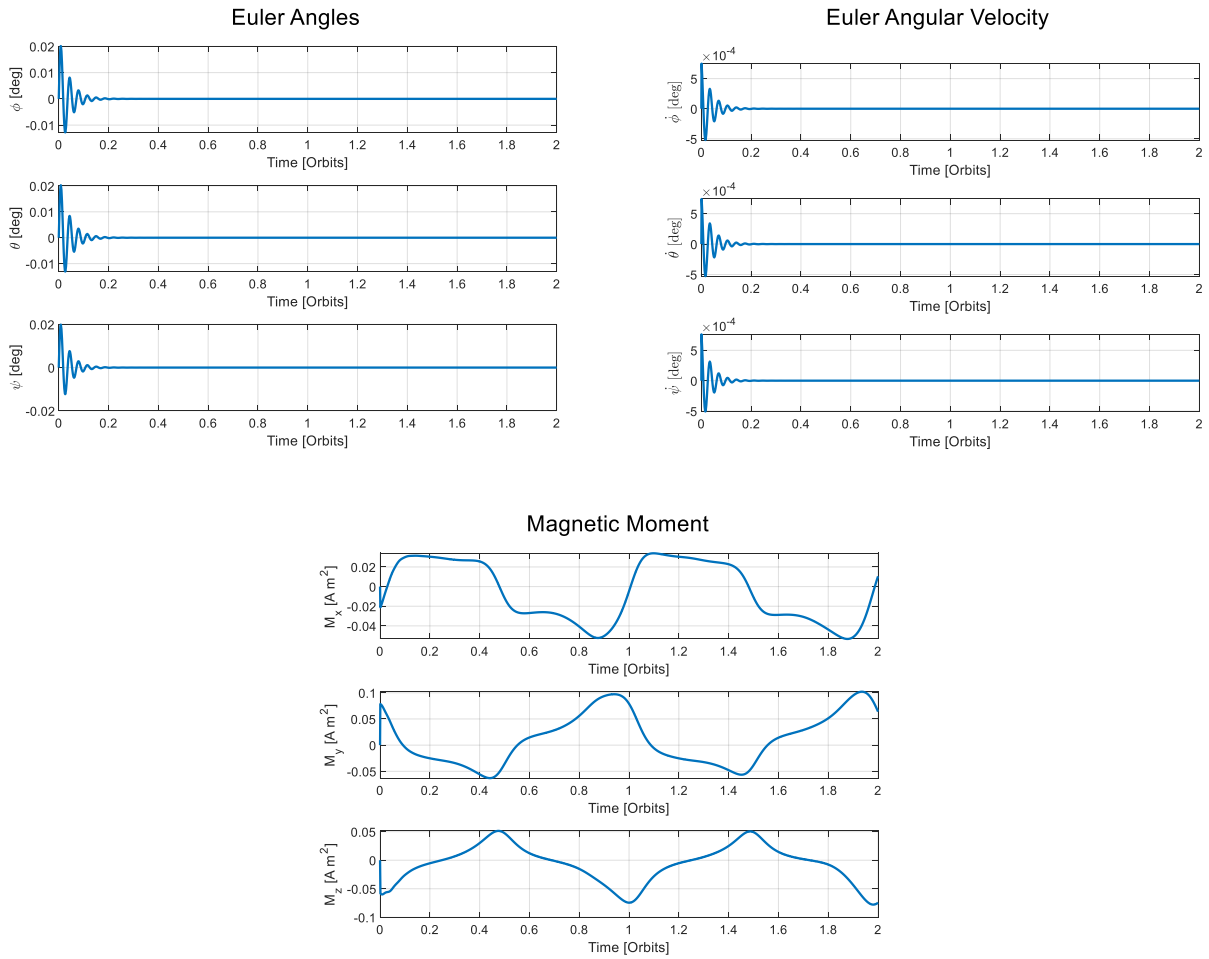


Figure 7.4: Euler angles (upper left), Euler angular velocity (upper right), magnetic moment (bottom) for the **PID controller**.

7.3.2. Three axis magnetic ACS + 1 Reaction Wheel

For a desired torque $\boldsymbol{T_B} = \boldsymbol{T_c}$ to be applied on the spacecraft, we must generate the magnetic moment vector $\boldsymbol{M} = [M_x\ M_y\ M_z.]^T$. However, we cannot invert the matrix in Eq. 7.3.2 because this matrix is singular. The solution to our problem lies in replacing one of the magnetic torqrods with, for instance, a reaction wheel. By exchanging the $\boldsymbol{Y_B}$ magnetic torqrod with a RW, Eq. 7.3.2 is accordingly changed to:

$$\begin{bmatrix} T_{Bx} \\ T_{By} \\ T_{Bz} \end{bmatrix} = \begin{bmatrix} 0 & B_z & -B_y \\ -B_z & 1 & B_x \\ B_y & -B_x & 0 \end{bmatrix} \begin{bmatrix} M_x \\ \dot{h}_{wy} \\ M_z \end{bmatrix} \tag{7.3.5}$$

The matrix in Eq. 7.3.5 does have an inverse, so

$$\begin{bmatrix} M_x \\ \dot{h}_{wy} \\ M_z \end{bmatrix} = \frac{1}{B_y^2} \begin{bmatrix} 0 & 0 & B_y \\ B_x B_y & B_y^2 & B_y B_z \\ -B_y & 0 & 0 \end{bmatrix} \begin{bmatrix} T_{cx} \\ T_{cy} \\ T_{cz} \end{bmatrix} \tag{7.3.6}$$

- **Simulation and results**

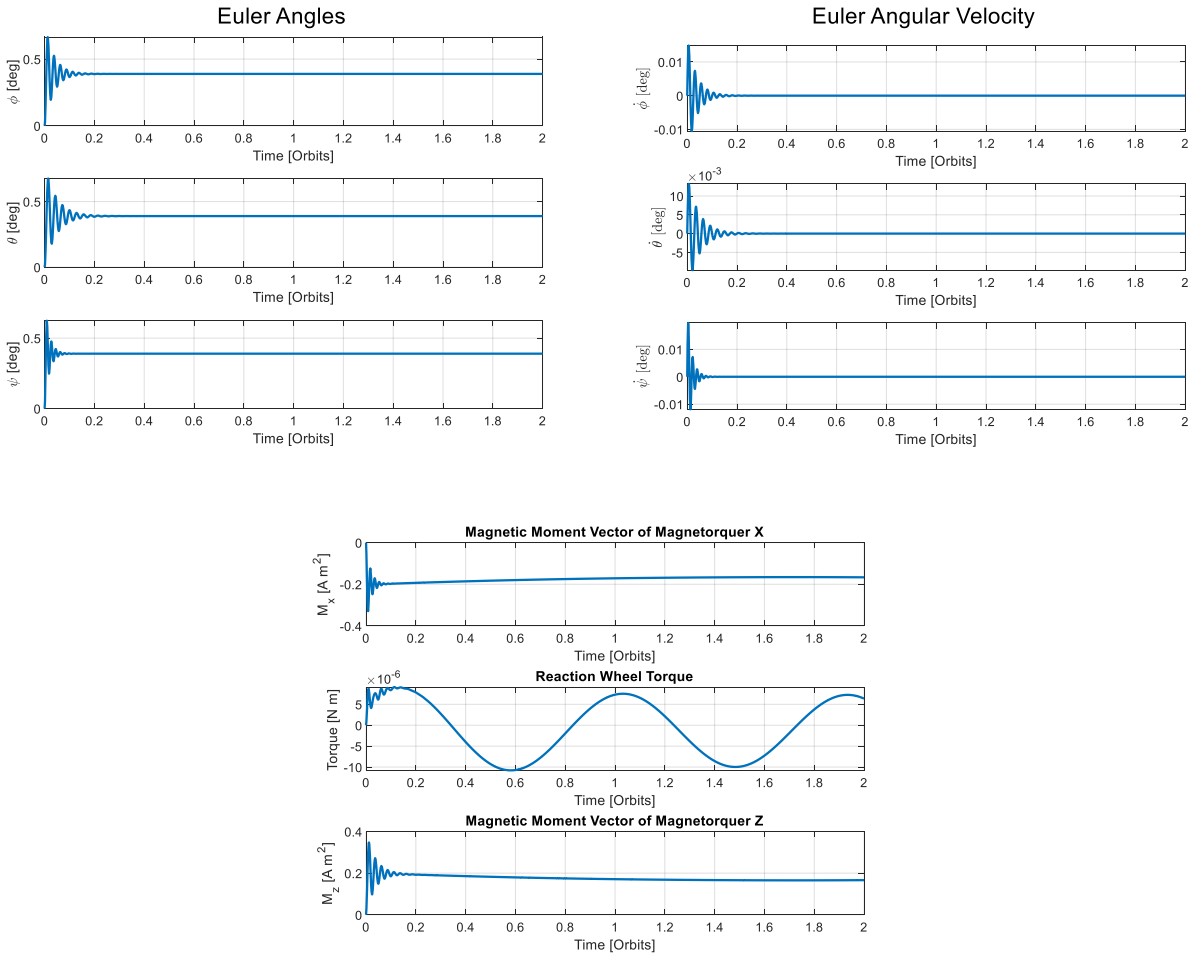
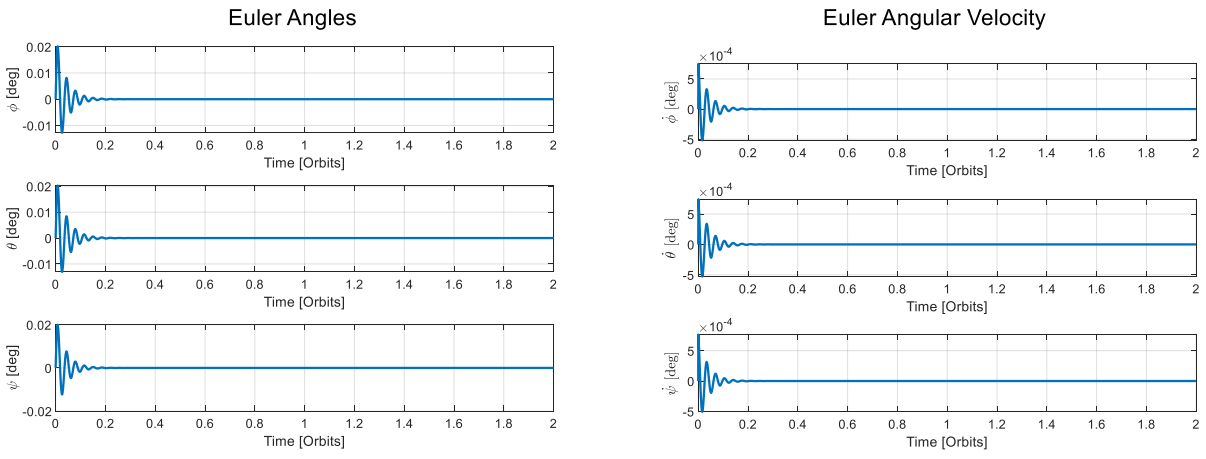


Figure 7.6: Euler angles (upper left), Euler angular velocity (upper right), magnetic moment and reaction wheel torque (bottom) for the **PD controller**.



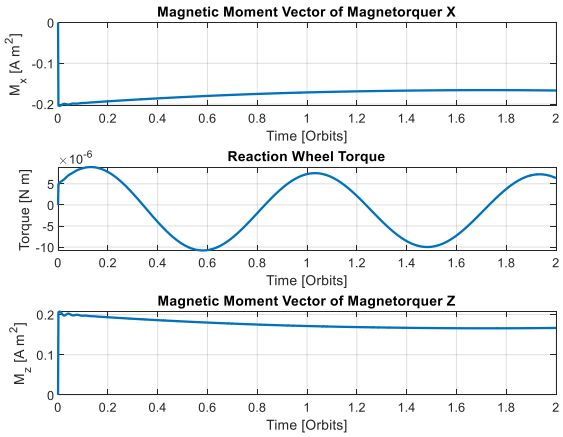


Figure 7.7: Euler angles (upper left), Euler angular velocity (upper right), magnetic moment and reaction wheel torque (bottom) for the **PID controller**.

7.3.3. Complete Reaction wheel-based ACS

Three reaction wheels, with each one's rotational axis parallel to one of the satellite's body axes, make up the simplest control system. As the three body axes' dynamics are separated, the design can be carried out independently for each axis. However, if one of the assemblies becomes damaged then the satellite's attitude can no longer be adequately controlled. For this reason, a fourth RWA is installed in order to increase the reliability of the entire system. The additional wheel is installed with its axis "off" the three principals six axes, enabling (reduced) torque control about anyone of those axes. Thus, the incapacity of anyone of the RWAs aligned with the satellite's principal axes can be compensated by the torque capabilities of the fourth wheel. In this section we have analysed a geometrical configuration of a control system based on four reaction wheels; see Figure 7.8.

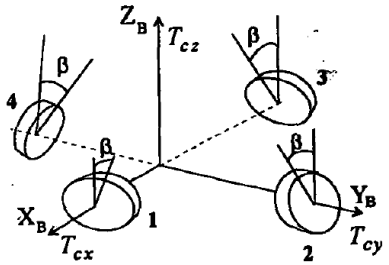


Fig. 7.8: Attitude control system with four reaction wheels

Four wheels are inclined to the $\boldsymbol{X_B - Y_B}$ plane by an angle β. Because of this inclination, each wheel can apply torques and momentum in the $\boldsymbol{Z_B}$ direction also. The

torques delivered by the wheels are called $T_i (i = 1, \dots, 4)$. The torques produced along the three body axes are $\hat{T}_{cx}$, $\hat{T}_{cx}$, $\hat{T}_{cx}$. Thus, we have the following relations:

$$\begin{bmatrix} \hat{T}_{cx} \\ \hat{T}_{cy} \\ \hat{T}_{cz} \end{bmatrix} = \begin{bmatrix} T_{cx}/\cos\beta \\ T_{cy}/\cos\beta \\ T_{cz}/\cos\beta \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{bmatrix} = [\boldsymbol{A_w}]\boldsymbol{T} \tag{7.3.7}$$

We need to calculate the components T_i which are the control torques to be applied by each one of the four wheels. Unfortunately, the matrix $[\boldsymbol{A_w}]$ in Eq. 7.3.7 is not square and cannot be inverted. To find the vector components of T_i, we must assume some optimizing criterion. For instance, we might wish to minimize the norm of the vector $\boldsymbol{T} = [T_1, T_2, T_3, T_4]^T$. Eq. 7.3.7 may be rewritten in the following matrix form:

$$\begin{bmatrix} \hat{T}_{cx} \\ \hat{T}_{cy} \\ \hat{T}_{cz} \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \end{bmatrix} \begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{bmatrix} \tag{7.3.8}$$

The square matrix in Eq. 7.3.8 has an inverse, which is easily found to be:

$$\begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 1 & \frac{1}{2} & -\frac{1}{2} \\ -1 & 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & -1 & \frac{1}{2} & -\frac{1}{2} \end{bmatrix} \begin{bmatrix} \hat{T}_{cx} \\ \hat{T}_{cy} \\ \hat{T}_{cz} \\ 0 \end{bmatrix} \tag{7.3.9}$$

Eq. 7.3.9 provides the needed transformation between the three body axes' command control torques and the four wheels' command control torques.

- **Simulation and results**

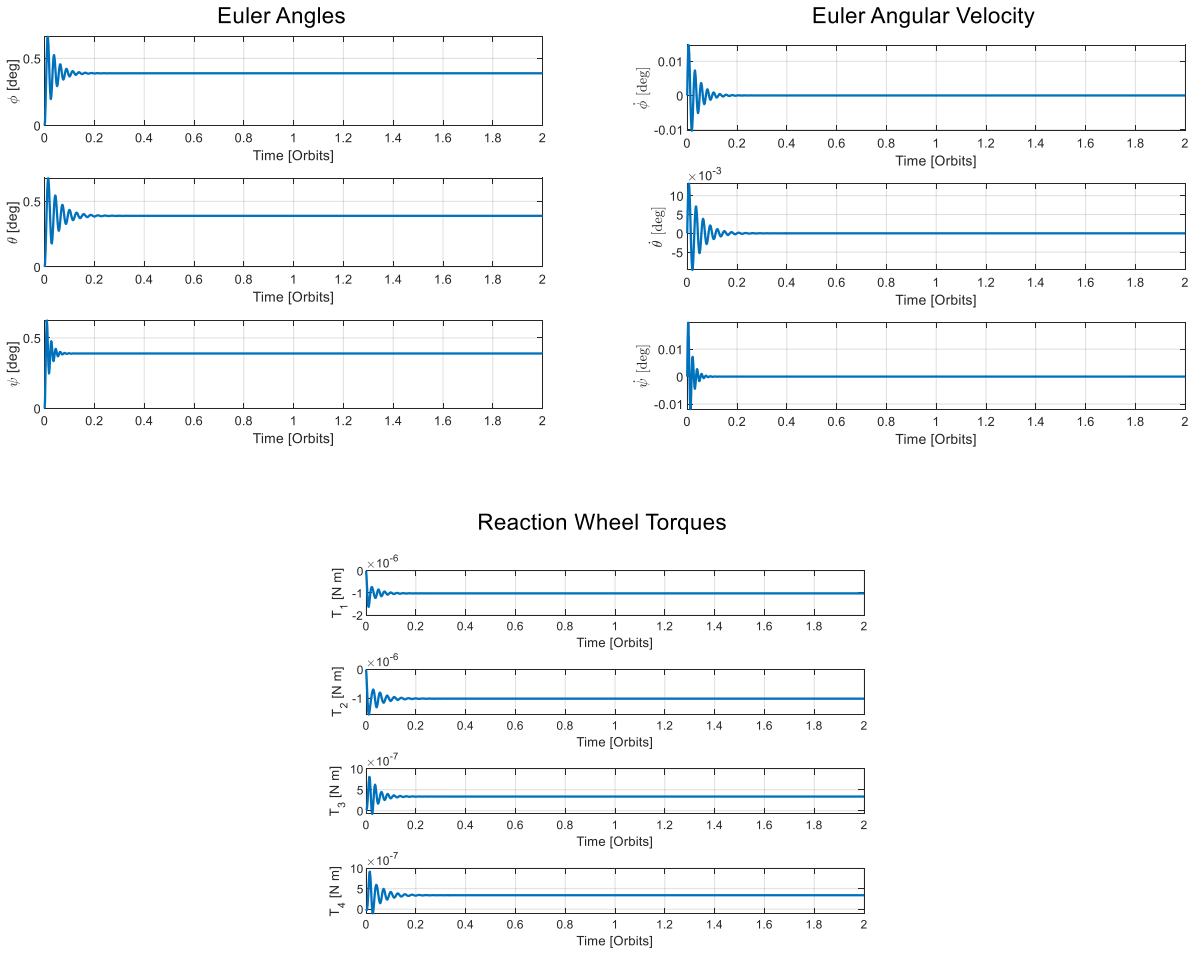
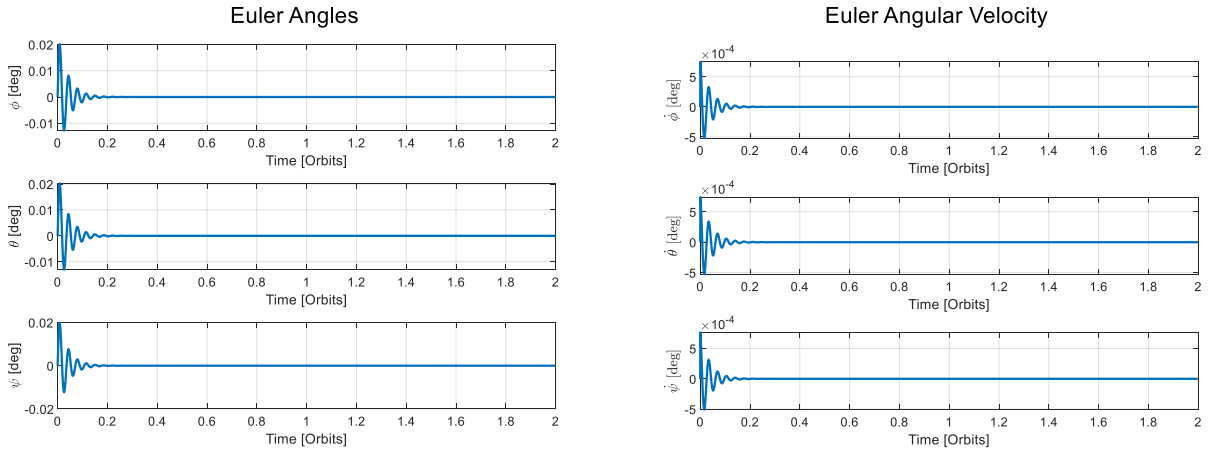


Figure 7.9: Euler angles (upper left), Euler angular velocity (upper right), reaction wheel torques (bottom) for the **PD controller**.



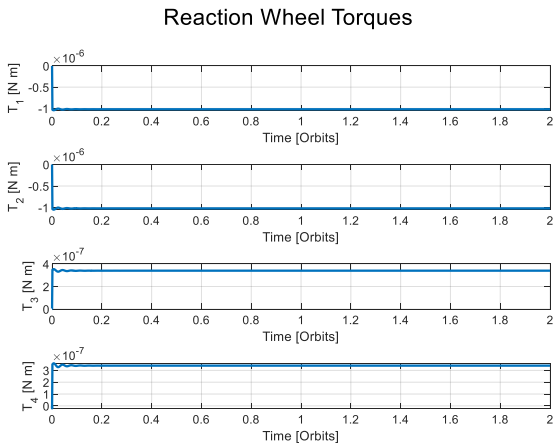


Figure 7.10: Euler angles (upper left), Euler angular velocity (upper right), reaction wheel torques (bottom) for the **PID controller**.

7.3.4. Magnetic momentum unloading of the RWs (desaturation)

One drawback of a momentum exchange device is that it cannot independently remove the angular momentum that accumulates, owing to external disturbances, in the satellite system. As we know, according to Euler's moment equations, any external torque disturbance acting on the body augments the angular momentum of the whole spacecraft system. With no active attitude control, the body accumulates an angular velocity as the angular momentum of the s/c increases, changing the attitude of the spacecraft. However, if reaction wheels are used to stabilize the s/c attitude, then the accumulated angular momentum will be transferred to the wheels. With constant external disturbances, the momentum of the wheels will increase without limit, thus saturating them and precluding their ability to provide the necessary control torques. The unwanted accumulated momentum in the wheel must be removed from the momentum exchange device, a process called dumping of the momentum.

The two primary control hardware items used to dump the wheels are magnetic torqrods and reaction thrusters. We will focus our attention on the first of these options. Magnetic torqrods generate magnetic dipole moments whose interactions with the Earth's magnetic field produce the torques necessary to remove the excess momentum. The basic control equation for momentum unloading is:

$$\boldsymbol{T} = -k(\boldsymbol{h} - \boldsymbol{h_N}) = -k\Delta\boldsymbol{h} \tag{7.3.10}$$

In this vectorial equation, k is the unloading control gain, $\boldsymbol{h}$ is the wheel's momentum vector, $\boldsymbol{h_N}$ is the desired and nominal wheel momentum vector, and $(\boldsymbol{h} - \boldsymbol{h_N}) = \Delta\boldsymbol{h}$ is

the excess momentum to be removed. The magnetic torque equation was previously stated to be $\boldsymbol{T} = \boldsymbol{M} \times \boldsymbol{B}$. Together with Eq. 7.3.10 this yields:

$$-k\boldsymbol{\Delta h} = \boldsymbol{M} \times \boldsymbol{B} \tag{7.3.11}$$

However, the control magnetic dipole vector $\boldsymbol{M}$ cannot be computed from Eq. 7.3.11. Using vector product by $\boldsymbol{B}$ on both sides, Eq. 7.3.11 becomes

$$\boldsymbol{B} \times (-k\boldsymbol{\Delta h}) = \boldsymbol{B} \times (\boldsymbol{M} \times \boldsymbol{B}) = B^2\boldsymbol{M} - \boldsymbol{B}(\boldsymbol{M} \cdot \boldsymbol{B}) \tag{7.3.12}$$

With some simplifying assumptions, we can find $\boldsymbol{M}$ from Eq. 7.3.12. Suppose that the applied $\boldsymbol{M}$ happens to be perpendicular to the Earth's magnetic field $\boldsymbol{B}$. In this case $\boldsymbol{M} \cdot \boldsymbol{B}$ (which is a scalar product) is zeroed, and from Eq. 7.3.12 we have:

$$\boldsymbol{M} = -\frac{k}{B^2}(\boldsymbol{B} \times \Delta\boldsymbol{h}) \tag{7.3.13}$$

This control magnetic moment produces a magnetic torque that is not exactly proportional to the excess momentum:

$$\boldsymbol{T} = -\frac{k}{B^2}\left(B^2\Delta\boldsymbol{h} - \boldsymbol{B}(\boldsymbol{B} \cdot \Delta\boldsymbol{h})\right) \tag{7.3.14}$$

Physically, Eq. 7.3.14 states that no torque can be obtained about the Earth's magnetic field $\boldsymbol{B}$ and, moreover, that if the excess momentum to be dumped is parallel to this vector, then the wheels cannot be unloaded. Fortunately, for most inclined orbits this condition does not continue indefinitely: both the direction and the amplitude of $\boldsymbol{B}$ with respect to body axes will vary during an orbit, and an average removal of the excess momentum takes place. In any case, the effectiveness of momentum unloading depends very much on the specific orbit in which the satellite is moving. For equatorial orbits, the efficiency is quite low.

Equation 7.3.13 can be put in more explicit form:

$$\begin{bmatrix} M_x \\ M_y \\ M_z \end{bmatrix} = -\frac{k}{B^2}\begin{bmatrix} B_y\Delta h_z - B_z\Delta h_y \\ B_z\Delta h_x - B_x\Delta h_z \\ B_x\Delta h_y - B_y\Delta h_x \end{bmatrix} \tag{7.3.15}$$

where B_x, B_y, B_z are the measured Earth magnetic field strength components in the body axis frame. Hence, a three-axis magnetometer is imperative. The excess momentum $\boldsymbol{\Delta h}$ is known from measuring the wheel's angular velocities and

transforming their components to body axes, so that the components of the magnetic dipole moment ***M*** can be computed also in body axes using Eq. 7.3.15. The saturation level of the magnetic torqrods must be such that they can provide torques larger than the external disturbing torques causing the momentum loading of the wheels.

- **Simulation and results**

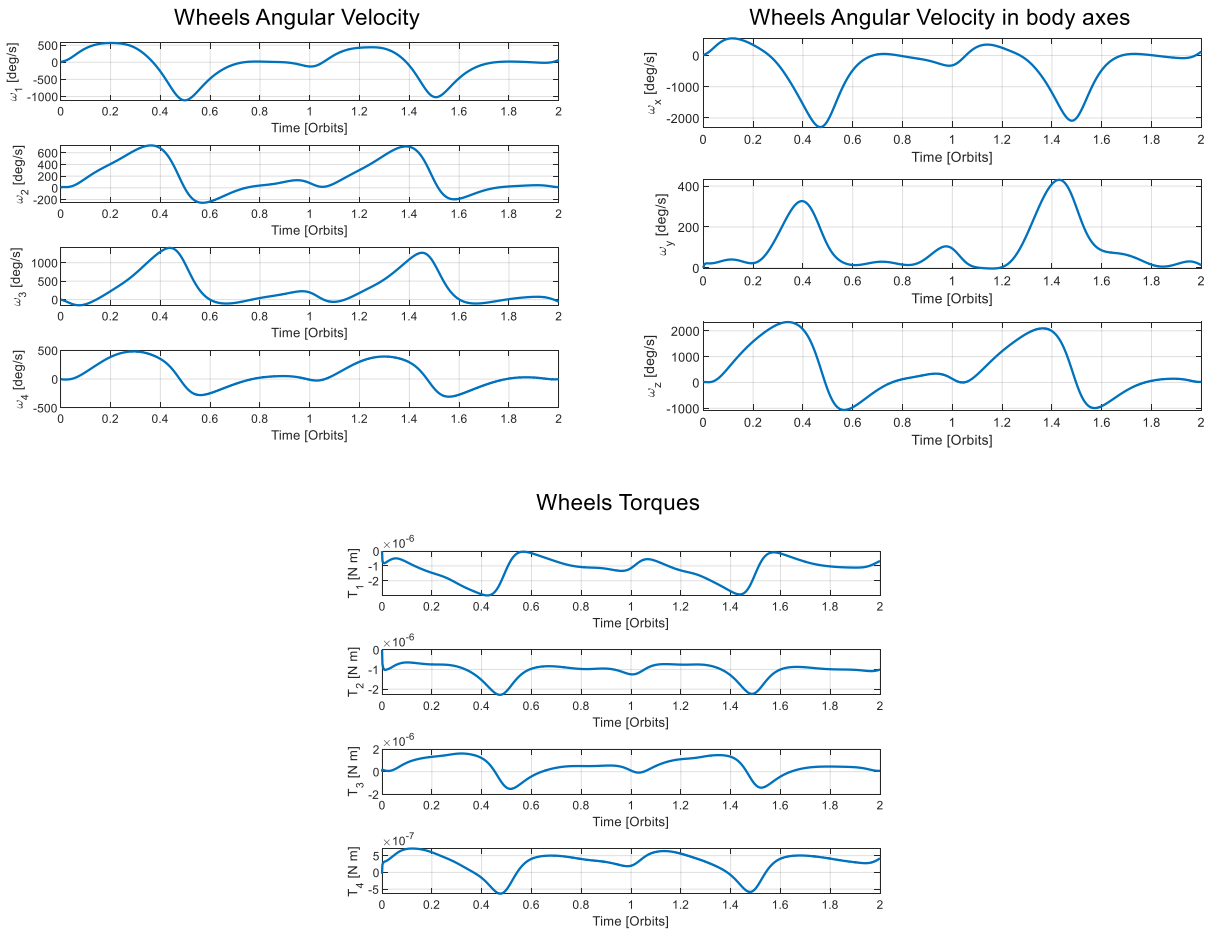


Figure 7.10: Wheels angular velocity on its own axis (upper left) and in body axes (upper right), reaction wheel torques (bottom).

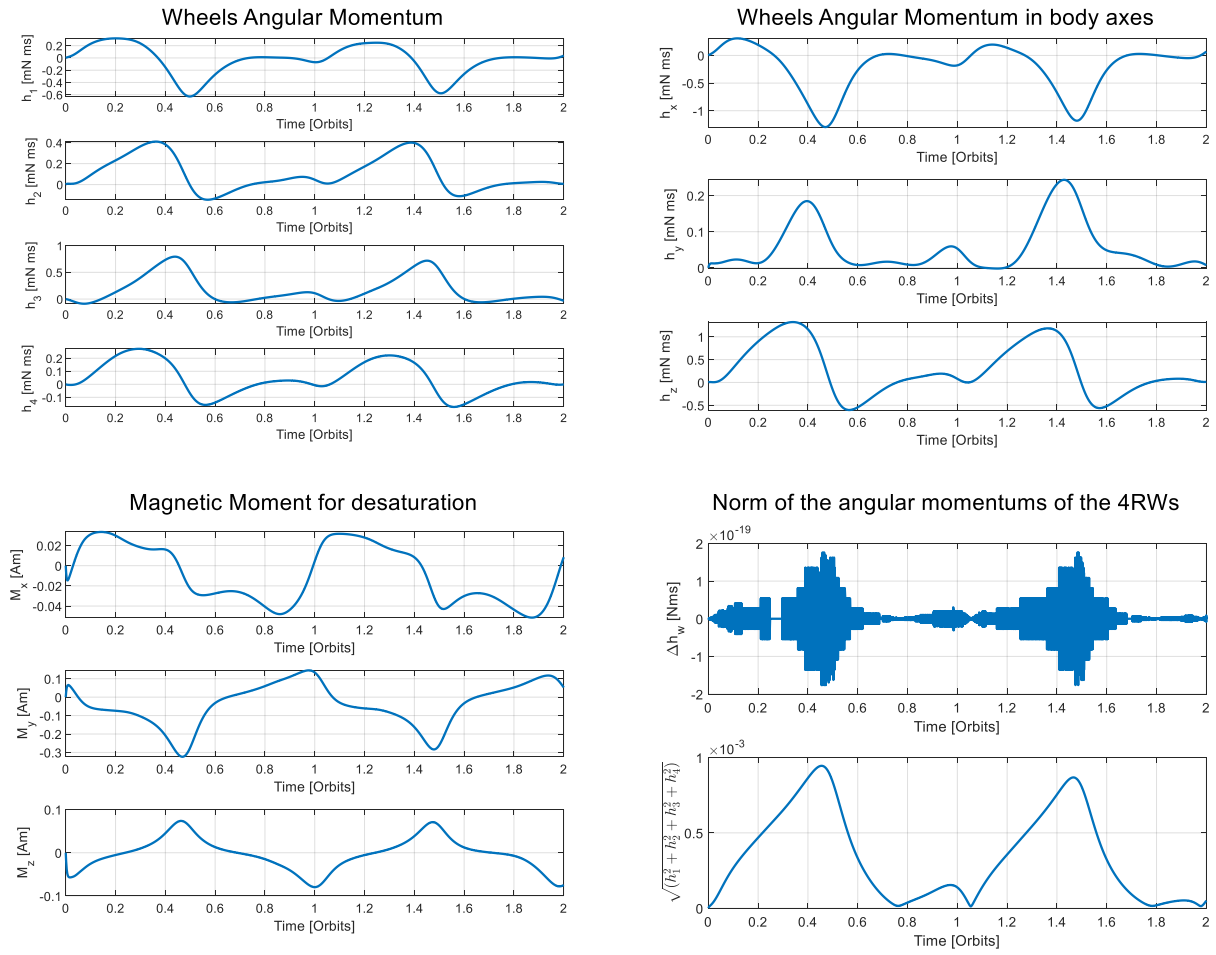


Figure 7.11: Wheels angular momentum on its own axis (upper left) and in body axes (upper right), magnetic moment (bottom left), norm of the angular momentum (bottom right).

7.4. Non-linear nadir pointing for RWs + desaturation

In order to control spacecraft attitude to maintain the nadir pointing, in addition to the just proposed linear control, a non-linear control has been developed.

It can be proven that the following feedback control law:

$$\underline{T_C} = \underline{\tilde{\omega}}\, J_C \underline{\omega} - \underline{M_C} + J_C \underline{\dot{\omega}_C} - J_C A^{-1} B \underline{\omega_D} - sgn\{q_{e0}(0)\} J_C A^{-1} \underline{q_e} \qquad (7.4.1)$$

brings the spacecraft to converge toward the commanded attitude in terms of angular rate and quaternions. In fact, Lyapunov method proves the convergence toward $\underline{\omega_D} = \underline{0}$, while the LaSalle's principle proves the convergence toward $q_{e0} = \pm 1$. It is important to remark that $q_{e0} = 1$ and $q_{e0} = -1$ correspond to the same attitude, what changes is the angular path to reach it: the term $sgn\{q_{e0}(0)\}$ is introduced to converge to the desired attitude through the shortest angular path.

The commanded attitude is expressed with respect to LVLH (Local Vertical Local Horizontal) frame $(\hat{r}, \hat{\theta}, \hat{h})$ (see figure 7.12).

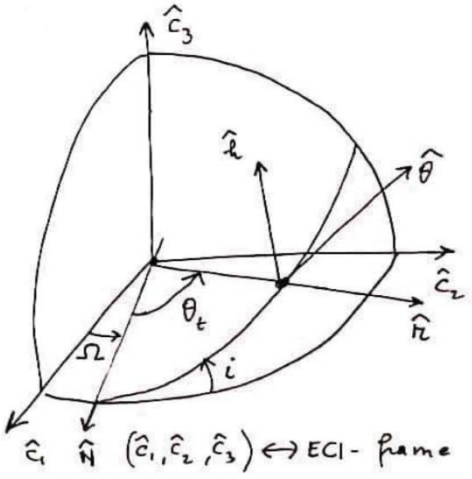


Figure 7.12: LVLH frame

Since it is required that $\hat{b}_{c_1} = \hat{\theta}$, $\hat{b}_{c_2} = -\hat{h}$ and $\hat{b}_{c_3} = -\hat{r}$ the rotation matrix from LVLH frame to commanded frame is

$$\begin{bmatrix} \hat{b}_{c_1} \\ \hat{b}_{c_2} \\ \hat{b}_{c_3} \end{bmatrix} = R_B \begin{bmatrix} \hat{r} \\ \hat{\theta} \\ \hat{h} \end{bmatrix} \qquad with \qquad R_B = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & -1 \\ -1 & 0 & 0 \end{bmatrix} \tag{7.4.2}$$

with $(\hat{b}_{c_1}, \hat{b}_{c_2}, \hat{b}_{c_3})$ that is the commanded frame. The LVLH frame is linked to ECI (Earth Centered Inertial) frame $(\hat{c}_1, \hat{c}_2, \hat{c}_3)$ by the following rotation matrices:

$$\begin{bmatrix} \hat{r} \\ \hat{\theta} \\ \hat{h} \end{bmatrix} = R_3(\theta_t) R_1(i) R_3(\Omega) \begin{bmatrix} \hat{c}_1 \\ \hat{c}_2 \\ \hat{c}_3 \end{bmatrix} \tag{7.4.3}$$

So, the commanded attitude at time $t_0 = 0\ s$ can be expressed with respect to ECI frame:

$$\begin{bmatrix} \hat{b}_{c_1} \\ \hat{b}_{c_2} \\ \hat{b}_{c_3} \end{bmatrix} = \underbrace{R_B R_3(\theta_t) R_1(i) R_3(\Omega)}_{C \overset{R}{\leftarrow} N} \begin{bmatrix} \hat{c}_1 \\ \hat{c}_2 \\ \hat{c}_3 \end{bmatrix} \tag{7.4.4}$$

The commanded quaternion $\left\{q_{c0}, \underline{q_c}\right\}$ at time t_0 can be easily obtained from $\underset{C\leftarrow N}{R}$. For the purpose of propagating $\left\{q_{c0}, \underline{q_c}\right\}$ but also for computing $\underline{T_C}$, $\underline{\omega_C}$ has to be found. It is the $3x1$ vector of components of $\underset{\longrightarrow}{{}^N\omega^C}$ written with respect to commanded frame. The following vectorial relation holds:

$$\underset{\longrightarrow}{{}^N\omega^C} = \underset{\longrightarrow}{{}^N\omega^L} + \underset{\longrightarrow}{{}^L\omega^C} \tag{7.4.5}$$

With N referred to ECI frame C referred to commanded frame L referred to LVLH frame, but $\underset{\longrightarrow}{{}^L\omega^C} = \underset{\rightarrow}{0}$ due to the choice of commanded frame. Since

$$\underset{\longrightarrow}{{}^N\omega^C} = \underset{\longrightarrow}{{}^N\omega^L} = \dot{\theta}_*\hat{h} = -\dot{\theta}_*\hat{b}_{c_2}$$

With θ_* the true anomaly. The sought $\underline{\omega_C}$ and its time derivative $\underline{\dot{\omega}_C}$ can be written as

$$\underline{\omega_C} = \begin{bmatrix} 0 \\ -\dot{\theta}_* \\ 0 \end{bmatrix} \tag{7.4.6}$$

$$\underline{\dot{\omega}_C} = \begin{bmatrix} 0 \\ -\ddot{\theta}_* \\ 0 \end{bmatrix} \tag{7.4.7}$$

The two quantities $\dot{\theta}_*$ and $\ddot{\theta}_*$ can be easily found from the following equations

$$\dot{\theta}_* = \frac{h}{r^2} \tag{7.4.8}$$

$$\ddot{\theta}_* = -\frac{2h}{r^3} \tag{7.4.9}$$

$$\dot{r} = -\frac{2h}{r^3} \tag{7.4.10}$$

$$v_r = -\frac{2h}{r^3}\sqrt{\frac{\mu}{p}}\, e\, \sin\theta_* \tag{7.4.11}$$

Now all the terms appearing in the feedback control law can be computed. The angular rate $\underline{\omega_D}$ is defined as

$$\underline{\omega_D} = \underline{\omega^{(B)}} - \underline{\omega_C^{(C)}} \tag{7.4.12}$$

The only missing variable term is the error quaternion. It is computed as

$$\begin{cases} q_{e_0} = q_{c_0} q_0 + \underline{q_e}^T \underline{q} \\ \underline{q_e} = -\underline{q_c} q_0 + q_{C_0} \underline{q} - \underline{\tilde{q}_c}\, \underline{q} \end{cases} \tag{7.4.13}$$

The body and commanded quaternions have to be computed through the integration of the following equations:

$$\begin{cases} \dot{q}_0 = -\frac{1}{2} \underline{\omega}^T \underline{q} \\ \underline{\dot{q}} = \frac{1}{2} \left(q_0 I_{3x3} + \underline{\tilde{q}} \right) \underline{\omega} \end{cases} \tag{7.4.14}$$

$$\begin{cases} \dot{q}_{c0} = -\frac{1}{2} \underline{\omega_c}^T \underline{q_c} \\ \underline{\dot{q_c}} = \frac{1}{2} \left(q_{c0} I_{3x3} + \underline{\tilde{q}_c} \right) \underline{\omega} \end{cases} \tag{7.4.15}$$

In order to complete the feedback control law, gain matrices have to be computed. The selection of the gain is accomplished in the following way. If one substitutes the feedback control law inside the Euler equation obtains

$$\underline{\dot{\omega_D}} = -A^{-1} B \underline{\omega_D} - A^{-1} \underline{q_e} \tag{7.4.16}$$

If the rotation is approximated as an eigenaxis rotation the following relations can be applied:

$$\begin{aligned} \underline{\dot{\omega_D}} &= \ddot{\Phi}_e \underline{a_D} \\ \underline{\omega_D} &= \dot{\Phi}_e \underline{a_D} \\ \underline{q_e} &= \sin\left(\frac{\Phi_e}{2}\right) \underline{a_D} \end{aligned} \tag{7.4.17}$$

and, choosing $A^{-1} = c_1 I_{3x3}$ and $B = c_2 I_{3x3}$, one obtains

$$\ddot{\Phi}_e + c_1 c_2 \dot{\Phi}_e - c_1 \sin\left(\frac{\Phi_e}{2}\right) = 0 \tag{7.4.18}$$

Passing to the canonical form, c_1 and c_2 can be derived from ω_n and ζ

$$c_1 = 2\omega_n^2$$
$$c_2 = \frac{\zeta}{\omega_n} \tag{7.4.19}$$

The parameters ζ and ω_n have to be chosen in accordance with the performances of the actuators and must respect their technological limits. Now the control torque can be inserted in the dynamical equation, written in the following form:

$$J_C^{(B)}\underline{\dot{\omega}} = -\underline{\tilde{\omega}}\, J_C^{(B)}\underline{\omega} + \underline{T_C} + \underline{M_C} \tag{7.4.20}$$

and integrate it in order to compute the body angular rate. $\underline{M_C}$ is the external torques vector. If we also consider the dynamic of actuators, we must distinguish between the theoretical control torque and the real torque provided by the reaction wheels. From the dynamical equations written for the spacecraft with the array of reaction wheels inside, one obtains that the torque provided by the actuators is:

$$\underline{T_a} = -\underline{\tilde{\omega}}A_w\underline{\omega_s} - A_w\underline{\dot{\omega}_s} \cong -A\underline{\dot{\omega}_s} \tag{7.4.21}$$

$$A_w = \begin{bmatrix} I_{s_1}\underline{a_1} & \cdots & I_{s_n}\underline{a_n} \end{bmatrix} \tag{7.4.22}$$

where $\underline{\omega_s}$ is the vector containing the wheels angular rates. It is possible to apply the previous approximation because $\underline{\omega}$ is small. By equating the actuators torque to the control torque while minimizing $\underline{\dot{\omega}_s}$, one obtains the pseudoinverse steering law.

$$\underline{\dot{\omega}_s} = -A_w^T(A_wA_w^T)^{-1}\underline{T_C} \tag{7.4.23}$$

So, substituting this equation in the complete expression of T_a one obtains

$$\underline{T_a} = -\underline{\tilde{\omega}}A_w\underline{\omega_s} + \underline{T_C} \tag{7.4.24}$$

that can be directly inserted in the Euler equation in place of $\underline{T_C}$. Because of the external disturbances, the wheels accumulate angular momentum that has to be desaturated. This is achieved by applying a desaturation torque with the magnetic torquers that have been chosen to accomplish the detumbling maneuver.

When one of the wheels reaches the maximum angular rate, the reactions wheels array stops providing the desired control torque and all the four wheels (because they all have similar values of angular rates) decelerate, providing the torque.

$$\underline{T_a} = -\underline{\tilde{\omega}} A_w \underline{\omega_s} - A_w \underline{\dot{\omega}_s} \tag{7.4.25}$$

with $\underline{\dot{\omega}_s}$ that now is the deceleration of the wheels (with negative components), which magnitude is chosen to be as large as possible in accordance with the possibilities of the magnetic torquers. So, the magnetic torquers shall provide the following torque:

$$\underline{T_{des}} = -\underline{T_a} + \underline{T_C} \tag{7.4.26}$$

that compensates the undesired torque provided by the reaction wheels and maintains the spacecraft close to the desired attitude, although the magnetic torquers can ensure a lower precision than reaction wheels. From $\underline{T_{des}}$ it is possible to compute the dipole moment that the torquerods shall provide. In fact,

$$\underline{T_{des}} = \underline{M_{des}} \times \underline{B} \tag{7.4.27}$$

and this relation can be inverted under the assumption that the three-axis magnetic torquers provides a dipole moment always near orthogonal to the magnetic field. So, one obtains that

$$\underline{B} \times \underline{T_{des}} = \underline{B} \times \left(\underline{M_{des}} \times \underline{B}\right) = \left(\underline{B} \cdot \underline{B}\right) \underline{M_{des}} - \left(\underline{B} \cdot \underline{M_{des}}\right) \underline{B} \tag{7.4.28}$$

and the dipole moment associated to $\underline{\mathrm{T_{des}}}$ is

$$\underline{M_{des}} = \frac{\underline{B} \times \underline{T_{des}}}{B^2} \tag{7.4.29}$$

This control law is then adapted to be applied to our CubeSat. After the detumbling phase the spacecraft has small angular rates, but its attitude is far from the desired one. So, there will be a first acquisition phase in which the spacecraft, starting from initial conditions, moves toward the desired attitude in terms of quaternions and angular rate.

In order to respect the maximum torque limit imposed by the reaction wheels chosen, a slightly different strategy is employed. The actuator torque during acquisition phase is

$$\underline{T_a} = -\underline{\tilde{\omega}} A_w \underline{\omega_s} + K \underline{T_C} \tag{7.4.30}$$

where K is a scalar function which increases linearly in time:

$$K = K_0 + \frac{t}{K_1} \tag{7.4.31}$$

with K_0 and K_1 chosen such that $0 \leq K \leq 1$ in the acquisition time interval $0 \leq t \leq 600$ s.Thanks to this strategy, the initial actuator torque (that is the maximum one) is lower than the maximum thrust available, but the attitude still converges to the desired one, as it can be clearly seen in the following figures.

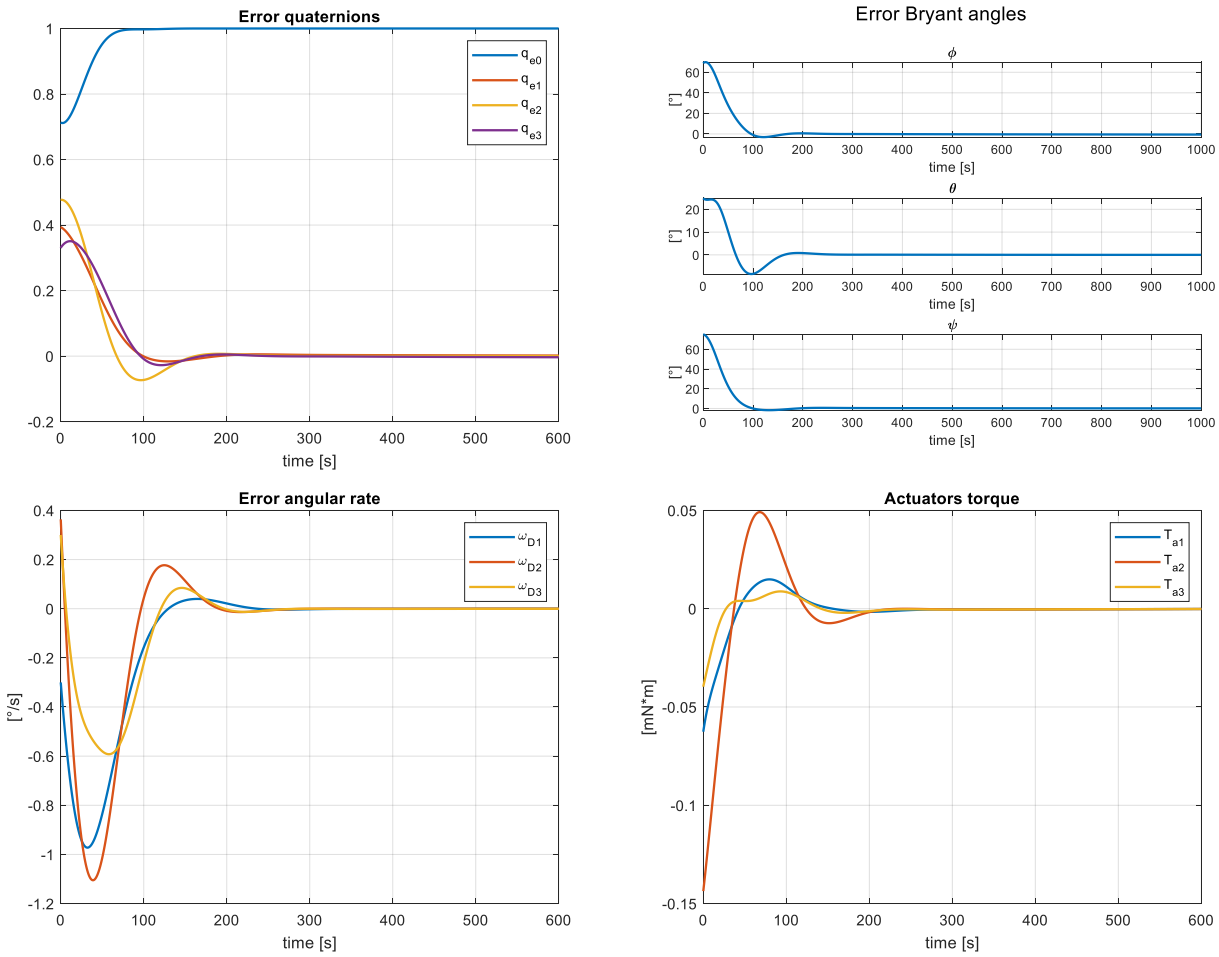


Figure 7.13: Euler quaternions (upper left), error Bryant errors (upper right), error angular rate (bottom left), actuators torque (bottom right)

Once convergence is achieved, the reaction wheels keep tracking the desired attitude until they reach their maximum angular rate. Every time it happens, they shall be desaturated using the strategy previously described. In terms of precision, during desaturation the payload does not observe, and it can increase the range of attitude requirements (better than 1° in pointing and 7° in desaturation mode). Anyway, the precision does not degrade excessively because the desaturation does not last long. The maximum dipole moment estimated must be lower than the maximum providable by the magnetic torquers chosen. The desaturation lasts approximatively 9 minutes and has to be accomplished every 3.1 hours (on average).

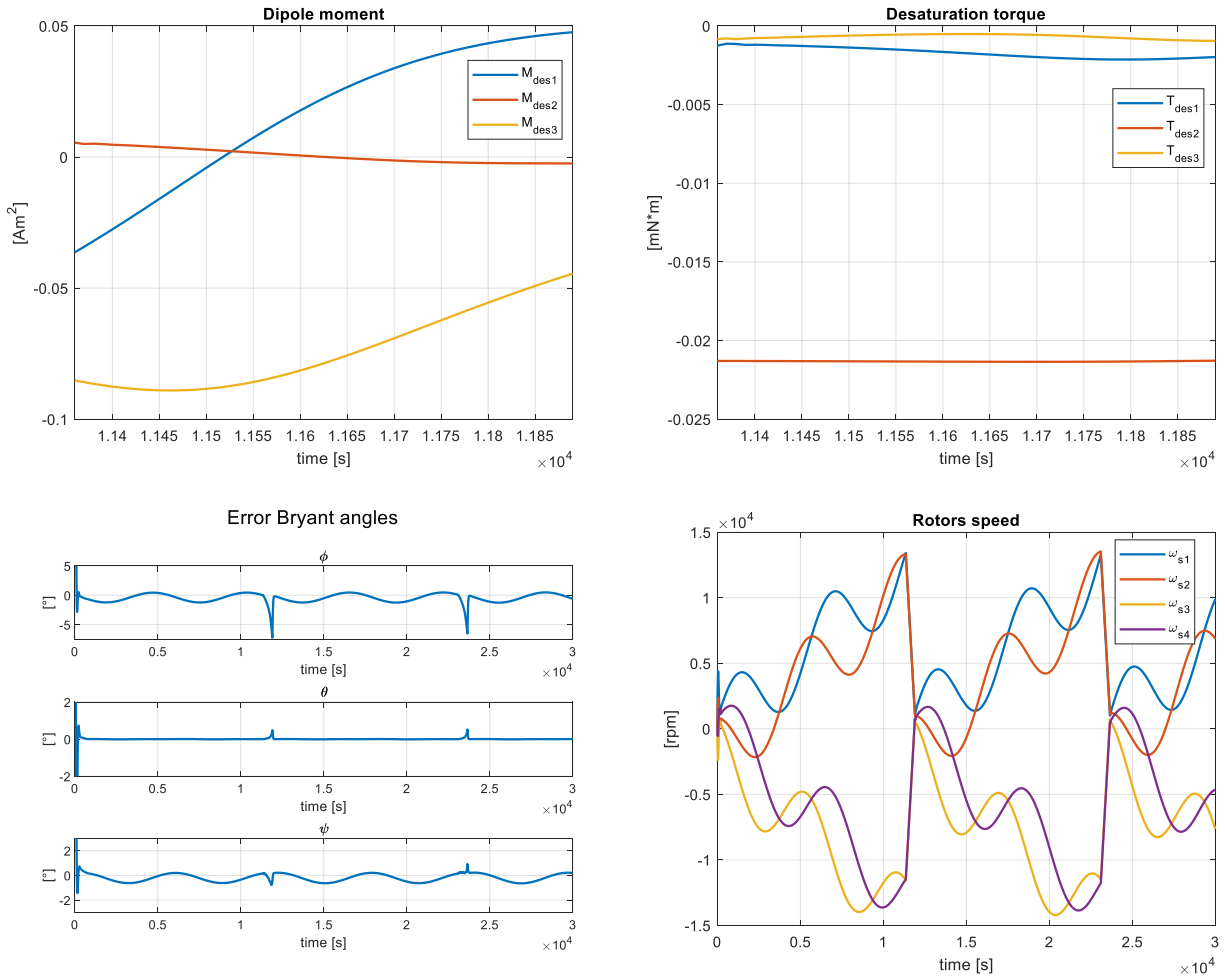


Figure 7.14: Dipole moment (upper left), desaturation torque (upper right), error Bryant angles (bottom left), rotors speed (bottom right).

8. Conclusions

Summarizing all the work carried out during Spacecraft Design 2020/2021, our subgroup analysed and studied the determination and control system for the orbit and attitude (AODCS) of the CubeSat 6U HYONOSAT. In this chapter we report what is the final configuration and the choice of the various methods implemented of all the simulations and analyses that we have developed in the previous chapters.

ORBIT	
Analysis	Propagator → J2 STK
	Magnetic Field →IGRF-13
	Disturbances → Worst case Analysis
Determination	GNSS Receiver
Control	Maintenance → Low-Thrust MULTIFEEP
	Deorbiting → Passive with Solar Panels
ATTITUDE	
Determination	Kalman Filter
Control	Detumbling →B-dot
	Nadir Pointing →Nonlinear three axis stabilisation control with 4 reaction wheels + desaturation

Table 8.1: Final simulation choices.

The analysis presented in this document were carried out in accordance with the requirements and restrictions imposed by the mission, linked to problems of cost, mass and power. Consequently, in some of the analysis we have accepted less precision to stay within the restrictions.

Future developments in this work could be the repetition of the simulations made with the Monte Carlo method in order to obtain a statistical approach of the results obtained instead of considering only the worst case which was the choice taken into consideration in this work. Another improvement to the analysis carried out can be the implementation of control software with control system diagrams through Simulink. Finally, more precise models could be obtained for the disturbing torques encountered in orbit over time and consequently have an orbit propagator based on this more accurate torque model.